CHAPTER III

Projector Resolved Seed Calculus III

Typed transfer, nonlinear packets, and obstruction theory

Typed operations, nonlinear coordinates, and correction maps give a common language for reconstruction, transport, and obstruction.

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Mathematical text

How to use this volume

Begin with typed edge words and exact coordinate transport. A chosen connection introduces defects, holonomy, and packet obstruction witnesses. Part II develops nonlinear packet jets, tree expansions, correction complexes, filtered lifting, elimination, and smooth transport.
Part I develops the finite algebraic core under its declaration-level formal contract. Part II gives the nonlinear and analytic development with ordinary proofs. The accompanying formal development verifies exact nonlinear coordinates, continuous packet jets on actual projector ranges, and the decorated composition and structural support laws in Lean. The integral Taylor remainder and the later analytic and obstruction results retain their ordinary proof status. Formal coverage stays attached to the identified declarations. The visual atlas at the end of this volume provides twenty-four worked plates. The separate Lean guide explains the formal contract and its evidence.

Overview

Finite projector packets turn linear transfer into typed walks and arbitrary maps into exact coordinate representations. Canonical coordinate transport satisfies an exact cocycle law. A chosen connection can fail to compose, producing holonomy and packet obstruction witnesses. Nonlinear packet jets lead to higher composition, convergent rooted tree expansions, graded operations, and nonlinear refinement defects. A specified correction map produces a canonical cokernel obstruction. Relative exact sequences, mapping cones, Hilbert splitting, nonlinear correction, and filtered lifting describe how these obstructions behave. Exact elimination produces Schur formulas, and finite transport reduces gluing to fundamental cycle holonomy. For smooth orthogonal packet families, direct packet rotations converge in operator norm to canonical unitary transport. The finite formal contract and the broader ordinary proof development retain their separate scopes throughout.

The chapter presents the mathematical development together with worked examples, a visual atlas, and a separate guide to the formal proof evidence. The packet norm estimate distinguishes the actual range jet from its ambient extension. The support statements keep external packet profiles and intermediate composition witnesses explicitly typed. The formal companion gives the exact source and scope of each formal verification.

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4 Second packet jets for $F(x, y) = (x + xy, y + x^2)$ at the origin. Coordinate axis projectors give $DF(0) = I$. The first output has mixed second derivatives equal to one, while the second has a repeated first input derivative equal to two. Each grid records the ordered pair of input packets. The mixed entries retain an interaction that diagonal restrictions would lose. 34

5 A specified correction range in $\mathbb{R}^2$. For $L(a, b) = (a, 0)$, the image is the horizontal axis. The defect $(2, 3)$ splits into $(2, 0)$ in that image and the remaining representative $(0, 3)$. Subtracting the image component leaves the same cokernel class. The functional $(x, y) \mapsto y$ vanishes on the correction range and detects the value three. The correction equation $Lu = -(2, 3)$ therefore has no solution. 46

6 A rotating rank-one packet and its orthogonal complement. With $u(\theta) = (\cos\theta, \sin\theta)$ and $P(\theta) = u(\theta)u(\theta)^{\mathsf{T}}$, the canonical generator is $[P', P] = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. For a positive step $\delta < \pi/2$, the polar unitary is the usual rotation $R(\delta)$. On each displayed mesh, the direct rotation product equals $R(\pi/3)$ exactly. This special planar example illustrates the construction. The general theorem supplies convergence for families whose step products need not already equal the continuous transport. 67

Part I Finite typed transfer and the formal contract

The finite algebraic core supplies the algebraic foundation and a contract for its individual declarations. Here, "roadmap" and "outside the contract" describe the scope of the finite Lean formalization. Ordinary mathematical results in Part II may lie beyond that scope. The finite contract covers thirty result environments. The source contract has 41 mapping rows because it also includes definitions and other interfaces.

The formal companion records the exact Lean and Mathlib dependencies [3, 17], compilation, kernel replay, and axiom checks. References to declarations from Chapters I–II retain their explicitly inherited status. The companion distinguishes the finite algebraic contract from the additional packet-coordinate and analytic interfaces.

1 Purpose and formal status

Chapter I fixed typed seeds, operator grammars, exact projector packets, generated hulls, and reconstruction [13]. Chapter II added coherent refinement coordinates, weighted transfer, arithmetic separation, and linear defect identities [14]. The present chapter begins the next layer: explicit edge words, optional connection data, and obstructions within individual packets.

Canonical coordinate transport and a connection play different mathematical roles. Canonical transport compares two exact descriptions of the *same* ambient object and is forced by analysis and synthesis. A connection chooses transports that may carry extra modeling, discretization, or gauge information. Canonical transport obeys a cocycle theorem. A connection may fail to obey a cocycle, and that failure is meaningful data. Keeping separate notation lets us distinguish the automatic coherence of exact coordinates from the possible holonomy of a chosen connection.

1.1 Statement labels

Every substantial theorem, lemma, corollary, proposition, definition, API, and roadmap item carries one or more of the three labels below. When separate clauses use a Chapter III result and an inherited bridge, the statement carries both applicable labels. Illustrative examples do not require a label.

- **S1. Lean-backed in Chapter III**: the displayed scope is no stronger than the named compiled declaration in the identified Chapter III module.
- **S2. Inherited Lean-backed**: the displayed result is supplied by the frozen formal contract of Chapter I or Chapter II.
- **S3. Roadmap only**: the text states a future interface or theorem target and is excluded from the completion denominator.

Boundary 1.1 (Finite theorem denominator). The completion denominator of the finite algebraic contract consists of the thirty result environments in Part I. Of these, 26 resolve to compiled Chapter III declarations and four resolve to the frozen Chapters I–II contracts. The authoritative theorem map names the exact declarations used by each statement. The kernel audit checks those targets and their supporting theorems. The namespace gate requires every declaration in the compiled `Komorebi.PaperIII`

namespace to pass its checks. Roadmap items are excluded from the denominator and remain visibly marked.

2 Finite typed packet graphs

2.1 Orientation and matrix convention

Let V be a finite packet type, E a finite edge type, and R a semiring. A directed multigraph is a pair of endpoint functions

$$\mathrm{src}, \mathrm{tgt} : E \longrightarrow V.$$

Parallel edges remain distinct elements of E. Rows are targets and columns are sources. Thus an edge $e : i \to j$ is represented by a matrix supported at row j, column i.

Definition 2.1 (Packet graph and scalar edge system)**.** For $a, b \in V$ and $r \in R$, define

$$\mathrm{packetMatrix}(a, b; r)_{yx} = \begin{cases} r, & y = b \text{ and } x = a, \\ 0, & \text{otherwise.} \end{cases}$$

An edge system is a coefficient function $c : E \to R$. Its elementary edge operator and total transfer operator are

$$L_e = \mathrm{packetMatrix}(\mathrm{src}(e), \mathrm{tgt}(e); c(e)), \qquad T = \sum_{e \in E} L_e.$$

Formal contract: finite algebraic core.

`Komorebi.PaperIII.EdgeOperators.PacketGraph;Komorebi.PaperIII.EdgeOperators.PacketGraph.packetMatrix;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp`

Words are written in execution order. In $L_f L_e$, the rightmost operator L_e acts first. For a written word $(e_1, \dots, e_n)$, e_1 acts first and the operator product is $L_{e_n} \cdots L_{e_1}$.

Reading the execution order

Read the edge list in the order of execution. The matrix product writes later actions farther to the left. The target packet of one edge must equal the source packet of the next before multiplication can contribute. Over a noncommutative semiring, this order also fixes the order of the scalar coefficients.

Lemma 2.2 (Declared support)**.** *For every edge e, $(L_e)_{yx} = 0$ unless $y = \mathrm{tgt}(e)$ and $x = \mathrm{src}(e)$.*

Formal contract: finite algebraic core.

`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp_eq_zero_off_support`

Theorem 2.3 (Typed interface annihilation)**.** *If $\mathrm{tgt}(e) \neq \mathrm{src}(f)$, then*

$$L_f L_e = 0.$$

Formal contract: finite algebraic core.

`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp_mul_edgeOp_eq_zero`

Proof. For a matrix entry (y, x),

$$(L_f L_e)_{yx} = \sum_{z \in V} (L_f)_{yz} (L_e)_{zx}.$$

A potentially nonzero summand would require simultaneously $z = \mathrm{src}(f)$ and $z = \mathrm{tgt}(e)$, contrary to the hypothesis. □

Theorem 2.4 (Collapse of a matched edge pair). *If* $\mathrm{tgt}(e) = \mathrm{src}(f)$, *then*

$$L_f L_e = \mathrm{packetMatrix}(\mathrm{src}(e), \mathrm{tgt}(f); c(f)c(e)).$$

Together with theorem 2.3, this gives both cases for the composition of two edge operators.

Formal contract: finite algebraic core.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp_mul_edgeOp_of_composable`

Proof. The sum defining the matrix product has exactly one potentially nonzero intermediate index, namely $\mathrm{tgt}(e) = \mathrm{src}(f)$. At that index the coefficients occur in operator order as $c(f)c(e)$, and the outer source and target are $\mathrm{src}(e)$ and $\mathrm{tgt}(f)$. □

Boundary 2.5 (Current fiber scope). The compiled Chapter III model above has one scalar coordinate per packet. For a family of modules $(X_i)_{i \in V}$, an intrinsic edge map would instead have type $\ell_e : X_{\mathrm{src}(e)} \to X_{\mathrm{tgt}(e)}$, and its totalization on $\bigoplus_i X_i$ would be $\iota_{\mathrm{tgt}(e)} \ell_e \pi_{\mathrm{src}(e)}$. The same support proof applies, but this generalization with dependent fibers is not attributed to the released Chapter III modules for scalar edge systems.

Figure 1 summarizes the support convention and shows how composable words differ from words with a broken interface. The next section uses this distinction.

3 Operator words and directed walks

3.1 Typed words

Definition 3.1 (Composable word and execution operator). A finite word $w = (e_1, \ldots, e_n)$ is composable when

$$\mathrm{tgt}(e_r) = \mathrm{src}(e_{r+1}) \qquad (1 \le r < n).$$

Set

$$L_w = L_{e_n} \cdots L_{e_1}, \qquad L_\emptyset = I.$$

This definition records the written order of execution even though matrix composition is read right to left.

Formal contract: finite algebraic core.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.Composable;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp`

Theorem 3.2 (First broken interface). *If a word begins* $(e, f, \ldots)$ *and* $\mathrm{tgt}(e) \neq \mathrm{src}(f)$, *then its word operator is zero.*

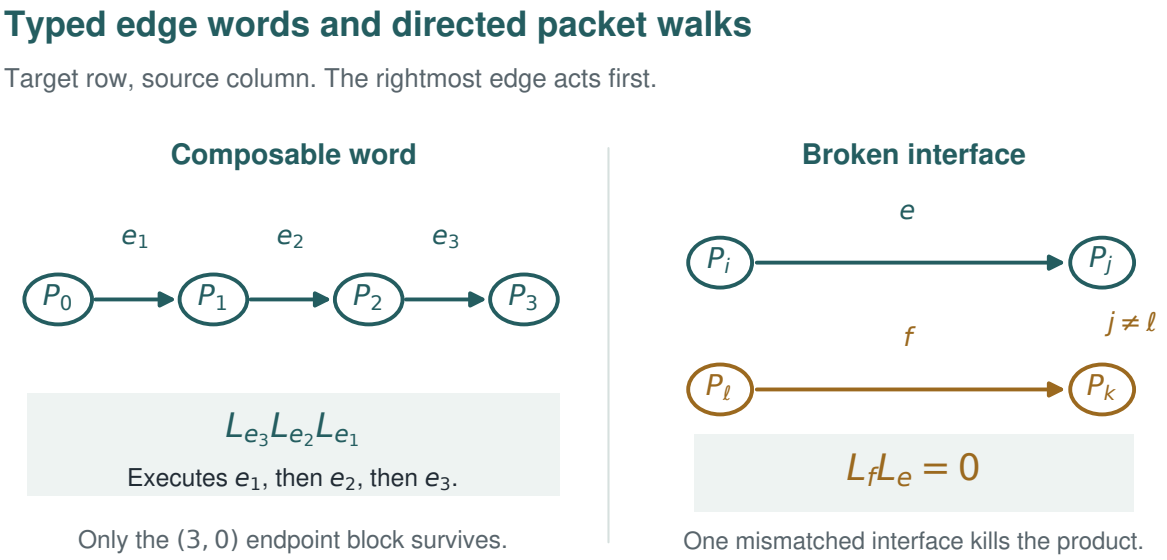


Figure 1. Rows record target packets and columns record source packets. This convention fixes execution order. Composable words collapse to endpoint blocks, while one broken packet interface annihilates the entire operator product.

Formal contract: finite algebraic core.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp_eq_zero_of_first_break`

Theorem 3.3 (Noncomposable words vanish). *If w is not composable, then $L_w = 0$.*

Formal contract: finite algebraic core.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp_eq_zero_of_not_composable`

Proof. Choose the first broken adjacent pair. The corresponding consecutive factor is zero by theorem 2.3, and associativity absorbs that zero into the complete product. The Lean proof implements the same argument by induction on the list. □

3.2 The checked recursive expansion

Definition 3.4 (Recursive word expansion). Define

$$\mathcal{W}_0 = I, \qquad \mathcal{W}_{n+1} = T\mathcal{W}_n.$$

Formal contract: finite algebraic core.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordExpansion`

Theorem 3.5 (Transfer power equals recursive word expansion). *For every $n \in \mathbb{N}$,*

$$T^n = \mathcal{W}_n.$$

Formal contract: finite algebraic core.

`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_eq_wordExpansion`

Proof. The zero case is the identity matrix. If $T^n = \mathcal{W}_n$, then the left-successor power convention gives $T^{n+1} = TT^n = T\mathcal{W}_n = \mathcal{W}_{n+1}$. □

Remark 3.6. The formal module also defines an explicit sum over tuples $w : \mathrm{Fin}(n) \to E$ and a second sum in which noncomposable tuples are replaced by zero. It proves that the recursive matrix, the explicit tuple sum, and the sum filtered by composability agree. The separate declarations make the finite enumeration claim directly checkable.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.explicitWordExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.walkExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordExpansion_eq_explicitWordExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.explicitWordExpansion_eq_walkExpansion`

3.3 Exact expansion from words to walks

For a nonempty composable word $w = (e_1, \dots, e_n)$, define its reverse coefficient product

$$c^{\leftarrow}(w) = c(e_n)c(e_{n-1}) \cdots c(e_1).$$

The reversal is essential over a noncommutative semiring.

Lemma 3.7 (One word is one endpoint block). *If $w = (e_1, \dots, e_n)$ is nonempty and composable, then*

$$L_w = \mathrm{packetMatrix}(\mathrm{src}(e_1), \mathrm{tgt}(e_n); c^{\leftarrow}(w)).$$

Formal contract: finite algebraic core.

`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.reverseCoefficient;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.terminalTarget;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp_eq_packetMatrix_of_composable`

Proof. Induct on n. The case $n = 1$ is the definition of L_e. Suppose the formula holds for the prefix $(e_1, \dots, e_n)$ and append f. Composability gives $\mathrm{tgt}(e_n) = \mathrm{src}(f)$. Multiplying the two packet matrices leaves one intermediate index and changes the coefficient to $c(f)c(e_n) \cdots c(e_1)$, with source $\mathrm{src}(e_1)$ and target $\mathrm{tgt}(f)$. □

Write $\mathrm{Walk}_n(i, j)$ for the finite set of composable edge tuples of length n whose first source is i and whose final target is j. For $n = 0$, it has one empty walk when $i = j$ and is empty otherwise. Assign the empty walk the coefficient 1.

The empty walk

The empty walk records doing nothing at a chosen packet. It contributes only to a diagonal entry. Thus the convention for length zero matches the identity matrix and lets the walk sum include every nonnegative power.

Theorem 3.8 (Finite operator expansion from words to walks)**.** *For every* $n \in \mathbb{N}$,

$$T^n = \sum_{w \in E^n} L_w = \sum_{w \in E^n} \begin{cases} L_w, & w \text{ composable}, \\ 0, & w \text{ not composable}. \end{cases}$$

Parallel edges occur as distinct summands.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_eq_explicitWordExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_eq_walkExpansion`

Proof. Induction and finite distributivity give

$$\left(\sum_{e \in E} L_e \right)^n = \sum_{(e_1, \ldots, e_n) \in E^n} L_{e_n} \cdots L_{e_1}.$$

Every noncomposable summand is zero by theorem 3.3, so inserting the composability filter does not change the sum. In Lean, tuple concatenation and the successor sum are handled by `tupleWordOp_snoc` and `explicitWordExpansion_succ`. The two bridge equalities then yield the displayed theorem. □

Corollary 3.9 (Endpoint coefficient walk sum)**.** *For all* $n \in \mathbb{N}$ *and* $i, j \in V$,

$$(T^n)_{ji} = \sum_{w \in \mathrm{Walk}_n(i,j)} c^{\leftarrow}(w).$$

Formal contract: finite algebraic core.

`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.Connects;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.endpointWalkCoefficientSum;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_apply_eq_sum_walks`

Proof. Take the (j, i) entry in theorem 3.8. By lemma 3.7, a surviving word contributes $c^{\leftarrow}(w)$ exactly when its endpoints are i and j, and contributes zero otherwise. At $n = 0$, the identity matrix gives the unique empty diagonal walk. □

Corollary 3.10 (Reachability obstruction at fixed length)**.** *If* $\mathrm{Walk}_n(i, j) = \emptyset$, *then* $(T^n)_{ji} = 0$.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_apply_eq_zero_of_no_walk`

Proof. This is immediate from the empty sum in the preceding corollary. □

Corollary 3.11 (Orbit support obstruction). *Let $S \subseteq V$, and let $x \in R^V$ be supported on S. If j is not reachable by a directed path from any vertex of S, then*

$$(T^n x)_j = 0 \qquad \text{for every } n \in \mathbb{N}.$$

Hence the orbit under T of x cannot generate a vector with a nonzero coordinate at j.

Formal contract: finite algebraic core.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.Reachable;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_mulVec_apply_eq_zero_of_unreachable_support`

Inherited formal contract: Chapters I–II.
`Komorebi.PacketNaturalityCompletion.Reachability.algebraicOrbitHull_le_reachable_support;Komorebi.PacketNaturalityCompletion.Reachability.unreachable_nonzero_packet_obstruction`

Proof. Expand $(T^n x)_j = \sum_i (T^n)_{ji} x_i$. Terms with $i \notin S$ vanish by the support of x, and terms with $i \in S$ vanish by corollary 3.10. The final claim follows because the j-coordinate also vanishes on every finite linear combination of orbit vectors. □

4 Canonical exact coordinate transport

4.1 The Chapter III frame model

Let every chart a carry a linear frame equivalence $F_a : M \xrightarrow{\sim} M$. The formal Chapter III module defines transport from a to b by

$$J_{b \leftarrow a} = F_b^{-1} F_a.$$

This frame formula means *synthesize in chart a, then analyze in chart b.*

Theorem 4.1 (Exact frame transport is coherent). *For every a, b, c,*

$$J_{a \leftarrow a} = \mathrm{id}, \qquad J_{a \leftarrow b} J_{b \leftarrow a} = \mathrm{id}, \qquad J_{c \leftarrow b} J_{b \leftarrow a} = J_{c \leftarrow a},$$

and therefore

$$J_{a \leftarrow c} J_{c \leftarrow b} J_{b \leftarrow a} = \mathrm{id}.$$

Formal contract: finite algebraic core.
`Komorebi.PaperIII.Transport.exactTransport_self;Komorebi.PaperIII.Transport.exactTransport_roundTrip;Komorebi.PaperIII.Transport.exactTransport_cocycle;Komorebi.PaperIII.Transport.exactTransport_triangleLoop`

Proof. Every identity is cancellation of adjacent $F_r F_r^{-1}$ factors. The Lean definitions use `LinearEquiv.trans`. The displayed formulas use the equivalent right-to-left notation for function composition fixed in this chapter. □

Theorem 4.2 (Every finite canonical path telescopes). *Let $p = (a_0, a_1, \dots, a_n)$ be any finite vertex list and put*

$$J_p = J_{a_n \leftarrow a_{n-1}} \cdots J_{a_1 \leftarrow a_0}, \qquad J_{(a_0)} = \mathrm{id}.$$

Then

$$J_p = J_{a_n \leftarrow a_0}.$$

In particular, if $a_n = a_0$*, then* $J_p = \mathrm{id}$.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.pathEndpoint;Komorebi.PaperIII.Transport.exactPathTransport;Komorebi.PaperIII.Transport.exactPathTransport_eq_endpoint;Komorebi.PaperIII.Transport.exactPathTransport_loop`

Proof. Induct on the list of successive vertices and apply the cocycle in theorem 4.1 at each step. If the endpoint is the start, the endpoint transport is the identity. □

4.2 Exact packet coordinates

The graph model above resolves a fixed scalar matrix. Coordinate transport is a separate construction. Let X be an R-module, and let

$$\mathcal{P}^a = (P_i^a)_{i \in I_a}$$

be a finite complete family of pairwise orthogonal linear idempotents:

$$P_i^a P_j^a = \delta_{ij} P_i^a, \qquad \sum_{i \in I_a} P_i^a = I_X.$$

Define the exact packet coordinate module

$$\mathrm{Coord}(\mathcal{P}^a) = \left\{ (x_i)_{i \in I_a} : P_i^a x_i = x_i \right\}.$$

Its analysis and synthesis maps are

$$\mathcal{A}_a(x) = (P_i^a x)_i, \qquad \mathcal{S}_a((x_i)_i) = \sum_i x_i.$$

Completeness and orthogonality give

$$\mathcal{S}_a \mathcal{A}_a = \mathrm{id}_X, \qquad \mathcal{A}_a \mathcal{S}_a = \mathrm{id}_{\mathrm{Coord}(\mathcal{P}^a)}.$$

What a packet coordinate contains

A packet coordinate is a whole vector fixed by its projector. Analysis splits an ambient vector into these components, and synthesis adds them back. The transport between exact systems keeps the ambient vector and changes its packet presentation.

Definition 4.3 (Canonical exact coordinate transport)**.** For two exact packet systems on the same ambient module, define

$$J^X_{b \leftarrow a} = \mathcal{A}_b \mathcal{S}_a : \mathrm{Coord}(\mathcal{P}^a) \longrightarrow \mathrm{Coord}(\mathcal{P}^b).$$

The arrow points from the source coordinate system a to the target coordinate system b.

Inherited formal contract: Chapters I–II.

`Komorebi.PaperII.NestedRefinement.refineCoordinates;Komorebi.PaperII.NestedRefinement.coordinateRefinementEquiv`

Theorem 4.4 (Identity, inverse, and cocycle for canonical transport)**.** *For any exact packet systems a, b, c on X,*

$$J^X_{a\leftarrow a} = \mathrm{id}, \qquad \left(J^X_{b\leftarrow a}\right)^{-1} = J^X_{a\leftarrow b}, \qquad J^X_{c\leftarrow b}J^X_{b\leftarrow a} = J^X_{c\leftarrow a}.$$

Inherited formal contract: Chapters I–II.

`Komorebi.PaperII.NestedRefinement.refineCoordinates_self;Komorebi.PaperII.NestedRefinement.coarsenCoordinates_refineCoordinates;Komorebi.PaperII.NestedRefinement.refineCoordinates_coarsenCoordinates;Komorebi.PaperII.NestedRefinement.refineCoordinates_comp;Komorebi.PaperII.NestedRefinement.coordinateRefinementEquiv_trans`

Proof. Insert $\mathcal{S}_b\mathcal{A}_b = \mathrm{id}_X$:

$$J^X_{c\leftarrow b}J^X_{b\leftarrow a} = \mathcal{A}_c\mathcal{S}_b\mathcal{A}_b\mathcal{S}_a = \mathcal{A}_c\mathcal{S}_a = J^X_{c\leftarrow a}.$$

The identity statement is $\mathcal{A}_a\mathcal{S}_a = \mathrm{id}$. Setting $c = a$ in the cocycle gives $J^X_{a\leftarrow b}J^X_{b\leftarrow a} = \mathrm{id}$. Interchanging a and b gives the other composition, so the two transports are inverses. □

If $F : X \to Y$ is linear and $\mathcal{P}^a, \mathcal{P}^b$ are source packet systems while $\mathcal{Q}^r, \mathcal{Q}^s$ are target packet systems, put

$$\widehat{F}_{r\leftarrow a} = \mathcal{A}^Y_r F \mathcal{S}^X_a.$$

Theorem 4.5 (Canonical coordinate naturality)**.** *The exact coordinate representation preserves typed composition and satisfies*

$$J^Y_{s\leftarrow r}\widehat{F}_{r\leftarrow a} = \widehat{F}_{s\leftarrow b}J^X_{b\leftarrow a}.$$

Inherited formal contract: Chapters I–II.

`Komorebi.PaperII.NestedRefinement.coordinateOperator_comp;Komorebi.PaperII.NestedRefinement.coordinateOperator_naturality`

Proof. Both sides reduce to $\mathcal{A}^Y_s F \mathcal{S}^X_a$ after inserting the appropriate synthesis–analysis identity. The composition law is the same calculation with the middle ambient module inserted. □

Corollary 4.6 (Canonical triangle holonomy is trivial)**.** *For every triple a, b, c,*

$$J^X_{a\leftarrow c}J^X_{c\leftarrow b}J^X_{b\leftarrow a} = \mathrm{id}.$$

Inherited formal contract: Chapters I–II.

`Komorebi.PaperII.NestedRefinement.refineCoordinates_comp;Komorebi.PaperII.NestedRefinement.refineCoordinates_self`

Boundary 4.7 (Refinement is not reduction). Although Chapter II uses the name *refineCoordinates*, J is an equivalence between two complete coordinate presentations. It neither truncates packets nor projects to a smaller state space. Any lossy restriction, interpolation, or learned identification is extra data and cannot inherit theorem 4.4 without proof.

5 A distinct noncanonical connection API

5.1 The minimal compiled API

The compiled Chapter III API assumes only normalization at each chart. All chart transports act on one module M, so chart labels distinguish routes but not Lean types.

Definition 5.1 (Noncanonical connection normalized at each chart)**.** A minimal connection is a family

$$\Gamma_{b\leftarrow a} \in \mathrm{Aut}_R(M) \qquad (a, b \in B)$$

with only

$$\Gamma_{a\leftarrow a} = \mathrm{id}_M.$$

Neither inversion on reversed edges nor the cocycle law is a field of the structure.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.Connection`

At a triple (a, b, c), the compiled cocycle predicate is

$$\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} = \Gamma_{c\leftarrow a}.$$

Flatness means this equality holds for every triple. The formal multiplicative relative defect and triangle holonomy are

$$D^{\times}_{c,b,a} = (\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a})\Gamma^{-1}_{c\leftarrow a}, \qquad H_{c,b,a} = \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a}.$$

The defect is based at the target c. The holonomy is based at a.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.Connection.CocycleAt;Komorebi.PaperIII.Transport.Connection.Flat;Komorebi.PaperIII.Transport.Connection.cocycleDefect;Komorebi.PaperIII.Transport.Connection.triangleHolonomy`

Theorem 5.2 (Checked flatness consequences)**.** *For the minimal connection API, flatness implies*

$$D^{\times}_{c,b,a} = \mathrm{id}_M, \qquad H_{c,b,a} = \mathrm{id}_M$$

for every a, b, c.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.Connection.cocycleDefect_eq_refl_of_flat;Komorebi.PaperIII.Transport.Connection.triangleHolonomy_eq_refl_of_flat`

Proof. Flatness replaces the two-edge route in $D^{\times}$ by the direct route, which cancels its inverse. For H, flatness first collapses $a \to b \to c$ to $a \to c$, then collapses $a \to c \to a$ to the normalized transport $a \to a$. □

Proposition 5.3 (The minimal API admits nontrivial holonomy)**.** *There is a connection on three points over $\mathbb{Q}$ for which transport $0 \to 1$ is multiplication by 2, every other transport between distinct endpoints is the identity, and*

$$H_{2,1,0}(1) = 2.$$

In particular $H_{2,1,0} \neq \mathrm{id}$.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.twistedThreePointConnection;Komorebi.PaperIII.Transport.twisted_triangleHolonomy_apply_one;Komorebi.PaperIII.Transport.twisted_triangleHolonomy_ne_refl`

Boundary 5.4 (Why the minimal defect and loop are distinct). Without reverse-edge inversion, $\Gamma_{a\leftarrow c}$ need not equal $\Gamma^{-1}_{c\leftarrow a}$. Therefore the multiplicative relative defect $D^{\times}_{c,b,a}$ and triangle holonomy $H_{c,b,a}$ are different comparisons. The formal module proves that global flatness trivializes both. It does not identify them for an arbitrary connection.

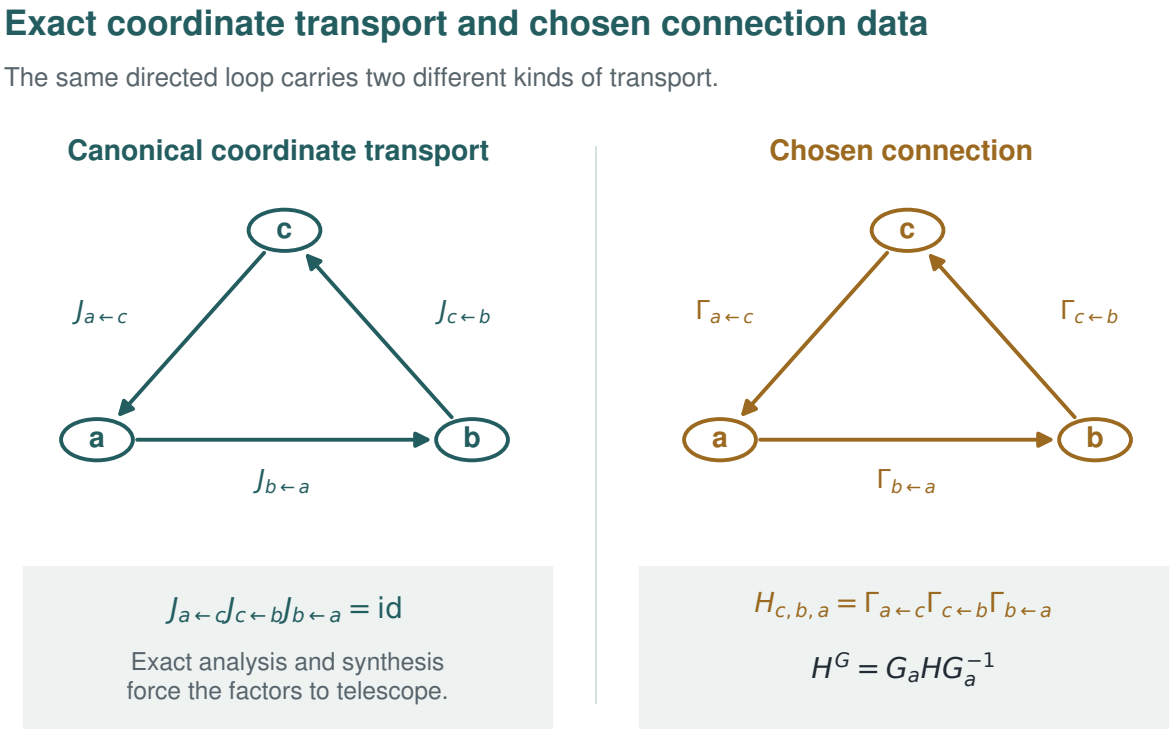


Figure 2. Exact coordinate transport and connection transport are distinct. The canonical loop telescopes to the identity. A chosen connection may carry nontrivial holonomy, which transforms by conjugation under gauge.

Figure 2 records this separation visually: the exact coordinate route telescopes, whereas the chosen connection route may carry a nonidentity loop.

Coordinate transport and connection choice

Canonical transport compares exact coordinate descriptions of the same object. Its loops telescope. A connection supplies a separate family of comparison maps, so its loops can carry information about that choice. The distinction keeps coordinate identities separate from connection assumptions.

5.2 A stronger reversible pair API

Let R be a commutative ring, let B be a type of charts or refinement states, and let each C_a be an R-module. The chart type is not assumed finite. The following definition is deliberately not derived from analysis and synthesis.

Definition 5.5 (Normalized reversible connection)**.** A normalized reversible connection is a family of linear equivalences

$$\Gamma_{b\leftarrow a} : C_a \xrightarrow{\sim} C_b \qquad (a, b \in B)$$

such that

$$\Gamma_{a\leftarrow a} = \mathrm{id}, \qquad \Gamma_{a\leftarrow b} = \Gamma_{b\leftarrow a}^{-1}.$$

No cocycle equation is included in the structure.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection`

This adds heterogeneous fibers and inversion on reversed edges to the homogeneous minimal connection. The connection is still a separate choice of maps, rather than a definition of canonical coordinate transport.

Definition 5.6 (Calibration and flatness)**.** When the C_a are exact packet coordinates of one ambient module, the connection is *canonically calibrated* if

$$\Gamma_{b\leftarrow a} = J_{b\leftarrow a} \quad \text{for every } a, b.$$

It is *flat* if

$$\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} = \Gamma_{c\leftarrow a} \quad \text{for every } a, b, c.$$

The canonical connection is flat. A general flat connection is proved in theorem 6.3 to be related to it by a gauge change.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection.CocycleAt;Komorebi.PaperIII.Transport.ReversibleConnection.Flat;Komorebi.PaperIII.Transport.ReversibleConnection.canonicalConnection;Komorebi.PaperIII.Transport.ReversibleConnection.canonicalConnection_flat`

The words *calibrated* and *flat* have only the algebraic meanings displayed above.

Remark 5.7 (A connection on a chart graph). One may specify Γ_e only on the oriented edges of a chart graph, together with inverse maps on reversed edges. Path transport and loop holonomy below still make sense. A direct cocycle defect for (a, b, c), however, requires a declared direct transport from a to c. The pairwise API in definition 5.5 keeps this comparison total.

5.3 Cocycle defect and triangle holonomy

The commutative ring hypothesis makes subtraction of linear maps available from this point through section 7.

Definition 5.8 (Cocycle defect)**.** For $a, b, c \in B$, define the additive cocycle defect

$$\Delta_{c,b,a} = \Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} - \Gamma_{c\leftarrow a} \in \mathrm{Hom}_R(C_a, C_c).$$

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection.cocycleDefect`

The manuscript subscript order is target, intermediate, source. The Lean arguments are supplied in traversal order `(a,b,c)`.

Definition 5.9 (Triangle holonomy)**.** The triangle holonomy based at a is

$$H_{c,b,a} = \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} \in \mathrm{Aut}_R(C_a).$$

It traverses $a \to b \to c \to a$.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection.triangleHolonomy;Komorebi.PaperIII.Transport.ReversibleConnection.triangleObstruction`

Holonomy is attached to the chosen connection Γ, never to an unspecified coordinate change.

Theorem 5.10 (Identity between defect and holonomy)**.** *For every triple* (a, b, c)*,*

$$\boxed{H_{c,b,a} - \mathrm{id}_{C_a} = \Gamma_{a\leftarrow c}\Delta_{c,b,a}.}$$

Consequently,

$$\Delta_{c,b,a} = 0 \quad \Longleftrightarrow \quad H_{c,b,a} = \mathrm{id}_{C_a}.$$

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection.triangleHolonomy_sub_id;Komorebi.PaperIII.Transport.ReversibleConnection.cocycleDefect_eq_zero_iff_triangleHolonomy_eq_refl`

Proof. Using reversibility,

$$\begin{aligned}\Gamma_{a\leftarrow c}\Delta_{c,b,a} &= \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} - \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow a}\\ &= H_{c,b,a} - \mathrm{id}_{C_a}.\end{aligned}$$

Left composition by the equivalence $\Gamma_{a\leftarrow c}$ is injective, which gives the equivalence of the two vanishing statements. □

Corollary 5.11 (Canonical transport has zero connection defect)**.** *If* $\Gamma_{b\leftarrow a} = J_{b\leftarrow a}$*, then*

$$\Delta_{c,b,a} = 0, \qquad H_{c,b,a} = \mathrm{id}$$

for all triples.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection.canonicalTransport_zero_connectionDefect;Komorebi.PaperIII.Transport.ReversibleConnection.canonicalTransport_triangleHolonomy_eq_refl`

Proof. The defect vanishes by the canonical cocycle in theorem 4.4. Apply theorem 5.10. □

5.4 Paths and endpoint independence

Definition 5.12 (Connection transport along a path)**.** For a vertex sequence $p = (a_0, a_1, \dots, a_n)$, define

$$\Gamma_p = \Gamma_{a_n \leftarrow a_{n-1}} \cdots \Gamma_{a_1 \leftarrow a_0}, \qquad \Gamma_{(a_0)} = \mathrm{id}_{C_{a_0}}.$$

If $a_n = a_0$, then Γ_p is the holonomy of the loop p.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection.pathTransport;Komorebi.PaperIII.Transport.ReversibleConnection.VertexPath;Komorebi.PaperIII.Transport.ReversibleConnection.indexedPathTransport`

Products follow path order: the first traversed edge acts on the right.

Proposition 5.13 (Flat pair transport collapses to endpoints)**.** *If Γ is flat, then every path satisfies*

$$\Gamma_p = \Gamma_{a_n \leftarrow a_0}.$$

In particular, every loop has identity holonomy.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection.pathTransport_eq_direct_of_flat;Komorebi.PaperIII.Transport.ReversibleConnection.indexedPathTransport_eq_direct_of_flat;Komorebi.PaperIII.Transport.ReversibleConnection.indexedLoopTransport_eq_refl_of_flat`

Proof. Induct on the path length. Appending a_{n+1} left-multiplies the inductive expression by $\Gamma_{a_{n+1} \leftarrow a_n}$, and flatness replaces the two factors by $\Gamma_{a_{n+1} \leftarrow a_0}$. For a loop, normalization gives $\Gamma_{a_0 \leftarrow a_0} = \mathrm{id}$. □

Remark 5.14. The converse *trivial holonomy implies flatness* depends on which loops are present in an API defined on a chart graph. In the complete pair API, triviality of every triangle holonomy is equivalent to flatness by theorem 5.10. No theorem about topological simple connectivity is being invoked.

6 Gauge covariance and comparison with canonical transport

Definition 6.1 (Gauge change)**.** A gauge is a family of automorphisms $G_a \in \mathrm{Aut}_R(C_a)$. It acts on a connection by

$$\Gamma^G_{b \leftarrow a} = G_b \Gamma_{b \leftarrow a} G_a^{-1}.$$

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection.Gauge;Komorebi.PaperIII.Transport.ReversibleConnection.gauge`

A gauge changes coordinate representatives at each chart. The chart graph stays the same.

Theorem 6.2 (Gauge covariance)**.** *The defect and triangle holonomy transform as*

$$\Delta^G_{c,b,a} = G_c \Delta_{c,b,a} G_a^{-1}, \qquad H^G_{c,b,a} = G_a H_{c,b,a} G_a^{-1}.$$

Thus flatness and triviality of triangle holonomy are gauge invariant.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection.gauge_cocycleDefect;Komorebi.PaperIII.Transport.ReversibleConnection.gauge_triangleHolonomy;Komorebi.PaperIII.Transport.ReversibleConnection.gauge_flat_iff;Komorebi.PaperIII.Transport.ReversibleConnection.gauge_triangleHolonomy_eq_refl_iff`

Proof. Substitute the definition of Γ^G. Adjacent factors $G_b^{-1}G_b$ and $G_c^{-1}G_c$ cancel. In the defect, both the composite and direct terms acquire the same outer factors G_c and G_a^{-1}. In the loop, all inner factors cancel and only conjugation at the base remains. □

Theorem 6.3 (Flat connections are gauges of the canonical pair transport)**.** *Suppose the C_a are exact packet coordinate modules for one ambient module, so the canonical $J_{b\leftarrow a}$ is available. If Γ is a flat normalized reversible pair connection, choose $o \in B$ and set*

$$G_a = \Gamma_{a\leftarrow o} J_{o\leftarrow a}.$$

Then every pair transport satisfies

$$\boxed{\Gamma_{b\leftarrow a} = G_b J_{b\leftarrow a} G_a^{-1}.}$$

Formal contract: finite algebraic core.

`Komorebi.PaperIII.Transport.ReversibleConnection.flatGauge;Komorebi.PaperIII.Transport.ReversibleConnection.flat_isGaugeOfCanonical`

Proof. Reversibility gives $G_a^{-1} = J_{a\leftarrow o}\Gamma_{o\leftarrow a}$. Therefore

$$\begin{aligned} G_b J_{b\leftarrow a} G_a^{-1} &= \Gamma_{b\leftarrow o} J_{o\leftarrow b} J_{b\leftarrow a} J_{a\leftarrow o} \Gamma_{o\leftarrow a} \\ &= \Gamma_{b\leftarrow o}\Gamma_{o\leftarrow a} \\ &= \Gamma_{b\leftarrow a}. \end{aligned}$$

The first reduction uses the canonical cocycle twice. The second uses flatness of Γ. □

Boundary 6.4 (Flat does not mean calibrated). The theorem produces a gauge relating Γ to J. It does not prove $\Gamma = J$ before that gauge is chosen. Conversely, nontrivial triangle holonomy cannot be removed by a gauge, because conjugation cannot turn a nonidentity automorphism into the identity.

7 Obstruction packets: local certificates and holonomy blocks

7.1 The generic compiled certificate

Let Candidate, Packet, and Residual be arbitrary types, with a distinguished $0 \in$ Residual, and let

$$r : \mathsf{Candidate} \longrightarrow \mathsf{Packet} \longrightarrow \mathsf{Residual}$$

be a residual function.

Definition 7.1 (Solution and obstruction packet for a candidate)**.** A candidate x is a solution when

$$\mathrm{IsSolution}_r(x) \quad \Longleftrightarrow \quad \forall p,\ r(x,p) = 0.$$

An obstruction packet for x is a pair consisting of a named packet p and a certificate $r(x, p) \neq 0$.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.NonlinearObstructions.IsSolution;Komorebi.PaperIII.NonlinearObstructions.ObstructionPacket`

Theorem 7.2 (Exact logical force of a local obstruction)**.** *Every obstruction packet for x proves that x is not a solution, and*

$$\neg\,\mathrm{IsSolution}_r(x) \quad \Longleftrightarrow \quad \textit{there exists an obstruction packet for } x.$$

The reverse implication from failure to a packet witness is classical and does not require the packet type to be finite.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.NonlinearObstructions.ObstructionPacket.not_isSolution;Komorebi.PaperIII.NonlinearObstructions.obstructionPacket_of_not_isSolution;Komorebi.PaperIII.NonlinearObstructions.not_isSolution_iff_nonempty_obstructionPacket`

Proof. A purported solution would make the residual vanish at the certified packet, contradicting its nonzero field. Conversely, if no packet carries a nonzero residual, then every residual is zero and the candidate satisfies the definition of solution. □

What one obstruction certifies

The certificate answers one question: does this specified candidate satisfy every declared packet condition? One nonzero residual settles that question for this candidate. A statement that no candidate can solve the problem would require an obstruction for every candidate.

Boundary 7.3 (Candidate-specific means candidate-specific). The theorem rules out the named x. It neither constructs a solution nor shows that every candidate is obstructed. It provides no minimality, stability, computability, or classification theorem for the packet witness.

7.2 Specialization to holonomy blocks

For a reversible connection, specialize the generic residual at a fixed triangle by taking a packet label to be a pair (μ, ν) and the residual to be the corresponding block of $H - \mathrm{id}$. A nonzero block, together with its label and a proof of nonvanishing, gives an obstruction packet certificate in the sense of definition 7.1. The reconstruction theorem below supplies the further step: vanishing of all these local residuals is equivalent to the cocycle equation at that triangle.

Fix a triangle (a, b, c) and resolve the base fiber C_a by a finite complete family of idempotents $(\Pi^a_\mu)_{\mu \in I_a}$. Orthogonality may also hold, as it does for the exact range coordinate projectors below, but the compiled reconstruction theorem does not require it. In exact range coordinates, Π^a_μ is simply the projector that retains component μ and zeros the other components.

Definition 7.4 (Holonomy obstruction and packet residuals)**.** Define

$$\mathrm{Ob}_{c,b,a} = H_{c,b,a} - \mathrm{id}_{C_a}, \qquad \mathrm{Ob}^{\mu\nu}_{c,b,a} = \Pi^a_\mu \,\mathrm{Ob}_{c,b,a}\, \Pi^a_\nu.$$

The map $\mathrm{Ob}^{\mu\nu}_{c,b,a}$ is the obstruction block, or packet residual, from input packet ν to output packet μ. If it is nonzero, then (μ, ν) and that nonvanishing proof form an obstruction packet certificate. Diagonal blocks detect holonomy within one packet. Off-diagonal blocks detect packet mixing induced by holonomy.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.HolonomyBlocks.PacketResolution;Komorebi.PaperIII.HolonomyBlocks.PacketResolution.block`

The certificate records a specified holonomy defect, the label of an input and output packet, and a proof that the corresponding block is nonzero.

Theorem 7.5 (Exact reconstruction and detection by obstruction blocks)**.** *The total obstruction has the exact finite resolution*

$$\boxed{\mathrm{Ob}_{c,b,a} = \sum_{\mu,\nu \in I_a} \mathrm{Ob}^{\mu\nu}_{c,b,a}\,.}$$

Moreover, the following are equivalent:

(i) $\Delta_{c,b,a} = 0$;

(ii) $H_{c,b,a} = \mathrm{id}$;

(iii) $\mathrm{Ob}_{c,b,a} = 0$;

(iv) $\mathrm{Ob}^{\mu\nu}_{c,b,a} = 0$ *for every* μ, ν.

Formal contract: finite algebraic core.

`Komorebi.PaperIII.HolonomyBlocks.obstruction_sum_blocks;Komorebi.PaperIII.HolonomyBlocks.obstructionBlocks_zero_iff;Komorebi.PaperIII.HolonomyBlocks.obstructionDetection_equivalences`

Proof. Insert completeness on both sides:

$$\mathrm{Ob} = \left(\sum_{\mu} \Pi^a_\mu\right) \mathrm{Ob} \left(\sum_{\nu} \Pi^a_\nu\right) = \sum_{\mu,\nu} \Pi^a_\mu \,\mathrm{Ob}\, \Pi^a_\nu.$$

Thus (iv) implies (iii), and (iii) implies (iv) by composition with the projectors. Conditions (ii) and (iii) are equivalent by the definition of Ob and additive cancellation. Conditions (i) and (ii) are equivalent by theorem 5.10. □

Why both packet indices matter

For the declared finite complete resolution, reconstruction uses both packet indices. Inspecting only diagonal blocks can miss transport from one packet to another. The full family records that mixing and reconstructs the total obstruction exactly.

Figure 3 depicts the finite block resolution on both sides of the obstruction and the off-diagonal blocks that witness packet mixing.

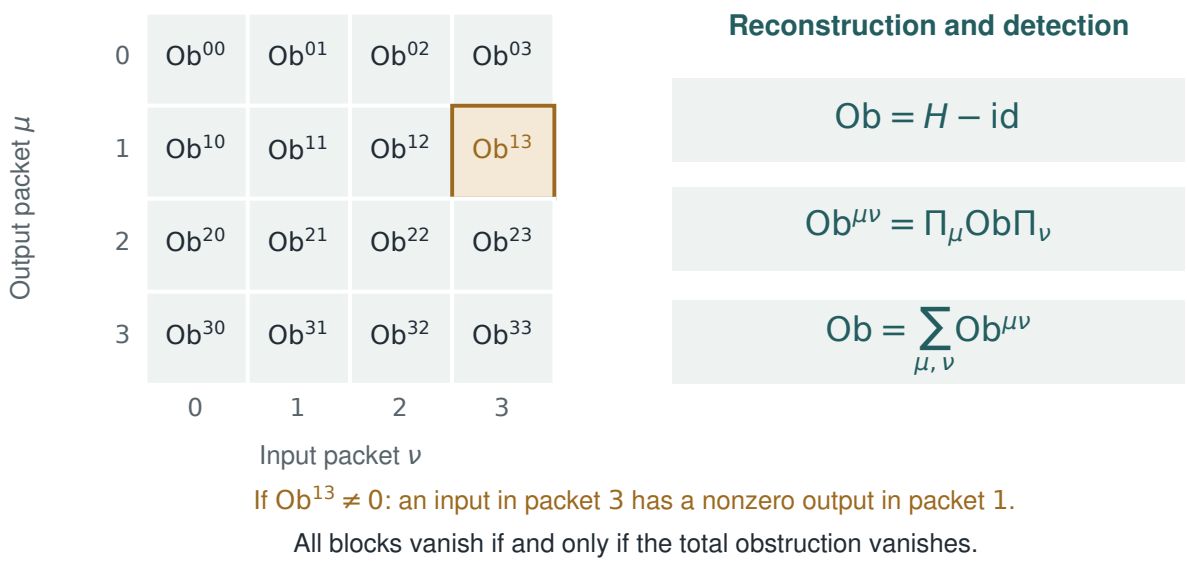


Figure 3. A finite complete packet resolution reconstructs the total holonomy obstruction from blocks indexed by input and output packets. A nonzero off-diagonal block is an exact witness of packet mixing.

Corollary 7.6 (Witness supported in one packet)**.** *If* $\mathrm{Ob}^{\mu\nu}_{c,b,a} \neq 0$*, then there is a vector* x *supported in packet* ν *such that*

$$\Pi^a_\mu \,\mathrm{Ob}_{c,b,a}\, x \neq 0.$$

Such an x *is an exact finite witness of nonflatness.*

Formal contract: finite algebraic core.
`Komorebi.PaperIII.HolonomyBlocks.exists_packetWitness_of_obstructionBlock_ne_zero`

Proof. Classically choose y on which the nonzero linear map $\Pi^a_\mu \,\mathrm{Ob}\, \Pi^a_\nu$ is nonzero and set $x = \Pi^a_\nu y$. Idempotence gives $\Pi^a_\nu x = x$, and the required output is the chosen nonzero value. □

7.3 Gauge behavior of packet blocks

Vanishing or nonvanishing of the total triangle obstruction is gauge invariant by theorem 6.2, but a fixed packet pattern need not be. A general G_a can mix the projectors Π^a_μ.

Proposition 7.7 (Gauge covariance when packet projectors are preserved)**.** *If* $G_a \Pi^a_\mu = \Pi^a_\mu G_a$ *for every* μ*, then*

$$\left(\mathrm{Ob}^G\right)^{\mu\nu} = G_a \,\mathrm{Ob}^{\mu\nu}\, G_a^{-1}.$$

Consequently the zero/nonzero status of each obstruction block is preserved by gauges that preserve the packet projectors.

Formal contract: finite algebraic core.
`Komorebi.PaperIII.HolonomyBlocks.obstructionBlock_gauge_of_commutes;Komorebi.PaperIII.HolonomyBlocks.obstructionBlock_gauge_eq_zero_iff`

Proof. Use $\mathrm{Ob}^G = G_a \,\mathrm{Ob}\, G_a^{-1}$ and commute G_a and G_a^{-1} through the two packet projectors. Composition by an equivalence is injective in both directions. □

Boundary 7.8 (Packet patterns are not absolute). Without the commutation hypothesis in proposition 7.7, only the vanishing of the total obstruction is gauge invariant. The location of individual nonzero blocks may change under a gauge that mixes packets.

8 Products supported in packets and holonomy walks

The passage from words to walks also applies to products of different matrices. Let V be one finite packet type, let R be a semiring, and let $B_1, \dots, B_m \in R^{V\times V}$ be a finite sequence of possibly different packet matrices. Rows are targets and columns are sources as before.

Proposition 8.1 (Packet path expansion for different matrices)**.** *For $i, j \in V$ and $m \geq 1$, set $i_0 = i$ and $i_m = j$. Then*

$$\boxed{(B_m \cdots B_1)_{ji} = \sum_{(i_1,\dots,i_{m-1})\in V^{m-1}} B_m[i_m, i_{m-1}] \cdots B_1[i_1, i_0].}$$

When $m = 1$, the sum is over the unique empty tuple and its summand is $B_1[j, i]$. For $m = 0$, the product is the identity and the entry is the Kronecker delta δ_{ji}. If every $B_r[q, p]$ vanishes outside a declared relation $\mathcal{R}_r(p, q)$, then the endpoint entry is zero whenever there is no packet path compatible with the successive relations.

Formal contract: finite algebraic core.
`Komorebi.PaperIII.EdgeOperators.heterogeneousProduct;Komorebi.PaperIII.EdgeOperators.heterogeneousPathSum;Komorebi.PaperIII.EdgeOperators.heterogeneousPacketPathExpansion;Komorebi.PaperIII.EdgeOperators.SupportedStep;Komorebi.PaperIII.EdgeOperators.SupportedPath;Komorebi.PaperIII.EdgeOperators.heterogeneousProduct_apply_eq_zero_of_no_supportedPath`

Proof. Expand each matrix product entry and distribute the finite sums. Induction on the factor list yields the displayed path sum, with the empty list evaluated as the identity matrix. If no supported intermediate packet exists, every summand contains an entry that is zero by its declared support relation. □

Boundary 8.2 (Changing packet types along a chain). The compiled theorems `heterogeneousPacketPathExpansion_two` and `heterogeneousPacketPathExpansion_three` allow different packet types at the successive stages of two- and three-factor products. The arbitrary-length theorem above permits different factors but uses one common finite packet type V. A fully dependent arbitrary-length chain of changing modules and packet types remains outside the finite algebraic contract.

Remark 8.3 (Checked specialization). When every B_r is one repeated endomorphism T, the recursive specialization to packet powers is inherited from Chapter I.

Inherited formal contract: Chapters I–II.

`Komorebi.PacketNaturalityCompletion.Packets.transferBlock_pow_eq_packetPathSum`

For a loop represented on one common finite scalar packet coordinate type, the proposition resolves its successive transport matrices into packet walks. If R is a ring, as in the reversible connection API, then $(H-I)_{ji} = H_{ji}-\delta_{ji}$. In particular, off-diagonal entries are unchanged. Thus a nonzero off-diagonal holonomy obstruction in this model with a common index type requires a transport word along the closed chart loop whose packet path runs from its input packet to its output packet. This conclusion requires declared block supports for the connection matrices. An arbitrary connection need not respect the transfer graph from section 2.

9 A minimal exact example

Example 9.1 (A scalar triangle)**.** Let $B = \{0, 1, 2\}$ and $C_0 = C_1 = C_2 = \Bbbk$ for a field $\Bbbk$. Choose $u, v, w \in \Bbbk^\times$ and set

$$\Gamma_{1\leftarrow 0}(x) = ux, \qquad \Gamma_{2\leftarrow 1}(x) = vx, \qquad \Gamma_{2\leftarrow 0}(x) = wx,$$

with reverse transports given by inverse scalars. Then

$$\Delta_{2,1,0}(x) = (vu - w)x,$$

and the triangle holonomy based at 0 is

$$H_{2,1,0}(x) = w^{-1}vux.$$

Therefore

$$H_{2,1,0} - \mathrm{id} = w^{-1}\Delta_{2,1,0}.$$

The connection is flat on this triangle exactly when $vu = w$. If the three coordinate systems are literally the same and canonical transport is used, then $u = v = w = 1$ and the obstruction vanishes. Choosing arbitrary nonzero u, v, w is a connection choice, not a consequence of changing exact packet coordinates.

Example 9.2 (A transfer walk with two steps)**.** Let a packet graph have edges $e : 0 \to 1$ and $f : 1 \to 2$, with no other edges. Then

$$T = L_e + L_f, \qquad (T^2)_{20} = c(f)c(e).$$

The products L_eL_e, L_eL_f, and L_fL_f vanish by typed mismatch. Only L_fL_e, corresponding to the walk $0 \to 1 \to 2$, contributes to the $(2, 0)$ entry. This example also fixes the noncommutative coefficient order.

10 Canonical coordinates for arbitrary maps

The analysis and synthesis identities are identities of functions, not only of linear maps. Consequently exact coordinate change does not become curved merely because the represented operation is nonlinear.

Proposition 10.1 (Exact naturality for arbitrary functions)**.** *Let $\mathcal{A}^X_a : X \xrightarrow{\sim} X_a$ and $\mathcal{A}^X_b : X \xrightarrow{\sim} X_b$ be exact coordinate equivalences for X, with inverses $\mathcal{S}^X_a$ and $\mathcal{S}^X_b$. Likewise, let $\mathcal{A}^Y_r : Y \xrightarrow{\sim} Y_r$ and $\mathcal{A}^Y_s : Y \xrightarrow{\sim} Y_s$ be*

exact coordinate equivalences for Y, with inverses $\mathcal{S}_r^Y$ and $\mathcal{S}_s^Y$. Let $F : X \to Y$ be any function, with no linearity or differentiability hypothesis. Set

$$J^X_{b\leftarrow a} = \mathcal{A}^X_b \circ \mathcal{S}^X_a, \qquad J^Y_{s\leftarrow r} = \mathcal{A}^Y_s \circ \mathcal{S}^Y_r,$$

and define

$$\widehat{F}_{r\leftarrow a} = \mathcal{A}^Y_r \circ F \circ \mathcal{S}^X_a.$$

Then

$$J^Y_{s\leftarrow r} \circ \widehat{F}_{r\leftarrow a} = \widehat{F}_{s\leftarrow b} \circ J^X_{b\leftarrow a}.$$

If $\mathcal{A}^Z_t : Z \xrightarrow{\sim} Z_t$ is another exact coordinate equivalence and $G : Y \to Z$ is any function, then exact coordinate representation also preserves composition:

$$\widehat{G \circ F}_{t\leftarrow a} = \widehat{G}_{t\leftarrow r} \circ \widehat{F}_{r\leftarrow a}.$$

Formal contract: finite algebraic core.
`Komorebi.PaperIII.Transport.exactCoordinateMap;Komorebi.PaperIII.Transport.exactCoordinateMap_comp;Komorebi.PaperIII.Transport.exactCoordinateMap_naturality`

Proof. Both sides of the naturality equation reduce to $\mathcal{A}^Y_s \circ F \circ \mathcal{S}^X_a$ using $\mathcal{S} \circ \mathcal{A} = \mathrm{id}$ in X and Y. For composition, the middle $\mathcal{S}^Y \circ \mathcal{A}^Y$ cancels in the same way. □

Boundary 10.2 (Where nonlinear defects can enter). A nonlinear *coordinate* defect cannot be blamed on the exact canonical J of definition 4.3. By proposition 10.1, that defect is identically zero. Nonzero defects require extra structure such as a noncanonical connection, truncation, approximation, restricted domains, or failure of an operation to respect a declared packet model.

11 Nonlinear operation trees: locality and the formalization frontier

11.1 The compiled foundation of unary and binary trees

Definition 11.1 (Finite grammar for operation trees)**.** For types of input labels I, constants K, unary symbols U, and binary symbols B, the finite formal grammar has four constructors:

$$\begin{aligned} t ::= \ & \mathrm{input}(i) \mid \mathrm{constant}(k) \mid \mathrm{unary}(u, t) \\ & \mid \mathrm{binary}(b, t_1, t_2). \end{aligned}$$

Given an arbitrary value type Y, evaluation takes completely explicit maps

$$I \to Y, \qquad K \to Y, \qquad U \to Y \to Y, \qquad B \to Y \to Y \to Y.$$

No algebraic, continuity, or differentiability law is imposed on these semantics.

Formal contract: finite algebraic core.
`Komorebi.PaperIII.NonlinearObstructions.OperationTree;Komorebi.PaperIII.NonlinearObstructions.OperationTree.eval`

When I has decidable equality, let $\mathrm{inputs}(t)$ be the finite set of input labels actually occurring in t.

Theorem 11.2 (Input locality for operation trees)**.** *Let $\eta_1, \eta_2 : I \to Y$ be two input environments. If*

$$\eta_1(i) = \eta_2(i) \qquad \textit{for every } i \in \mathrm{inputs}(t),$$

then the two evaluations of t are equal under the same constant, unary, and binary semantics. In particular, if $\mathrm{inputs}(t) = \emptyset$, evaluation is independent of the input environment.

Formal contract: finite algebraic core.
`Komorebi.PaperIII.NonlinearObstructions.OperationTree.eval_congr_on_inputs;Komorebi.PaperIII.NonlinearObstructions.OperationTree.eval_eq_of_inputs_eq_empty`

Proof. Structural induction on t. The input case is the agreement hypothesis. The constant case is reflexive. The unary case applies the induction hypothesis to its child. The binary case applies it independently to the left and right subtrees, whose input sets form a union. □

Boundary 11.3 (Exact force of locality). The theorem says evaluation reads only syntactically named inputs. It does not prove that the operations are linear or nonlinear in any analytic sense, that the tree is stable, that two different trees are equivalent, or that a solution to a tree equation exists.

11.2 Typed colored extension beyond the finite contract

Definition 11.4 (Finitary colored operation signature)**.** A finitary colored signature consists of:

- a finite type V of colors;
- a type Σ of operation symbols;
- an arity $r(\sigma) \in \mathbb{N}$ for every $\sigma \in \Sigma$;

- input colors $\mathrm{in}_\sigma : \mathrm{Fin}(r(\sigma)) \to V$;
- an output color $\mathrm{out}(\sigma) \in V$.

A realization assigns a space X_v to each color and a map

$$\Phi_\sigma : \prod_{q \in \mathrm{Fin}(r(\sigma))} X_{\mathrm{in}_\sigma(q)} \longrightarrow X_{\mathrm{out}(\sigma)}.$$

Outside the finite formal contract. The intended Lean target is a dependent inductive typed tree. No such declaration is part of the finite algebraic completion contract.

Definition 11.5 (Typed operation tree and evaluation)**.** A typed operation tree is recursively either a leaf of declared color or a node labeled by σ, with one correctly colored child for each input slot of σ. Evaluation is recursive:

$$\mathrm{ev}(\mathrm{node}(\sigma; t_1, \dots, t_{r(\sigma)})) = \Phi_\sigma(\mathrm{ev}(t_1), \dots, \mathrm{ev}(t_{r(\sigma)})).$$

Outside the finite formal contract. Candidate API: `Komorebi.PaperIII.NonlinearObstructions.ColoredOperationTree` and `ColoredOperationTree.eval`. This is stronger than the compiled uncolored unary–binary grammar.

11.3 Packet labelings and the multilinear first target

Suppose every X_v has a finite complete packet family $(P_i^v)_{i \in I_v}$. A packet labeling assigns an admissible packet index to each edge or node of a typed tree. For multilinear Φ_σ, inserting

$$\mathrm{id}_{X_v} = \sum_{i \in I_v} P_i^v$$

at every input and output yields a finite sum over trees with packet labels. This is the branching analogue of inserting intermediate projectors in proposition 8.1.

Formalization target 11.6 (Finite multilinear tree expansion). The first milestone for the nonlinear module is an exact theorem saying that the evaluation of a finite tree of multilinear nodes equals the finite sum of its packet-resolved labeled evaluations. Required proof obligations include:

(a) a dependent well-typed tree syntax;

(b) finite packet-label types at every color;

(c) an explicit convention for nullary and unary nodes;

(d) multilinearity in every input slot;

(e) preservation of the order of input slots without assuming commutativity of operations.

Outside the finite formal contract. Candidate target: `Komorebi.PaperIII.NonlinearObstructions.ColoredOperationTree.eval_eq_sum_packetLabellings`.

11.4 Connection defects at nonlinear nodes

For coordinate models of the source and target colors, an additive target coordinate space, and a chosen noncanonical connection, a pointwise node defect can be defined by

$$\mathfrak{D}^{\Gamma}_{b,a}(\Phi) = \Gamma^{Y}_{b\leftarrow a} \circ \widehat{\Phi}_a - \widehat{\Phi}_b \circ \left(\prod_q \Gamma^{X_q}_{b\leftarrow a} \right).$$

This expression compares two *chosen* routes. With canonical exact transport in every slot it vanishes by proposition 10.1.

Formalization target 11.7 (Operation-tree defect propagation). For multilinear nodes, one expects an exact finite expansion obtained by replacing one or more nodes by their defects and transporting the remaining branches. For smooth nonlinear nodes, the corresponding chain rule is indexed by rooted partitions and Faà di Bruno trees. A formal theorem will require a fixed differentiability category, base points, derivative conventions, symmetry factors, and remainder policy.

Outside the finite formal contract. Tree-level defect identities, higher chain rules, and nonlinear norm bounds are outside the finite Lean contract. Ordinary mathematical results in these directions appear in Part II under their stated hypotheses.

Boundary 11.8 (Analytic and geometric exclusions). The finite formal contract does not include convergence of an infinite tree series, existence of derivatives, a Banach-space Faà di Bruno theorem, a Baker–Campbell–Hausdorff expansion, a continuum limit, or an identification of algebraic holonomy with curvature of a smooth bundle. Part II supplies ordinary proofs for its explicit nonlinear and analytic statements. The theory does not derive a connection from data, optimization, physics, or probability.

12 Finite algebraic formalization ledger

Topic	Status	Declaration or roadmap target	Exact boundary
Packet matrix and edge operator	Lean-backed III	`PacketGraph.packetMatrix`; `EdgeSystem.edgeOp`	Finite scalar matrices over a semiring, with one scalar coordinate per packet.
Edge support and pair dichotomy	Lean-backed III	`edgeOp_eq_zero_off_support`; `edgeOp_mul_edgeOp_eq_zero`; `edgeOp_mul_edgeOp_of_composable`	Broken interfaces vanish. Matched pairs collapse to the endpoint block with coefficient in operator order.
Typed word evaluation	Lean-backed III	`Composable`; `wordOp`; `wordOp_append`; `wordOp_eq_zero_of_not_composable`	List order is execution order. Concatenation reverses operator-product order. Nonwalks vanish.

Topic	Status	Declaration or roadmap target	Exact boundary
Finite word and walk expansions	Lean-backed III	`explicitWordExpansion`; `walkExpansion`; `wordExpansion_eq_explicitWordExpansion`; `transferOp_pow_eq_walkExpansion`	Actual finite tuple sum and zero-filtered composable-walk sum over finite graph and packet types.
Endpoint coefficient walk sum	Lean-backed III	`reverseCoefficient`; `Connects`; `transferOp_pow_apply_eq_sum_walks`	Exact scalar endpoint entry, including the empty-walk case and noncommutative coefficient order.
Fixed reachability and orbit support	Lean-backed III, with an inherited hull bridge	`transferOp_pow_apply_eq_zero_of_no_walk`; `transferOp_pow_mulVec_apply_eq_zero_of_unreachable_support`	Finite powers and supported seeds. The orbit-hull span conclusion uses the frozen Chapter I contract.
Exact frame transport	Lean-backed III	`exactTransport_cocycle`; `exactPathTransport_eq_endpoint`; `exactPathTransport_loop`	One frame family on a single module. Arbitrary finite paths telescope and closed paths are trivial.
Exact coordinate cocycle	Inherited Lean-backed II	`NestedRefinement.refineCoordinates_comp`; `NestedRefinement.coordinateRefinementEquiv_trans`	Complete finite projector coordinates, not reduction.
Typed coordinate naturality	Inherited Lean-backed II	`NestedRefinement.coordinateOperator_comp`; `NestedRefinement.coordinateOperator_naturality`	Linear maps over semiring modules.
Minimal connection API	Lean-backed III	`Transport.Connection`; `cocycleDefect`; `triangleHolonomy`; `triangleHolonomy_eq_refl_of_flat`	Only self-normalization is assumed. Reverse inversion and the cocycle are not structure fields.
Twisted three-point loop	Lean-backed III	`twistedThreePointConnection`; `twisted_triangleHolonomy_ne_refl`	Concrete rational connection. The selected loop sends 1 to 2.
Reversible additive defect	Lean-backed III	`ReversibleConnection`; `triangleHolonomy_sub_id`; `cocycleDefect_eq_zero_iff_triangleHolonomy_eq_refl`	Heterogeneous module fibers, reverse-edge inversion, and an additive defect over a commutative ring.

Topic	Status	Declaration or roadmap target	Exact boundary
Reversible finite paths	Lean-backed III	`pathTransport_eq_direct_of_flat`; `indexedPathTransport_eq_direct_of_flat`; `indexedLoopTransport_eq_refl_of_flat`	Finite pair-connection paths collapse to endpoints under algebraic flatness.
Gauge covariance	Lean-backed III	`gauge_cocycleDefect`; `gauge_triangleHolonomy`; `gauge_flat_iff`	Fiberwise algebraic conjugation, without smooth gauge theory.
Flat versus canonical transport	Lean-backed III	`flatGauge`; `flat_isGaugeOfCanonical`	A chosen base chart and ambient frames exhibit every flat pair connection as a gauge transform of canonical transport.
Operation-tree locality	Lean-backed III	`OperationTree`; `OperationTree.eval`; `OperationTree.eval_congr_on_inputs`	Arbitrary unary and binary semantics. Only dependence on named inputs is proved.
Generic obstruction packet	Lean-backed III	`ObstructionPacket.not_isSolution`; `not_isSolution_iff_nonempty_obstructionPacket`	Exact candidate-specific logical certificate, without global classification.
Holonomy block obstruction	Lean-backed III	`obstruction_sum_blocks`; `obstructionBlocks_zero_iff`; `obstructionDetection_equivalences`	Finite complete idempotent packet family on the base fiber. Orthogonality is not required by the compiled reconstruction.
Packet witness and block gauge	Lean-backed III	`exists_packetWitness_of_obstructionBlock_ne_zero`; `obstructionBlock_gauge_of_commutes`; `obstructionBlock_gauge_eq_zero_iff`	Witness extraction is classical. Blockwise covariance requires the gauge to commute with every packet projector.
Heterogeneous matrix paths	Lean-backed III	`heterogeneousPacketPathExpansion`; `heterogeneousProduct_apply_eq_zero_of_no_supportedPath`	Arbitrarily many different matrices over one common finite packet type. Fully typed changing-index chains compile only at lengths two and three.

Topic	Status	Declaration or roadmap target	Exact boundary
Arbitrary-function coordinates	Lean-backed III	`exactCoordinateMap_comp`; `exactCoordinateMap_naturality`	Exact set-level equivalences and arbitrary functions, without a linearity or differentiability assumption.
Colored and smooth trees	Roadmap only	roadmap `ColoredOperationTree.eval_eq_sum_packetLabellings`	Dependent colors, multilinear packet expansion, derivatives, and Faà di Bruno identities remain outside the contract.

12.1 Kernel-backed Chapter III surface

The authoritative declaration-level contract is `formalization/lean/LEAN_THEOREM_MAP_v1.0.1.md`. It groups the compiled surface as follows.

- **EdgeOperators and EdgeExtensions:** scalar packet matrices, typed edge and word support, finite word/walk expansions, endpoint coefficient sums, fixed-length reachability, orbit support, and common-index heterogeneous products with declared supports.
- **Transport and ReversibleConnections:** exact frame transport, the minimal connection, heterogeneous reversible pair transport, additive defect and triangle holonomy, finite paths, canonical connections, and the fiberwise gauge action.
- **CoordinateNaturality and HolonomyBlocks:** composition and naturality for arbitrary functions between exact coordinate systems, finite two-sided block reconstruction, equivalent obstruction criteria, packet-supported witnesses, and packet-preserving block gauges.
- **NonlinearObstructions:** finite unary–binary operation trees, input locality, and candidate-specific obstruction certificates.

The formal companion records the checker-generated declaration and audit totals. The parity gate requires every fully qualified Chapter III target printed beside a statement to occur in the theorem map, rejects stale or duplicate targets, and checks it against the compiled environment. `KernelAudit.lean` reports dependencies for the mapped and supporting theorems, and its fail-closed gate scans every compiled `Komorebi.PaperIII` declaration. The four inherited statement contracts remain identified separately by their inherited declaration names.

Boundary 12.1 (No scope promotion by prose). The displayed endpoint coefficient formula, reachability results, reversible connection theorems, additive defect–holonomy identity, gauge covariance, holonomy-block reconstruction, packet witnesses, common-index heterogeneous path expansion, and arbitrary-function coordinate naturality are all part of the finite algebraic kernel-backed surface. What remains outside is the dependent fiber-valued edge-system generalization, a fully dependent arbitrary-length chain whose module and packet type change at every stage, dependent colored operation trees, packet-labeled multilinear tree expansions, nonlinear tree-defect and Faà di Bruno identities, and the analytic and geometric extensions listed above.

13 The finite contract and the nonlinear transition

Typed packet support turns the expansion of a transfer power into a walk calculus. Incompatible words vanish because their interfaces do not match. Canonical packet coordinate transport is the exact equivalence obtained from analysis and synthesis. It satisfies the cocycle law, so every finite canonical loop telescopes to the identity. Holonomy enters after a separate connection is supplied. The compiled minimal API proves that flatness makes both the relative defect and holonomy trivial, and it exhibits a nontrivial rational loop. Under the compiled reversible API, a nonzero additive triangle defect is equivalent to nonidentity holonomy, transforms covariantly under gauge, and is detected exactly by a finite family of packet blocks. Finite flat paths collapse to their endpoints. Every flat reversible pair connection is explicitly related to canonical transport by a gauge change through a chosen base chart.

These facts define the boundary of the finite formal contract. At that boundary, exact coordinates preserve composition and satisfy naturality for arbitrary functions in Chapter III, while the typed linear packet coordinate counterpart remains inherited from Chapter II. Nontrivial nonlinear defects must therefore be attached to additional modeling choices or to packet compatibility of the particular operation. Colored operation trees, multilinear packet labelings, and Faà di Bruno expansions remain outside this finite Lean contract. Part II develops the corresponding ordinary mathematics, including packet jets, rooted tree expansions, and explicit nonlinear obstruction theorems.

Part II Nonlinear packet calculus, obstruction complexes, and holonomy

This part moves from exact coordinates for arbitrary functions to differentiable interactions, specified correction problems, and finite or smooth transport. Each ordinary proof retains its stated hypotheses. These results are not asserted to belong to the finite Lean contract of Part I.

The notational change from $\mathcal{A}, \mathcal{S}$ to $\mathcal{A}, \mathcal{S}$ is typographical: both denote analysis and synthesis for the declared packet system. A connection remains additional data. Smooth canonical transport is constructed from a moving projector family, whereas the algebraic exact coordinate transport in Part I compares fixed exact descriptions of one ambient object.

14 Introduction

A finite family of pairwise annihilating idempotents gives exact coordinates. If

$$P_\alpha P_\beta = 0 \quad \text{for } \alpha \neq \beta, \qquad \sum_\alpha P_\alpha = I,$$

then every vector is the sum of its packet components. Linear operators become block matrices in these coordinates. A nonlinear map cannot be reconstructed from its diagonal restrictions, since mixed interactions between packet components are usually present. It can still be represented exactly if its input is the complete packet tuple.

This part develops nonlinear packet representation and the obstruction theory that follows from it. Exact coordinates preserve a nonlinear problem, but they do not solve it. Once the equation is written in packet coordinates, its derivatives, compositions, correction maps, refinement failures, and gluing defects can be separated without discarding mixed terms.

The first stage of Part II treats the local nonlinear calculus. Packet analysis and synthesis conjugate arbitrary maps. Fréchet derivatives split into multilinear packet jets. The higher chain rule then becomes a Faà di Bruno formula whose terms carry intermediate packet labels. Standard background on Fréchet calculus and the higher chain rule may be found in [1, 2, 4].

The next stage studies nonlinear iteration. Repeated substitution is indexed by rooted trees rather than operator words. A scalar majorant gives an absolute convergence theorem for polynomial fixed-point equations. The quadratic case recovers the Catalan series. Rooted tree combinatorics are discussed in [16, 11, 6].

A linear correction map

$$D : U \longrightarrow V$$

places the obstruction to solving $Du = -r$ in the canonical quotient coker D. This quotient has a universal property and behaves naturally under commuting squares. Mapping cones extend the same construction to cochain complexes. In a Hilbert space with closed range, the quotient is represented by the orthogonal projection onto Ker D^*. The Moore–Penrose inverse then gives the least norm correction. Relevant background appears in [18, 15].

A vanishing linear obstruction is only the first step. We give a contraction theorem for nonlinear correction, followed by a local Lyapunov–Schmidt reduction. In a complete filtered problem, every lifting step has its own cokernel class. Vanishing at all depths and compatible choices of corrections produce an exact inverse limit solution. A filtered correction complex carries a spectral sequence. Its first page records graded obstruction candidates, while later differentials record interactions that can cancel them. We use the standard spectral sequence of a filtered complex from [18, 12].

The final stage studies elimination and gluing. The implicit function theorem gives an exact Schur derivative after hidden variables are removed. Finite transport systems reduce, after a spanning tree gauge, to one holonomy element for each fundamental cycle. The same idea extends to smooth projector families through Kato transport. We use standard material from [4, 5, 7, 8, 10].

The chapter does not claim that every nonlinear problem has a finite obstruction space. The cokernel is canonical only after a correction map has been specified. It may be infinite dimensional. A first-order obstruction may vanish while a higher-order obstruction remains. The purpose of the calculus is to identify the exact object in which each failure lives and to show how those objects change under packets, refinement, filtration, and transport.

15 A map of the nonlinear calculus

Four questions organize the nonlinear calculus: representation, interaction, correction, and gluing.

The first question is representation. Given a nonlinear map F, can its packet form be written without approximation? The answer is yes. Analysis and synthesis are inverse coordinate maps, so conjugation preserves the full equation.

The second question is interaction. Which input packets influence a given output packet, and at what derivative order? Packet jets answer this question. They form a directed hypergraph rather than an ordinary block matrix because one nonlinear interaction can have several inputs.

The third question is correction. Suppose an approximate solution leaves a defect r. A declared linearized correction map D removes the part of r lying in im D. The quotient class

$$[r] \in \operatorname{coker} D$$

is the part that no allowed first-order correction can remove. This quotient class is the obstruction. Its vanishing must be supplemented by a nonlinear theorem.

The fourth question is gluing. Local coordinate changes can solve every edge problem and still fail around a loop. The remaining cycle product is holonomy. It measures the failure of a global frame even when each local equation has been solved.

These four layers fit into the sequence

$$\begin{aligned}\text{exact coordinates} &\longrightarrow \text{packet interactions} \longrightarrow \text{correction classes}\\ &\longrightarrow \text{global gluing.}\end{aligned}$$

Each arrow needs hypotheses. Exact coordinates need a complete packet system. Differential interactions need regularity. A cokernel obstruction needs a specified correction map. Holonomy needs invertible local transports.

A second distinction concerns scale. A refinement defect compares a fine nonlinear model with a coarse one. It is a point in the coarse output space. An obstruction class is the image of such a defect, or

of another residual, in a quotient by allowed corrections. The correction map must be declared before the defect determines an obstruction class. A small defect can have a nonzero obstruction, and a large defect can have zero obstruction if it lies entirely in the correction range.

The chapter uses finite packet systems in the coordinate and graph sections. Banach space hypotheses enter only where norms, contractions, or Fréchet derivatives are used. Module and complex language is used for cokernels, filtered lifting, and cohomological obstructions. This separation keeps each result at the level of generality supported by its proof.

Standing conventions

All packet systems are finite unless a statement says otherwise. In a normed setting every packet projector and every coarsening operator is bounded. Differentiability of operator-valued maps is taken in operator norm. A Banach space isomorphism is a bounded linear bijection with bounded inverse. For a decreasing filtration $(F^m X)$, completeness means that the canonical map

$$X \longrightarrow \varprojlim_m X/F^m X$$

is an isomorphism and separatedness means $\bigcap_m F^m X = 0$. Statements about congruence classes in a filtered nonlinear problem also assume that the nonlinear map preserves filtration congruences at the stated depth. These assumptions are repeated where they are essential to a proof.

16 Exact nonlinear packet coordinates

Let X be a module over a unital ring. A finite packet system on X is a family

$$\mathcal{P} = (P_\alpha)_{\alpha \in I} \subseteq \mathrm{End}(X)$$

of nonzero idempotents such that

$$P_\alpha P_\beta = 0 \quad \text{when } \alpha \neq \beta, \qquad \sum_{\alpha \in I} P_\alpha = I_X.$$

Write

$$X_\alpha = P_\alpha X, \qquad X^{\mathcal{P}} = \bigoplus_{\alpha \in I} X_\alpha.$$

The analysis and synthesis maps are

$$\mathcal{A}_{\mathcal{P}} x = (P_\alpha x)_\alpha, \qquad \mathcal{S}_{\mathcal{P}} (x_\alpha)_\alpha = \sum_\alpha x_\alpha.$$

They are inverse isomorphisms.

Definition 16.1 (Nonlinear packet coordinate map)**.** Let $\mathcal{P}$ be a packet system on X and let $\mathcal{Q}$ be a packet system on Y. For any map $F : X \to Y$, define

$$F_{\mathcal{Q} \leftarrow \mathcal{P}} = \mathcal{A}_{\mathcal{Q}} F \mathcal{S}_{\mathcal{P}}.$$

Thus

$$F_{\mathcal{Q}\leftarrow\mathcal{P}}((x_\alpha)_\alpha) = \left(Q_\gamma F\left(\sum_\alpha x_\alpha\right)\right)_\gamma.$$

Theorem 16.2 (Exact nonlinear coordinate theorem)**.** *For every map $F : X \to Y$,*

$$\mathcal{S}_{\mathcal{Q}} F_{\mathcal{Q}\leftarrow\mathcal{P}} \mathcal{A}_{\mathcal{P}} = F.$$

If $G : Y \to Z$ and $\mathcal{R}$ is a packet system on Z, then

$$(G \circ F)_{\mathcal{R}\leftarrow\mathcal{P}} = G_{\mathcal{R}\leftarrow\mathcal{Q}} \circ F_{\mathcal{Q}\leftarrow\mathcal{P}}.$$

No linearity, continuity, or differentiability assumption is needed.

Proof. The identities $\mathcal{S}_{\mathcal{P}}\mathcal{A}_{\mathcal{P}} = I_X$ and $\mathcal{S}_{\mathcal{Q}}\mathcal{A}_{\mathcal{Q}} = I_Y$ give

$$\mathcal{S}_{\mathcal{Q}} F_{\mathcal{Q}\leftarrow\mathcal{P}} \mathcal{A}_{\mathcal{P}} = \mathcal{S}_{\mathcal{Q}}\mathcal{A}_{\mathcal{Q}} F \mathcal{S}_{\mathcal{P}}\mathcal{A}_{\mathcal{P}} = F.$$

For composition,

$$G_{\mathcal{R}\leftarrow\mathcal{Q}} F_{\mathcal{Q}\leftarrow\mathcal{P}} = \mathcal{A}_{\mathcal{R}} G \mathcal{S}_{\mathcal{Q}}\mathcal{A}_{\mathcal{Q}} F \mathcal{S}_{\mathcal{P}} = \mathcal{A}_{\mathcal{R}}(G \circ F)\mathcal{S}_{\mathcal{P}}.$$

□

Corollary 16.3 (Dynamical equivalence)**.** *Suppose $X = Y$ and the same packet system is used on both sides. Packet coordinates preserve fixed points, periodic points, injectivity, surjectivity, and invertibility. In particular,*

$$F(x) = x \quad \Longleftrightarrow \quad F_{\mathcal{P}\leftarrow\mathcal{P}}(\mathcal{A}_{\mathcal{P}} x) = \mathcal{A}_{\mathcal{P}} x.$$

Proof. The packet map is conjugate to F through the isomorphism $\mathcal{A}_{\mathcal{P}}$. Every listed property is invariant under conjugacy. □

Proposition 16.4 (Change of packet coordinates)**.** *Let $\mathcal{P}$ and $\mathcal{P}'$ be packet systems on X. Let $\mathcal{Q}$ and $\mathcal{Q}'$ be packet systems on Y. For any two packet systems $\mathcal{U}$ and $\mathcal{V}$ on the same space, write*

$$J_{\mathcal{U}\leftarrow\mathcal{V}} = \mathcal{A}_{\mathcal{U}}\mathcal{S}_{\mathcal{V}}.$$

Then

$$F_{\mathcal{Q}'\leftarrow\mathcal{P}'} = J_{\mathcal{Q}'\leftarrow\mathcal{Q}} F_{\mathcal{Q}\leftarrow\mathcal{P}} J_{\mathcal{P}\leftarrow\mathcal{P}'}.$$

Proof. Insert $\mathcal{S}_{\mathcal{Q}}\mathcal{A}_{\mathcal{Q}} = I_Y$ and $\mathcal{S}_{\mathcal{P}}\mathcal{A}_{\mathcal{P}} = I_X$ into the definition of $F_{\mathcal{Q}'\leftarrow\mathcal{P}'}$. □

Exactness by cancellation

Analysis and synthesis undo each other on the complete packet tuple. In a composition of packet descriptions, the intermediate synthesis and analysis maps cancel. This compares exact descriptions of the same ambient object. A connection adds data for transport along parameters or paths, with its own hypotheses.

Example 16.5 (A mixed quadratic map)**.** Let $X = Y = \Bbbk^2$ with coordinate packets. For

$$F(x_1, x_2) = (x_1 x_2, x_1^2 + x_2),$$

the first output packet depends on both input packets. The second output packet contains a quadratic contribution from the first input packet and a linear contribution from the second. Packet coordinates record these interactions exactly. Replacing F by the two diagonal restrictions would lose the product $x_1 x_2$.

Remark 16.6. Exact packet coordinates do not imply packet independence. They retain the complete input tuple. Every output packet may depend on every input packet.

17 Packet jets and higher composition

Let X and Y be Banach spaces over the same field $\mathbb{R}$ or $\mathbb{C}$ and let the packet projectors on both spaces be bounded. In the complex case, derivatives are complex Fréchet derivatives and the Taylor integral below has a real parameter. Assume that $F : X \to Y$ is m times continuously Fréchet differentiable. The same local statements hold on open domains containing the points and segments in question. For $1 \le r \le m$, the derivative

$$D^r F(x) : X^r \longrightarrow Y$$

is a continuous symmetric multilinear map.

The range X_α is closed because $X_\alpha = \ker(I - P_\alpha)$. It is therefore Banach with its inherited norm. The same holds for every Y_γ. Give the finite product of ranges the maximum norm. Packet analysis and synthesis are bounded, since they are formed from finitely many bounded projectors and a finite sum. Their algebraic inverse identities from section 16 make them bounded linear isomorphisms. For a nonempty input family, $\|\mathcal{A}_{\mathcal{P}}\| \le \max_\alpha \|P_\alpha\|$ and $\|\mathcal{S}_{\mathcal{P}}\| \le |I|$. An empty complete family can occur only on the zero space. Its coordinate product is the zero vector space with norm zero, and the coordinate identities hold directly.

Definition 17.1 (Packet jet)**.** For $1 \le r \le m$, define

$$J^r_{\gamma|\alpha_1,\dots,\alpha_r} F(x) = Q_\gamma D^r F(x)\big|_{X_{\alpha_1} \times \cdots \times X_{\alpha_r}}.$$

This is a continuous multilinear map

$$X_{\alpha_1} \times \cdots \times X_{\alpha_r} \longrightarrow Y_\gamma.$$

Theorem 17.2 (Exact reconstruction of derivatives)**.** *For $h_1, \dots, h_r \in X$,*

$$D^r F(x)[h_1, \dots, h_r] = \sum_\gamma \sum_{\alpha_1,\dots,\alpha_r} J^r_{\gamma|\alpha_1,\dots,\alpha_r} F(x)[P_{\alpha_1} h_1, \dots, P_{\alpha_r} h_r].$$

Proof. Insert $I_Y = \sum_\gamma Q_\gamma$ on the output and $I_X = \sum_\alpha P_\alpha$ in every input slot. Multilinearity expands the finite sums. □

An exact jet model with two packets

$F(x, y) = (x + xy,\ y + x^2)$ at $(0, 0)$. Coordinate packets in both input and output.

$$DF(0) = I: \quad DF(0)e_x = (1, 0), \qquad DF(0)e_y = (0, 1).$$

Output 1

H_1

	x	y
x	0	**1**
y	**1**	0

$D^2F_1(0)[e_x, e_y] = D^2F_1(0)[e_y, e_x] = 1$

Output 2

H_2

	x	y
x	**2**	0
y	0	0

$D^2F_2(0)[e_x, e_x] = 2$

$$D^2F(0)[h, k] = (h_x k_y + h_y k_x,\ 2h_x k_x)$$

$$\tfrac{1}{2}D^2F(0)[h, h] = (h_x h_y,\ h_x^2)$$

$F(h) = h + \frac{1}{2}D^2F(0)[h, h]$ exactly. All higher jets vanish.

Cells are ordered Hessian responses: the two mixed slots and the Taylor factor 1/2 are both retained.

Figure 4. Second packet jets for $F(x, y) = (x + xy, y + x^2)$ at the origin. Coordinate axis projectors give $DF(0) = I$. The first output has mixed second derivatives equal to one, while the second has a repeated first input derivative equal to two. Each grid records the ordered pair of input packets. The mixed entries retain an interaction that diagonal restrictions would lose.

Proposition 17.3 (Jet norm bound)**.** *Give every packet range its inherited subspace norm. The restricted jet satisfies the sharper estimate*

$$\left\|J^r_{\gamma|\alpha_1,\dots,\alpha_r}F(x)\right\| \le \|Q_\gamma\|\,\|D^rF(x)\|.$$

Its ambient extension inserts P_{α_j} in the jth slot and has norm at most $\|Q_\gamma\|\,\|D^rF(x)\|\prod_j\|P_{\alpha_j}\|$. Consequently the operator norm of the restricted packet jet also satisfies

$$\left\|J^r_{\gamma|\alpha_1,\dots,\alpha_r}F(x)\right\| \le \|Q_\gamma\|\,\|D^rF(x)\|\prod_{j=1}^{r}\|P_{\alpha_j}\|.$$

When all packets are orthogonal projections on Hilbert spaces, every nonzero projector has norm one and the bound reduces to the norm of the full derivative.

Proof. For $h_j \in X_{\alpha_j}$, the inherited norm gives

$$\|Q_\gamma D^rF(x)[h_1,\dots,h_r]\| \le \|Q_\gamma\|\,\|D^rF(x)\|\prod_j\|h_j\|.$$

Taking the multilinear operator norm proves the sharper restricted estimate. For ambient inputs, replace h_j by $P_{\alpha_j}h_j$ and apply the operator norm bound to each projector. Every nonzero idempotent has norm at least one, since it fixes a nonzero vector in its range. The displayed product bound therefore also follows for the restricted jet. If a zero packet is retained as a redundant index, multilinearity makes any jet with that input or output packet zero, so both estimates remain valid. □

Corollary 17.4 (Packet Taylor formula)**.** *Assume F is $m+1$ times continuously Fréchet differentiable on an open set containing the segment $x+th$ for $0\le t\le 1$. Then*

$$\begin{aligned}Q_\gamma F(x+h) = Q_\gamma F(x) &+ \sum_{r=1}^{m}\frac{1}{r!}\sum_{\alpha_1,\dots,\alpha_r} J^r_{\gamma|\alpha_1,\dots,\alpha_r}F(x)[P_{\alpha_1}h,\dots,P_{\alpha_r}h]\\ &+ Q_\gamma R_{m+1}(x,h),\end{aligned}$$

where the Bochner integral remainder is

$$R_{m+1}(x,h) = \frac{1}{m!}\int_0^1 (1-t)^m D^{m+1}F(x+th)[h,\dots,h]\,dt.$$

Proof. Set $g(t) = F(x+th)$ on the open preimage of the domain under $t\mapsto x+th$. It contains $[0,1]$, and repeated differentiation gives $g^{(r)}(t) = D^rF(x+th)[h,\dots,h]$. The derivatives are continuous on the compact interval, so their Bochner integrals exist in the Banach codomain. Put

$$T_j = \frac{1}{j!}\int_0^1 (1-t)^j g^{(j+1)}(t)\,dt \qquad (0\le j\le m).$$

The fundamental theorem of calculus gives $T_0 = g(1)-g(0)$. For $j\ge 1$, integration by parts gives $T_j = T_{j-1} - g^{(j)}(0)/j!$. Iterating this relation proves the Taylor identity with remainder $T_m = R_{m+1}(x,h)$ and its exact factorial normalization.

Apply Q_γ to the identity and use theorem 17.2. Pairwise annihilation and idempotence retain exactly the output packet γ. Boundedness of Q_γ allows it to pass through the Bochner integral. In particular, if $\|D^{m+1}F(x+th)\| \le M$ along the segment, then

$$\|Q_\gamma R_{m+1}(x,h)\| \le \frac{\|Q_\gamma\| M \|h\|^{m+1}}{(m+1)!}.$$

For $m = 0$ the Taylor sum is empty. For $h = 0$, multilinearity makes every positive-order term and the remainder zero. □

The relation to derivatives of the complete coordinate representation is also literal. For an open domain $U \subseteq X$, put $\widehat{F} = \mathcal{A}_\mathcal{Q} \circ F \circ \mathcal{S}_\mathcal{P}$ on the open coordinate set $\mathcal{A}_\mathcal{P}(U)$. Boundedness and linearity of analysis and synthesis imply, for $1 \le r \le m$,

$$D^r \widehat{F}(u)[v_1, \dots, v_r] = \mathcal{A}_\mathcal{Q}(D^r F(\mathcal{S}_\mathcal{P} u)[\mathcal{S}_\mathcal{P} v_1, \dots, \mathcal{S}_\mathcal{P} v_r]).$$

Indeed, the first derivative follows from the chain rule. Repeating it adds no higher derivatives of the fixed linear coordinate maps. If each v_j is supported in one coordinate slot α_j, synthesis includes that slot into X. Its γ output component is exactly the range jet defined above, including when a chosen direction is zero.

At a fixed base point, the nonzero packet jets form a directed hypergraph. An order r interaction has r input packet labels and one output packet label. The support therefore records interactions relative to the chosen packet system.

For a block $B = \{i_1 < \cdots < i_s\}$ and an intermediate packet γ, write

$$G_{\gamma,B}(x \mid h) = J^s_{\gamma|\alpha_{i_1},\dots,\alpha_{i_s}} G(x)[h_{i_1}, \dots, h_{i_s}].$$

Arity of packet jets

A packet jet has one output slot and as many input slots as its derivative order. In the composition formula, each partition block collects inputs for one derivative of the inner map. The number of blocks determines the arity of the derivative of the outer map. Intermediate packet labels record the outputs passed between them.

Theorem 17.5 (Packet Faà di Bruno formula). *Let $n \ge 1$, let X, Y, Z be Banach spaces over the same real or complex field, and let $G : X \to Y$ and $F : Y \to Z$ be n times continuously Fréchet differentiable. Let $\mathcal{P}$, $\mathcal{Q}$, and $\mathcal{R}$ be packet systems on X, Y, and Z. Fix input packets $\alpha_1, \dots, \alpha_n$, an output packet δ, and directions $h_i \in X_{\alpha_i}$. Then*

$$\begin{aligned} &J^n_{\delta|\alpha_1,\dots,\alpha_n}(F \circ G)(x)[h_1, \dots, h_n] \\ &= \sum_{\pi=\{B_1,\dots,B_k\}} \sum_{\gamma_1,\dots,\gamma_k} J^k_{\delta|\gamma_1,\dots,\gamma_k} F(Gx)[G_{\gamma_1,B_1}(x \mid h), \dots, G_{\gamma_k,B_k}(x \mid h)]. \end{aligned}$$

The first sum runs over all set partitions of $\{1, \dots, n\}$.

Proof. First derive the unprojected partition formula. At order one it is the ordinary chain rule. Suppose it holds at order n and the maps are C^{n+1}. Differentiating a summand in direction h_{n+1} has two kinds

of contribution. A derivative of the outer map creates the new singleton block $\{n+1\}$. A derivative of an inner block jet adjoins $n+1$ to that block. Every partition of $\{1,\dots,n+1\}$ arises exactly once: if $\{n+1\}$ is a block, remove that block. Otherwise remove $n+1$ from the unique block that contains it. The chain and product rules for bounded multilinear evaluation justify the differentiation. This proves the unprojected formula by induction.

Insert $I_Y = \sum_\gamma Q_\gamma$ in each argument of the outer derivative and expand the finite sums by multilinearity. Apply R_δ to the output. The directions already lie in the declared input ranges, and each intermediate projection lands in its literal output range. The resulting summands are exactly the displayed packet jets. List directions within a block in increasing index order and blocks in increasing least-element order. Symmetry of the higher derivative makes the final value independent of the block ordering. □

Proposition 17.6 (Number of labeled composition terms)**.** *For $n \geq 1$, if Y has M packets, then the number of packet labeled terms at order n before zero jets are removed is*

$$N_{n,M} = \sum_{k=1}^{n} S(n,k) M^k,$$

where $S(n,k)$ is a Stirling number of the second kind. Thus $N_{n,M}$ is the nth Touchard polynomial evaluated at M.

Proof. A set partition with k blocks receives one independent intermediate packet label on each block. It therefore gives M^k labeled terms. There are $S(n,k)$ such partitions. □

Corollary 17.7 (Structural pruning)**.** *For a map H with fixed input and output packet systems, let $\operatorname{supp}_r H(x)$ be the set of external profiles $(\delta;\alpha_1,\dots,\alpha_r)$ for which the restricted jet $J^r_{\delta|\alpha_1,\dots,\alpha_r} H(x)$ is nonzero. A profile belongs to the admissible composition set if it admits a partition $\pi = \{B_1,\dots,B_k\}$ and intermediate labels $\gamma_1,\dots,\gamma_k$ such that the outer jet of F and every inner block jet of G in theorem 17.5 are nonzero. Then $\operatorname{supp}_n (F \circ G)(x)$ is contained in this admissible set. In particular, a summand vanishes whenever one of its constituent packet jets vanishes.*

Proof. If a profile has no admissible decorated partition, every summand in theorem 17.5 has a zero constituent and therefore vanishes. The composed jet is then the zero multilinear map. This proves the support inclusion by contraposition. The inclusion can be strict. Nonzero constituent maps may compose to zero, and nonzero summands may cancel. Thus the admissible set is a structural upper bound rather than a claim of equality. □

Example 17.8 (The second-order composition law)**.** At order two, the partition formula has two shapes. For packet directions h_1 and h_2,

$$\begin{aligned} R_\delta D^2(F\circ G)(x)[h_1,h_2] = &\sum_{\gamma_1,\gamma_2} J^2_{\delta|\gamma_1,\gamma_2} F(Gx)[J^1_{\gamma_1|\alpha_1} G(x)h_1, J^1_{\gamma_2|\alpha_2} G(x)h_2] \\ &+ \sum_{\gamma} J^1_{\delta|\gamma} F(Gx)[J^2_{\gamma|\alpha_1,\alpha_2} G(x)[h_1,h_2]]. \end{aligned}$$

The first line records two first-order changes entering a quadratic interaction of F. The second records a genuine second-order interaction created inside G before F is applied.

Remark 17.9. Jet sparsity is not invariant under an arbitrary change of coordinates. A useful packet system normally comes from symmetry, scale, support, grading, valuation, or another structure that is already present in the problem. Every sparsity claim should therefore name the packet system with respect to which it is made.

18 Rooted tree expansions

Repeated nonlinear substitution is indexed by rooted trees. We begin with a polynomial fixed point equation on a Banach space.

Let X be a Banach space. Fix $s \in X$ and continuous multilinear maps

$$B_r : X^r \longrightarrow X, \qquad 2 \le r \le d.$$

Set

$$a = \|s\|, \qquad b_r = \|B_r\|, \qquad \phi(t) = a + \sum_{r=2}^{d} b_r t^r.$$

An ordered rooted tree has leaves labeled by s and every internal vertex of arity r labeled by B_r. Its value is obtained by evaluating from the leaves toward the root.

Theorem 18.1 (Polynomial tree expansion). *Suppose there is $R > 0$ such that*

$$\phi(R) \le R$$

and

$$q = \sum_{r=2}^{d} r b_r R^{r-1} < 1.$$

Then the map

$$\Phi(x) = s + \sum_{r=2}^{d} B_r(x, \dots, x)$$

has a unique fixed point in the closed ball $\overline{B}_R(0)$. The sum of the values of all finite ordered rooted trees converges absolutely to that fixed point.

Proof. For $x, y \in \overline{B}_R(0)$, multilinearity and telescoping in the r input positions give

$$\|B_r(x, \dots, x) - B_r(y, \dots, y)\| \le r b_r R^{r-1} \|x - y\|.$$

Hence Φ is q-Lipschitz on the ball. The inequality $\phi(R) \le R$ shows that Φ preserves the ball. The contraction mapping theorem gives a unique fixed point u in that ball.

Let $\mathfrak{T}_m$ be the finite set of ordered rooted trees of height less than m. A leaf has value s. If the root of τ has arity r and ordered subtrees $\tau_1, \dots, \tau_r$, define

$$B_\tau(s) = B_r(B_{\tau_1}(s), \dots, B_{\tau_r}(s)).$$

Induction on m shows that the Picard iterate $u_0 = 0$ and $u_{m+1} = \Phi(u_m)$ satisfies

$$u_m = \sum_{\tau \in \mathfrak{T}_m} B_\tau(s).$$

Define a nonnegative scalar weight by assigning weight a to a leaf and applying the scalar map $(t_1, \dots, t_r) \mapsto b_r t_1 \cdots t_r$ at a vertex of arity r. A second induction gives

$$\sum_{\tau \in \mathfrak{T}_m} \|B_\tau(s)\| \le \sum_{\tau \in \mathfrak{T}_m} w(\tau) = t_m,$$

where $t_0 = 0$ and $t_{m+1} = \phi(t_m)$. The sequence (t_m) is increasing because ϕ has nonnegative coefficients. It is bounded by R because $t_0 \le R$ and $\phi(R) \le R$. The sum of the norms of all tree values is therefore finite. The tree series converges absolutely. Its partial sums are the Picard iterates, so its sum is u. □

Corollary 18.2 (Quadratic Catalan expansion). *Let $B : X \times X \to X$ be bounded and consider*

$$u = s + B(u, u).$$

Put $a = \|s\|$ and $b = \|B\|$. If $4ab < 1$, then

$$u = \sum_{n \ge 1} \sum_{\substack{\tau \text{ full ordered binary tree} \\ \text{with } n \text{ leaves}}} B_\tau(s)$$

converges absolutely. It also satisfies

$$\|u\| \le \frac{1 - \sqrt{1 - 4ab}}{2b}$$

when $b > 0$.

Proof. If $b = 0$, the equation is $u = s$ and the tree series contains only its one leaf. If $a = 0$, every tree value is zero. Choose any R with $0 < R < 1/(2b)$ and apply theorem 18.1 to obtain uniqueness of the zero fixed point in that ball.

Assume now that $a > 0$ and $b > 0$. Take

$$R = \frac{1 - \sqrt{1 - 4ab}}{2b}.$$

Then $a + bR^2 = R$ and $2bR < 1$. Apply theorem 18.1. The number of full ordered binary trees with n leaves is the Catalan number C_{n-1}, which gives the displayed majorant. □

Theorem 18.3 (Packet labeled tree reconstruction). *Let X carry a packet system $\mathcal{P}$. At each internal vertex of a tree, insert the output identity and one input identity for every child. Summing over all internal packet labels reconstructs the original tree value exactly.*

Proof. At a vertex labeled by an r multilinear map B_r, insert

$$I_X = \sum_{\beta} P_\beta$$

on the output and in every input slot. Multilinearity gives an exact finite sum of packet blocks at that vertex. Apply this identity successively from the leaves toward the root. Since the tree and the packet set are finite, all sums are finite and their total reconstructs the original tree value. □

For an infinite tree expansion, the finite sum of packet decorations for each underlying tree is kept together. The unlabeled tree majorant does not imply absolute convergence when all decorations are summed separately for arbitrary oblique projectors. Finite tree reconstruction and the grouped interpretation remain exact.

> **Arity in a tree**
>
> At each internal vertex, the number of children is the number of input slots of its multilinear map. Their order is retained because symmetry is not assumed. Packet labels resolve each finite tree into a finite sum. In an infinite expansion these sums stay grouped by tree, as required by the convergence statement above.

Corollary 18.4 (Tree pruning)**.** *A packet labeled tree vanishes if one internal vertex carries a zero packet block of the corresponding multilinear map. Removing all such zero terms leaves every finite tree sum unchanged and therefore leaves the grouped infinite series unchanged.*

Proof. A zero vertex map makes the value of the full tree zero. □

The tree theorem separates combinatorial growth from analytic convergence. The number of trees may grow rapidly, while multilinear bounds control their values. The scalar map ϕ controls the total weight of the expansion. Packet pruning changes the combinatorics but not the proof. It removes trees whose values are already zero before any norm estimate is used.

Example 18.5 (Catalan parity packets)**.** Over a field of characteristic zero, the formal equation

$$u = z + u^2$$

has the branch

$$u(z) = \frac{1 - \sqrt{1 - 4z}}{2}.$$

Write $u = u_0 + u_1$, where u_0 contains the even powers of z and u_1 contains the odd powers. Exact packet projection gives

$$u_0 = u_0^2 + u_1^2, \qquad u_1 = z + 2u_0u_1.$$

The two equations remain coupled. Their sum is the original fixed point equation.

19 Graded nonlinear operations

Let G be a finite abelian group and let

$$A = \bigoplus_{g \in G} A_g$$

be a G graded module.

Definition 19.1 (Grade respecting operation)**.** An r multilinear map $\mu : A^r \to A$ respects the grading when

$$\mu(A_{g_1}, \dots, A_{g_r}) \subseteq A_{g_1+\cdots+g_r}.$$

Theorem 19.2 (Graded packet convolution)**.** *If $a_j = \sum_{g\in G} a_{j,g}$ and μ respects the grading, then*

$$P_k\mu(a_1, \dots, a_r) = \sum_{g_1+\cdots+g_r=k} \mu(a_{1,g_1}, \dots, a_{r,g_r}).$$

Proof. Expand every input into its homogeneous components. Each term lands in the component whose label is the sum of its input labels. Projection onto A_k retains exactly the displayed terms. □

Corollary 19.3 (Cyclic product formula)**.** *For a cyclically graded algebra*

$$A = \bigoplus_{j\in\mathbb{Z}/N\mathbb{Z}} A_j, \qquad A_iA_j \subseteq A_{i+j},$$

one has

$$P_k(ab) = \sum_{i+j=k} a_ib_j.$$

Proof. Apply theorem 19.2 to multiplication. □

Theorem 19.4 (Packet Maurer Cartan equations)**.** *Let*

$$\mathcal{L} = \bigoplus_{g\in G} \mathcal{L}_g$$

be a differential graded Lie algebra over a ring in which two is invertible. Assume

$$d\mathcal{L}_g \subseteq \mathcal{L}_g, \qquad [\mathcal{L}_g, \mathcal{L}_h] \subseteq \mathcal{L}_{g+h}.$$

For a degree one element $\eta = \sum_g \eta_g$, the Maurer Cartan equation

$$d\eta + \frac{1}{2}[\eta, \eta] = 0$$

is equivalent to

$$d\eta_k + \frac{1}{2}\sum_{g+h=k} [\eta_g, \eta_h] = 0$$

for every $k \in G$.

Proof. The differential preserves the group label and the bracket adds labels. Apply the projector onto $\mathcal{L}_k$ and use exact reconstruction. □

The same argument applies to higher brackets.

Proposition 19.5 (Packet equations for L_∞ operations)**.** *Let $\mathcal{L}$ be complete and carry operations ℓ_r that respect a finite abelian group grading. Assume the packet projectors are continuous for the topology in which the series below converges. Assume the Maurer Cartan series*

$$\sum_{r\geq 1}\frac{1}{r!}\ell_r(\eta,\dots,\eta)$$

converges and the required factorials are invertible. Then its packet of label k is

$$\sum_{r\geq 1}\frac{1}{r!}\sum_{g_1+\cdots+g_r=k}\ell_r(\eta_{g_1},\dots,\eta_{g_r}).$$

The full Maurer Cartan equation holds if and only if every packet equation vanishes.

Proof. Apply theorem 19.2 to each operation ℓ_r. Continuity of the packet projectors permits the convergent sums to be projected term by term. □

Definition 19.6 (Divided square)**.** A divided square on degree one elements is a map q satisfying

$$q(x+y)=q(x)+q(y)+[x,y],\qquad q(ax)=a^2q(x),$$

whenever the expressions are defined.

Proposition 19.7 (Divided square packet equations)**.** *Assume $q(\mathcal{L}_g)\subseteq\mathcal{L}_{2g}$. Choose an ordering of the finite label set. The equation*

$$d\eta+q(\eta)=0$$

is equivalent to

$$d\eta_k+\sum_{2g=k}q(\eta_g)+\sum_{\substack{g<h\\ g+h=k}}[\eta_g,\eta_h]=0$$

for every k.

Proof. Repeated use of the polarization identity for q gives

$$q\left(\sum_g\eta_g\right)=\sum_g q(\eta_g)+\sum_{g<h}[\eta_g,\eta_h].$$

Project onto the component of label k. □

Example 19.8 (Matrix difference grading)**.** Let $A=M_N(\Bbbk)$ and assign the matrix unit E_{rs} the label $r-s$ modulo N. Since

$$E_{rs}E_{uv}=\delta_{su}E_{rv},$$

the output label is

$$(r-s)+(u-v)=r-v$$

whenever the product is nonzero. Matrix multiplication therefore obeys the cyclic convolution law exactly. The same grading resolves commutators and polynomial matrix equations.

Remark 19.9. These results begin with a graded differential or higher algebra. They do not assert that every deformation problem has such a controlling object. Differential graded and infinity algebra methods are developed in [11].

20 Nonlinear refinement defects

Let

$$F_f : X_f \longrightarrow Y_f, \qquad F_c : X_c \longrightarrow Y_c$$

be fine and coarse maps. Let

$$C_X : X_f \longrightarrow X_c, \qquad C_Y : Y_f \longrightarrow Y_c$$

be linear coarsening maps. Whenever a norm, a Lipschitz constant, or a Fréchet derivative is used, the spaces are Banach spaces and the coarsening maps are bounded.

Definition 20.1 (Nonlinear refinement defect)**.** The pointwise defect of F is

$$\Delta_F(x) = C_Y F_f(x) - F_c(C_X x).$$

The two models are exactly compatible when Δ_F vanishes everywhere.

The defect is a vector in Y_c. It determines an obstruction class after a correction map into Y_c has been chosen and the defect has been passed to the corresponding cokernel. Norm estimates measure the size of the defect. The quotient determines whether an allowed linear correction can remove it.

Theorem 20.2 (Exact defect under composition)**.** *Let $G_f : Y_f \to Z_f$ and $G_c : Y_c \to Z_c$, with coarsening $C_Z : Z_f \to Z_c$. Then*

$$\Delta_{G\circ F}(x) = \Delta_G(F_f(x)) + G_c(F_c(C_X x) + \Delta_F(x)) - G_c(F_c(C_X x)).$$

If G_c is L-Lipschitz on the relevant set, then

$$\|\Delta_{G\circ F}(x)\| \le \|\Delta_G(F_f(x))\| + L\|\Delta_F(x)\|.$$

Proof. Add and subtract $G_c(C_Y F_f(x))$. Use

$$C_Y F_f(x) = F_c(C_X x) + \Delta_F(x).$$

The norm estimate follows from the Lipschitz bound. □

Corollary 20.3 (Linear product rule)**.** *If all maps in theorem 20.2 are linear, then*

$$\Delta_{GF} = \Delta_G F_f + G_c \Delta_F.$$

Proof. Linearity turns the last difference in theorem 20.2 into $G_c\Delta_F(x)$. □

Theorem 20.4 (Iterated orbit defect). *Let $F_f : X_f \to X_f$ and $F_c : X_c \to X_c$, with coarsening $C : X_f \to X_c$. Suppose F_c is q-Lipschitz. For a fine orbit starting at x, assume*

$$\|\Delta_F(F_f^j x)\| \le \varepsilon_j.$$

Then

$$\left\|CF_f^n(x) - F_c^n(Cx)\right\| \le \sum_{j=0}^{n-1} q^{n-1-j}\varepsilon_j.$$

If $0 \le q < 1$ and every $\varepsilon_j \le \varepsilon$, then

$$\left\|CF_f^n(x) - F_c^n(Cx)\right\| \le \varepsilon\frac{1-q^n}{1-q}.$$

Proof. Let

$$e_n = \left\|CF_f^n(x) - F_c^n(Cx)\right\|.$$

The definition of the defect and the Lipschitz property give

$$e_{n+1} \le \varepsilon_n + qe_n.$$

Iterating this recurrence proves the first estimate. The second is the geometric sum. □

Theorem 20.5 (Fixed point stability). *Assume F_c is a contraction with constant $q < 1$. Let x_f and x_c be fixed points of F_f and F_c. If*

$$\|\Delta_F(x_f)\| \le \varepsilon,$$

then

$$\|Cx_f - x_c\| \le \frac{\varepsilon}{1-q}.$$

When the defect at x_f is zero, the fixed points are compatible.

Proof. Using the fixed-point equations,

$$\begin{aligned}\|Cx_f - x_c\| &\le \|CF_f(x_f) - F_c(Cx_f)\| + \|F_c(Cx_f) - F_c(x_c)\| \\ &\le \varepsilon + q\|Cx_f - x_c\|.\end{aligned}$$

Rearrange. □

Corollary 20.6 (Several refinement levels). *Let $C_\ell : X_{\ell+1} \to X_\ell$ for $0 \le \ell < m$. Suppose x_ℓ is a fixed point at level ℓ, the coarse map at level ℓ has contraction constant $q_\ell < 1$, and the adjacent defect at $x_{\ell+1}$ is at most ε_ℓ. Write*

$$C_{a,b} = C_a C_{a+1} \cdots C_{b-1} \quad \text{for } a < b, \qquad C_{a,a} = I.$$

Then

$$\|C_{0,m}x_m - x_0\| \le \sum_{\ell=0}^{m-1} \|C_{0,\ell}\|\frac{\varepsilon_\ell}{1-q_\ell}.$$

Proof. Insert the intermediate vectors $C_{0,\ell}x_\ell$ and telescope. Apply theorem 20.5 to every adjacent pair. □

Proposition 20.7 (Derivative defects). *Assume F_f and F_c are r times Fréchet differentiable and the coarsening maps are bounded and linear. Then*

$$D\Delta_F(x) = C_Y DF_f(x) - DF_c(C_X x)C_X.$$

For $r \geq 2$,

$$D^r\Delta_F(x)[h_1, \dots, h_r] = C_Y D^r F_f(x)[h_1, \dots, h_r] - D^r F_c(C_X x)[C_X h_1, \dots, C_X h_r].$$

Proof. Differentiate the defining identity for Δ_F. The coarsening maps are linear, so their higher derivatives vanish. □

Example 20.8 (Sharp affine bound). Let $F_c(y) = qy + b$ with $0 < q < 1$. Choose a fine fixed point whose coarsening has constant defect ε in the direction of the scalar state. The fixed point mismatch is exactly $\varepsilon/(1-q)$. Thus the constant in theorem 20.5 cannot be improved without more information.

21 Canonical obstruction objects

We begin by isolating the part of a defect that cannot be removed by an allowed linear correction.

Definition 21.1 (Linear correction datum). A linear correction datum consists of a homomorphism

$$D : U \longrightarrow V$$

and a defect $r \in V$. A correction is an element $u \in U$ satisfying

$$Du = -r.$$

The obstruction class is

$$\mathrm{Ob}_D(r) = [r] \in \operatorname{coker} D = V / \operatorname{im} D.$$

Theorem 21.2 (Cokernel obstruction criterion). *A linear correction exists if and only if*

$$\mathrm{Ob}_D(r) = 0.$$

The quotient map

$$q_D : V \longrightarrow \operatorname{coker} D$$

has the following universal property. If $\varphi : V \to W$ is linear and $\varphi D = 0$, then there is a unique linear map

$$\overline{\varphi} : \operatorname{coker} D \longrightarrow W$$

with $\varphi = \overline{\varphi} q_D$.

Proof. The class of r is zero exactly when $r \in \operatorname{im} D$, which is equivalent to solvability of $Du = -r$. If $\varphi D = 0$, then φ vanishes on $\operatorname{im} D$. It therefore descends uniquely to the quotient. □

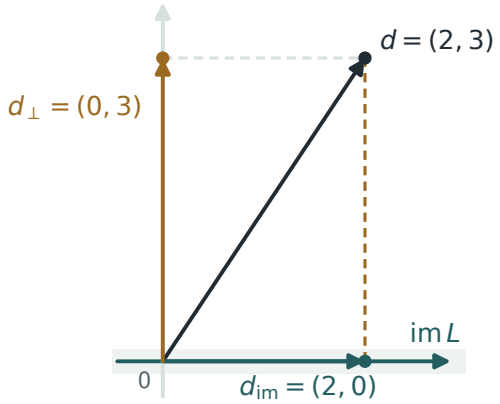
The universal property explains why the cokernel is the canonical obstruction object relative to D. Any linear obstruction test that vanishes on correctable defects factors through it.

Canonical obstruction data

The residual, the correction map, and the obstruction class are distinct data. The residual is a vector. The correction map specifies which changes are allowed, and its cokernel records the remaining class. Canonicality is relative to that declared correction map. A scalar or orthogonal representative of the class may require further choices or hypotheses.

Correctable defect and cokernel residual

$L : \mathbb{R}^2 \to \mathbb{R}^2$, $L(a, b) = (a, 0)$. Defect $d = (2, 3)$.



A chosen scalar detector

$$\ell(x, y) = y$$

$$\ell L = 0$$

$$\ell(d) = 3 \neq 0$$

$$u = (-2, 0): \quad d + Lu = (0, 3) \neq 0.$$

The class $[d] \in \mathbb{R}^2/\operatorname{im} L$ is represented by $(0, 3)$.

Standard Euclidean splitting is used. The scalar 3 depends on the displayed quotient identification.

Figure 5. A specified correction range in $\mathbb{R}^2$. For $L(a, b) = (a, 0)$, the image is the horizontal axis. The defect $(2, 3)$ splits into $(2, 0)$ in that image and the remaining representative $(0, 3)$. Subtracting the image component leaves the same cokernel class. The functional $(x, y) \mapsto y$ vanishes on the correction range and detects the value three. The correction equation $Lu = -(2, 3)$ therefore has no solution.

Theorem 21.3 (Naturality of obstruction classes)**.** *Suppose*

$$\begin{array}{ccc} U & \xrightarrow{D} & V \\ a\downarrow & & \downarrow b \\ U' & \xrightarrow{D'} & V' \end{array}$$

is a commuting square. Then there is a unique induced map

$$\overline{b} : \operatorname{coker} D \longrightarrow \operatorname{coker} D'$$

satisfying

$$\overline{b}\,\operatorname{Ob}_D(r) = \operatorname{Ob}_{D'}(br).$$

Proof. The relation $bD = D'a$ shows that the quotient map $q_{D'}b$ vanishes on $\operatorname{im} D$. Apply the universal property in theorem 21.2. □

Theorem 21.4 (Packet decomposition of an obstruction)**.** *Suppose*

$$U = \bigoplus_{\alpha\in I} U_\alpha, \qquad V = \bigoplus_{\alpha\in I} V_\alpha,$$

and $D(U_\alpha) \subseteq V_\alpha$ for every α. Let

$$D_\alpha : U_\alpha \longrightarrow V_\alpha$$

be the restriction. Then

$$\operatorname{coker} D \cong \bigoplus_{\alpha\in I} \operatorname{coker} D_\alpha.$$

Under this isomorphism, the obstruction class of $r = \sum_\alpha r_\alpha$ is the tuple

$$\big([r_\alpha]\big)_\alpha.$$

Hence r is correctable if and only if every packet obstruction vanishes.

Proof. The grading assumption gives

$$\operatorname{im} D = \bigoplus_{\alpha} \operatorname{im} D_\alpha.$$

Taking the quotient of the direct sum by this direct sum of images gives the displayed isomorphism. □

Remark 21.5. The conclusion of theorem 21.4 requires the correction map to preserve the packets. Off-diagonal correction blocks can couple packet defects. In that case the full cokernel remains canonical, but it need not split into the separate packet cokernels.

Proposition 21.6 (Projectors on the obstruction quotient)**.** *Let $(Q_\alpha)_{\alpha\in I}$ be a complete family of pairwise annihilating idempotents on V. Assume*

$$Q_\alpha(\operatorname{im} D) \subseteq \operatorname{im} D$$

for every α. Then each Q_α induces an idempotent $\overline{Q}_\alpha$ on $\operatorname{coker} D$*, and*

$$\operatorname{coker} D = \bigoplus_{\alpha\in I} \overline{Q}_\alpha(\operatorname{coker} D).$$

For every defect $r \in V$,

$$\overline{Q}_\alpha \operatorname{Ob}_D(r) = [Q_\alpha r].$$

No packet decomposition of the correction space U is required.

Proof. If $v - v' \in \operatorname{im} D$, then

$$Q_\alpha(v - v') \in \operatorname{im} D.$$

Hence $[v] \mapsto [Q_\alpha v]$ is well defined on the quotient. The idempotent, annihilation, and reconstruction identities descend from V to $V/\operatorname{im} D$. The formula for an obstruction class follows from the definition. □

21.1 Relative obstruction under coarsening

Coarsening can erase an obstruction that remains present in the fine problem. The next exact sequence identifies the part removed by coarsening.

Theorem 21.7 (Relative obstruction exact sequence)**.** *Suppose the following diagram of modules commutes and has exact rows:*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & U_0 & \xrightarrow{i_U} & U_f & \xrightarrow{p_U} & U_c & \longrightarrow & 0 \\ & & D_0 \downarrow & & D_f \downarrow & & D_c \downarrow & & \\ 0 & \longrightarrow & V_0 & \xrightarrow{i_V} & V_f & \xrightarrow{p_V} & V_c & \longrightarrow & 0. \end{array}$$

There is a natural exact sequence

$$\begin{aligned} 0 \longrightarrow \operatorname{Ker} D_0 \longrightarrow \operatorname{Ker} D_f \longrightarrow \operatorname{Ker} D_c \xrightarrow{\delta} \operatorname{coker} D_0 \\ \longrightarrow \operatorname{coker} D_f \longrightarrow \operatorname{coker} D_c \longrightarrow 0. \end{aligned}$$

For $u_c \in \operatorname{Ker} D_c$, choose $u_f \in U_f$ with $p_U u_f = u_c$. There is a unique $v_0 \in V_0$ such that

$$D_f u_f = i_V v_0,$$

and the connecting map is

$$\delta(u_c) = [v_0].$$

Proof. Since $D_c u_c = 0$, commutativity gives

$$p_V D_f u_f = D_c p_U u_f = 0.$$

Exactness of the lower row gives a unique v_0 with $D_f u_f = i_V v_0$. A different lift has the form $u_f + i_U u_0$. The corresponding element of V_0 changes by $D_0 u_0$, so $[v_0]$ is independent of the lift.

The map $\operatorname{Ker} D_0 \to \operatorname{Ker} D_f$ is injective because i_U is injective. If $u_f \in \operatorname{Ker} D_f$ maps to zero in U_c, then $u_f = i_U u_0$ for a unique u_0. The identity

$$i_V D_0 u_0 = D_f i_U u_0 = 0$$

and injectivity of i_V show that $u_0 \in \operatorname{Ker} D_0$. This proves exactness at $\operatorname{Ker} D_f$.

If u_c comes from an element of $\operatorname{Ker} D_f$, one may choose that element as its lift and obtain $\delta(u_c) = 0$. Conversely, if $\delta(u_c) = 0$, then $v_0 = D_0 u_0$ for some u_0. Replacing u_f by $u_f - i_U u_0$ gives an element of $\operatorname{Ker} D_f$ that maps to u_c. This proves exactness at $\operatorname{Ker} D_c$.

The class $[v_0] \in \operatorname{coker} D_0$ maps to zero in $\operatorname{coker} D_f$ whenever it equals $\delta(u_c)$, since the chosen lift satisfies $i_V v_0 = D_f u_f$. Conversely, if $i_V v_0 = D_f u_f$, then $u_c = p_U u_f$ lies in $\operatorname{Ker} D_c$ and $\delta(u_c) = [v_0]$. This proves exactness at $\operatorname{coker} D_0$.

If a class in $\operatorname{coker} D_f$ comes from $\operatorname{coker} D_0$, its image in $\operatorname{coker} D_c$ is zero. Conversely, suppose $[v_f]$ maps to zero in $\operatorname{coker} D_c$. Choose u_c with $D_c u_c = p_V v_f$ and lift it to $u_f \in U_f$. Then

$$p_V(v_f - D_f u_f) = 0,$$

so $v_f - D_f u_f = i_V v_0$ for some $v_0 \in V_0$. Thus $[v_f]$ comes from $[v_0]$. Surjectivity at the last term follows from surjectivity of p_V. □

Corollary 21.8 (Coarse vanishing and fine vanishing)**.** *Let $[r_f] \in \operatorname{coker} D_f$. If its image in $\operatorname{coker} D_c$ is zero, then $[r_f]$ comes from $\operatorname{coker} D_0$. Coarse vanishing therefore means that the fine obstruction is supported in information removed by coarsening. It does not imply $[r_f] = 0$.*

Proof. This is exactness of the sequence in theorem 21.7 at $\operatorname{coker} D_f$. □

Example 21.9 (A fine obstruction erased by coarsening)**.** Take all three correction spaces to be zero. Let

$$V_f = \Bbbk^2, \qquad V_c = \Bbbk, \qquad p_V(x, y) = x + y,$$

and let

$$V_0 = \operatorname{Ker} p_V = \operatorname{span}\{(1, -1)\}.$$

The fine defect $(1, -1)$ is a nonzero class in $\operatorname{coker} D_f = V_f$, while its coarse image is zero. The class comes from $\operatorname{coker} D_0 = V_0$, exactly as corollary 21.8 predicts.

The same construction has a cohomological form.

Definition 21.10 (Mapping cone)**.** Let $D : C^\bullet \to E^\bullet$ be a cochain map. Its mapping cone is

$$\operatorname{Cone}(D)^n = E^n \oplus C^{n+1}$$

with differential

$$d_{\operatorname{Cone}}(e, c) = (d_E e + Dc, -d_C c).$$

Theorem 21.11 (Cohomological obstruction through the cone)**.** *Let $e \in E^n$ satisfy $d_E e = 0$. The class of $(e, 0)$ in*

$$H^n(\operatorname{Cone}(D))$$

vanishes if and only if the class $[e] \in H^n(E)$ lies in the image of

$$H^n(D) : H^n(C) \longrightarrow H^n(E).$$

When C and E are concentrated in degree zero,

$$H^0(\mathrm{Cone}(D)) \cong \operatorname{coker} D.$$

Proof. The class of $(e, 0)$ vanishes exactly when there are $e' \in E^{n-1}$ and $c \in C^n$ with

$$(e, 0) = d_{\mathrm{Cone}}(e', c).$$

This means

$$e = d_E e' + Dc, \qquad d_C c = 0.$$

Equivalently, $[e] = H^n(D)[c]$. The degree zero statement follows directly from the definition of the cone differential. □

Corollary 21.12 (Packet obstruction complexes)**.** *If $C^\bullet$ and $E^\bullet$ split into finite packet subcomplexes and D preserves those packets, then* $\mathrm{Cone}(D)$ *splits into packet subcomplexes and*

$$H^n(\mathrm{Cone}(D)) \cong \bigoplus_\alpha H^n(\mathrm{Cone}(D_\alpha)).$$

Proof. Both the cone and its cohomology commute with finite direct sums. □

Example 21.13 (A two component obstruction)**.** Let

$$D : \Bbbk \longrightarrow \Bbbk^2, \qquad D(t) = (t, t).$$

The cokernel is one dimensional. The defect (a, b) is correctable exactly when $a = b$. The obstruction can be represented by $a - b$, but this scalar representative depends on a choice of isomorphism from the cokernel to $\Bbbk$. The quotient class itself is canonical.

22 Split Hilbert obstructions and nonlinear correction

The quotient obstruction has a concrete orthogonal representative when the correction map acts between Hilbert spaces and has closed range. The Hilbert and Banach space results below are analytic extensions of the finite calculus. They use the stated analytic hypotheses and lie outside the finite formal contract.

Theorem 22.1 (Orthogonal obstruction projector)**.** *Let $D : H_1 \to H_2$ be a bounded operator between Hilbert spaces with closed range. Let $D^\dagger$ be its Moore–Penrose inverse. Then*

$$P_{\mathrm{ob}} = I_{H_2} - DD^\dagger$$

is the orthogonal projector onto

$$(\operatorname{im} D)^\perp = \operatorname{Ker} D^*.$$

For $r \in H_2$, the following are equivalent:

1. *$Du = -r$ is solvable*
2. *$[r] = 0$ in* $\operatorname{coker} D$

3. $P_{\text{ob}} r = 0$.

When these conditions hold, $-D^{\dagger} r$ is the least norm correction. For every r,

$$\operatorname{dist}(r, \operatorname{im} D) = \|P_{\text{ob}} r\|.$$

Proof. Closed range gives the orthogonal decomposition

$$H_2 = \operatorname{im} D \oplus \operatorname{Ker} D^*.$$

The Moore–Penrose identities show that $DD^{\dagger}$ is the orthogonal projector onto $\operatorname{im} D$. The stated equivalences and distance formula follow from this decomposition. The least norm property is the standard characterization of $D^{\dagger} r$. □

Once the defect lies in the range, nonlinear correction still requires control of the nonlinear term. The next theorem gives that control through a bounded right inverse and a contraction estimate.

Theorem 22.2 (Right inverse correction theorem)**.** *Let $D : X \to Y$ be a bounded linear map between Banach spaces. Assume $Z = \operatorname{im} D$ is closed and there is a bounded linear right inverse*

$$R : Z \longrightarrow X, \qquad DR = I_Z.$$

Put $X_R = R(Z)$. Let

$$N : \overline{B}_{\rho}(0) \cap X_R \longrightarrow Z$$

be L-Lipschitz. Let $y \in Z$ and put

$$q = \|R\| L.$$

Assume $q < 1$ and

$$\|R(y - N(0))\| \leq (1 - q)\rho.$$

Then the equation

$$Dx + N(x) = y$$

has a unique solution in $\overline{B}_{\rho}(0) \cap X_R$.

Proof. The space X_R is closed. Indeed, if $Rz_n \to x$, then $z_n = DRz_n \to Dx$, so $x = R(Dx)$. Define

$$\Psi(x) = R(y - N(x)).$$

This map takes X_R to itself. For $x \in \overline{B}_{\rho}(0) \cap X_R$,

$$\begin{aligned} \|\Psi(x)\| &\leq \|R(y - N(0))\| + \|R\| \, \|N(x) - N(0)\| \\ &\leq (1 - q)\rho + q\rho = \rho. \end{aligned}$$

Also,

$$\|\Psi(x) - \Psi(x')\| \leq q \|x - x'\|.$$

The contraction mapping theorem gives a unique fixed point in the stated set. Applying D to the fixed point identity gives the equation. Conversely, a solution in X_R equals $R(Dx)$ and is therefore a fixed point of Ψ. □

The hypotheses behind correction

Closed range gives the obstruction an orthogonal representative in the Hilbert setting. A bounded right inverse chooses a correction slice. In the nonlinear theorem, the target and the nonlinear term both lie in the range, and the Lipschitz bound makes the correction map a contraction on the chosen ball. The uniqueness statement concerns the chosen ball within that correction slice.

Corollary 22.3 (Stability with respect to the target)**.** *Under the hypotheses of theorem 22.2, let x and x' be the fixed points associated with targets y and y'. If both fixed point maps preserve the same ball, then*

$$\|x - x'\| \leq \frac{\|R\|}{1-q}\|y - y'\|.$$

Proof. Subtract the two fixed-point equations and use the Lipschitz bound for N:

$$\|x - x'\| \leq \|R\|\|y - y'\| + q\|x - x'\|.$$

□

Lyapunov–Schmidt reduction separates the equation into two parts. The implicit function theorem solves the range component, leaving an equation in the obstruction space. Standard accounts of this reduction appear in [9, 19].

Theorem 22.4 (Local Lyapunov–Schmidt reduction)**.** *Let $\mathcal{U} \subseteq X$ be an open neighborhood of zero and let $F : \mathcal{U} \to Y$ be a C^r map between Banach spaces with $r \geq 1$, $F(0) = 0$, and $L = DF(0)$. Assume there are closed decompositions*

$$X = K \oplus X_1, \qquad Y = R \oplus O,$$

where $K = \operatorname{Ker} L$, $R = \operatorname{im} L$, and

$$L_1 = L|_{X_1} : X_1 \longrightarrow R$$

is a Banach space isomorphism. Let π_R and π_O be the two projections on Y. Then there are neighborhoods $U_K \subseteq K$ and $U_1 \subseteq X_1$ of zero and a unique C^r map

$$\psi : U_K \longrightarrow U_1$$

with $\psi(0) = 0$ such that

$$\pi_R F(k + \psi(k)) = 0$$

for every $k \in U_K$. After the neighborhoods are made smaller if needed, the solutions of $F(x) = 0$ near zero are exactly the points

$$x = k + \psi(k)$$

for which the reduced obstruction map

$$\kappa(k) = \pi_O F(k + \psi(k))$$

vanishes. At the base point,

$$D\psi(0) = 0, \qquad D\kappa(0) = 0.$$

Proof. Define

$$G(k, x_1) = \pi_R F(k + x_1).$$

The derivative of G in the X_1 variable at $(0, 0)$ is L_1, which is invertible. The implicit function theorem gives the unique map ψ. Every nearby x has a unique decomposition $x = k + x_1$. The range equation forces $x_1 = \psi(k)$, and the remaining equation is precisely $\kappa(k) = 0$.

Differentiate $G(k, \psi(k)) = 0$ at zero. Since L vanishes on K, one obtains $L_1 D\psi(0) = 0$, hence $D\psi(0) = 0$. Finally,

$$D\kappa(0) = \pi_O L|_K = 0.$$

□

Corollary 22.5 (Quadratic reduced obstruction)**.** *Assume the hypotheses of theorem 22.4 with $r \geq 2$. For $k_1, k_2 \in K$,*

$$D^2\kappa(0)[k_1, k_2] = \pi_O D^2 F(0)[k_1, k_2].$$

Proof. Differentiate $\kappa(k) = \pi_O F(k+\psi(k))$ twice. The terms containing $D\psi(0)$ vanish. The term containing $D^2\psi(0)$ is multiplied by $\pi_O L$, which is zero because $L(X) = R$. □

Example 22.6 (A purely quadratic obstruction)**.** Let

$$F(x, y) = (y, x^2)$$

on $\mathbb{R}^2$. At the origin,

$$L(h, k) = (k, 0).$$

The kernel is the x axis, the range is the first output axis, and the cokernel is the second output axis. The range equation forces $y = 0$. The reduced obstruction map is

$$\kappa(x) = x^2.$$

The linear obstruction vanishes for every kernel direction, but the nonlinear equation still has only the solution $x = 0$ near the origin.

Remark 22.7. The reduced map κ begins at quadratic order. This is why vanishing of the first-order cokernel class need not settle a nonlinear problem. Higher terms can produce a genuine obstruction on the kernel directions.

23 Filtered lifting

Let X and Y be modules with decreasing filtrations

$$X = F^0X \supseteq F^1X \supseteq \cdots, \qquad Y = F^0Y \supseteq F^1Y \supseteq \cdots.$$

Assume both are separated and complete for their filtration topologies. Let $\Phi : X \to Y$ be continuous and filtration congruence preserving in the sense that

$$h \in F^mX \quad \Longrightarrow \quad \Phi(x + h) - \Phi(x) \in F^mY$$

for every $x \in X$ and every m. This condition makes the statement $\Phi(x) \in F^m Y$ depend only on the class of x modulo $F^m X$.

Suppose x_m satisfies

$$\Phi(x_m) \in F^m Y.$$

Assume that corrections $h \in F^m X$ obey a first-order congruence

$$\Phi(x_m + h) \equiv \Phi(x_m) + L_m h \pmod{F^{m+1} Y},$$

where $L_m : F^m X \to F^m Y$ is a module homomorphism satisfying

$$L_m(F^{m+1} X) \subseteq F^{m+1} Y$$

and therefore induces

$$\overline{L}_m : F^m X / F^{m+1} X \longrightarrow F^m Y / F^{m+1} Y.$$

At depth m, only the associated graded correction is visible. The cokernel below has finite length whenever the target graded piece has finite length, even if the completed problem is infinite. Vanishing of the depth obstruction decides whether the current approximation can be improved by one level. When the graded pieces are finite, the branching count below applies. Finite length alone does not imply finite cardinality.

Definition 23.1 (Depth obstruction)**.** The depth m obstruction of x_m is

$$\mathrm{Ob}_m(x_m) = [\Phi(x_m)] \in \operatorname{coker} \overline{L}_m,$$

where $\Phi(x_m)$ is first viewed in $F^m Y / F^{m+1} Y$.

Theorem 23.2 (One-step lifting criterion)**.** *The approximation x_m has a correction $h_m \in F^m X$ such that*

$$\Phi(x_m + h_m) \in F^{m+1} Y$$

if and only if

$$\mathrm{Ob}_m(x_m) = 0.$$

When a correction exists, the set of its classes in

$$F^m X / F^{m+1} X$$

is a coset of $\operatorname{Ker} \overline{L}_m$*. If the graded pieces are vector spaces, this coset is an affine space.*

Proof. The next depth condition is equivalent to

$$\overline{L}_m[h_m] = -[\Phi(x_m)]$$

in $F^m Y / F^{m+1} Y$. This is solvable exactly when the displayed defect has zero cokernel class. If h_m and h'_m are two solutions, their difference lies in $\operatorname{Ker} \overline{L}_m$. Adding an element of the kernel to one solution gives another, so the solution set is a coset of the kernel. □

Corollary 23.3 (Finite branching count)**.** *Suppose the graded pieces are finite vector spaces over a finite field. If the depth obstruction vanishes, then the number of correction classes at depth m is*

$$|\operatorname{Ker} \overline{L}_m|.$$

If the obstruction does not vanish, the number is zero.

Proof. A nonempty affine space over a finite vector space has the same number of elements as that vector space. □

Lemma 23.4 (Well-defined lift classes)**.** *Under the filtration congruence assumption, the condition*

$$\Phi(x) \in F^m Y$$

depends only on the class of x in $X/F^m X$. Fix a representative x_m of a depth m class. The map

$$[h] \longmapsto [x_m + h]_{m+1}$$

identifies correction classes $[h] \in F^m X/F^{m+1}X$ with the classes at depth $m+1$ that reduce to $[x_m]_m$. Under this identification, the solutions of the associated graded correction equation in theorem 23.2 are exactly the children whose defect lies in $F^{m+1}Y$. A different representative of the same parent gives a translated parametrization of the same set of children.

Proof. If $x' - x \in F^m X$, then filtration congruence preservation gives

$$\Phi(x') - \Phi(x) \in F^m Y.$$

Thus one defect lies in $F^m Y$ exactly when the other does. If $h - h' \in F^{m+1}X$, then $x_m + h$ and $x_m + h'$ define the same depth $m+1$ class. Conversely, two depth $m+1$ classes reducing to $[x_m]_m$ differ from x_m by elements of $F^m X$, and they are equal precisely when those elements agree modulo $F^{m+1}X$. The first-order congruence shows that such a child has defect in $F^{m+1}Y$ exactly when its correction class solves the associated graded equation. Replacing x_m by another representative translates the correction coordinate by an element of $F^m X/F^{m+1}X$ and leaves the child classes themselves unchanged. □

Definition 23.5 (Lift tree)**.** Fix an initial approximation. A vertex at depth m is a class $[x]_m \in X/F^m X$ with $\Phi(x) \in F^m Y$ that reduces to the fixed initial class. A depth $m+1$ vertex is joined to its reduction in $X/F^m X$. By lemma 23.4, this rooted graph is independent of the chosen representatives. It is called the lift tree.

Theorem 23.6 (Finitely branching lift tree)**.** *Assume every level of the lift tree is nonempty and every vertex has only finitely many children. Then the tree contains an infinite branch. If X and Y are separated and complete and Φ is continuous, that branch determines an exact solution of $\Phi(x) = 0$.*

Proof. The first assertion is König's infinity argument. The root has finitely many children. Since vertices occur at arbitrarily large depths, at least one child has descendants at arbitrarily large depths. Choose such a child and repeat the argument at every chosen vertex. This gives one compatible vertex at each depth.

Choose representatives recursively. Once x_m represents the selected depth m vertex, choose a representative x_{m+1} of its selected child with

$$x_{m+1} - x_m \in F^m X.$$

The resulting sequence is Cauchy for the filtration topology. Each representative satisfies $\Phi(x_m) \in F^m Y$. The conclusion now follows from theorem 23.8. □

Corollary 23.7 (Unique lift branch)**.** *Suppose an initial approximation is fixed, every depth obstruction vanishes along its descendants, and every map $\overline{L}_m$ has zero kernel. Then at most one compatible lift branch extends the initial approximation.*

Proof. By theorem 23.2, the correction classes at each depth form a coset of $\operatorname{Ker} \overline{L}_m$. A zero kernel leaves at most one correction class, so induction gives at most one branch. □

Theorem 23.8 (Complete filtered lifting)**.** *Assume there is a compatible sequence*

$$x_{m+1} \equiv x_m \pmod{F^m X}$$

with

$$\Phi(x_m) \in F^m Y$$

for every m. Then x_m converges to an element $x_\infty \in X$ and

$$\Phi(x_\infty) = 0.$$

In particular, if the obstruction vanishes at every depth and compatible corrections are chosen, an exact solution exists.

Proof. For every fixed r, the congruences imply

$$x_n - x_m \in F^r X$$

whenever $n \geq m \geq r$. Thus (x_m) is Cauchy for the filtration topology. Completeness gives a limit x_∞.

Fix r. For every $m \geq r$, one has $\Phi(x_m) \in F^m Y \subseteq F^r Y$. The submodule $F^r Y$ is open and therefore closed in the filtration topology. Continuity of Φ gives $\Phi(x_m) \to \Phi(x_\infty)$, so $\Phi(x_\infty) \in F^r Y$. Since this holds for every r and Y is separated,

$$\Phi(x_\infty) \in \bigcap_r F^r Y = 0.$$

□

Theorem 23.9 (Naturality of filtered obstructions)**.** *Suppose a morphism of filtered lifting problems sends x_m to x'_m, intertwines L_m with L'_m, and sends $\Phi(x_m)$ to $\Phi'(x'_m)$ on the associated graded pieces. Then the induced cokernel map sends*

$$\mathrm{Ob}_m(x_m) \quad \textit{to} \quad \mathrm{Ob}'_m(x'_m).$$

Proof. Apply theorem 21.3 to the induced commuting square on the depth m graded pieces. □

Example 23.10 (Three lifting outcomes)**.** The scalar equation $x^2-2=0$ at $x=3$ modulo 7 has invertible derivative. It has one correction modulo 49 and gives $x=10$ modulo 49.

For $x^2-3=0$ at $x=0$ modulo 3, the derivative is zero and the defect class is nonzero. No correction exists.

For $x^2=0$ at $x=0$ modulo 3, the derivative and defect both vanish. Every correction digit lies in the kernel, so the lift branches.

Proposition 23.11 (Maurer Cartan lifting obstruction)**.** *Let $\mathcal{L}$ be a complete filtered differential graded Lie algebra over a ring in which two is invertible. Assume*

$$dF^i\mathcal{L} \subseteq F^i\mathcal{L}, \qquad [F^i\mathcal{L}, F^j\mathcal{L}] \subseteq F^{i+j}\mathcal{L}.$$

For a degree one element η, put

$$\Omega(\eta) = d\eta + \frac{1}{2}[\eta,\eta].$$

Suppose $\Omega(\eta) \in F^m\mathcal{L}^2$ with $m \geq 1$. For every $h \in F^m\mathcal{L}^1$,

$$\Omega(\eta+h) \equiv \Omega(\eta) + d_\eta h \pmod{F^{m+1}\mathcal{L}^2}, \qquad d_\eta = d + [\eta,\cdot].$$

The obstruction to improving η by one depth is therefore the class of $\Omega(\eta)$ in the cokernel of

$$\mathrm{gr}^m(d_\eta) : \mathrm{gr}^m\,\mathcal{L}^1 \longrightarrow \mathrm{gr}^m\,\mathcal{L}^2.$$

Proof. Direct expansion gives

$$\Omega(\eta+h) = \Omega(\eta) + dh + [\eta,h] + \frac{1}{2}[h,h].$$

The filtration rule places $[h,h]$ in $F^{2m}\mathcal{L}^2$, which is contained in $F^{m+1}\mathcal{L}^2$ because $m \geq 1$. The congruence follows. The last statement is theorem 23.2 applied to the induced graded correction map. □

24 Filtered obstruction complexes

A direct packet decomposition splits a complex into independent pieces, so cohomology splits packet by packet. A filtration retains relations between depths. Its associated graded cohomology can contain classes that disappear on later pages of a spectral sequence.

There are therefore two ways for fine obstruction data to give zero total obstruction. Each fine class may already vanish, or later differentials may remove nonzero classes through compatibility between depths. Only the first mechanism occurs in a direct sum of subcomplexes. A filtration can exhibit both.

Standard Fact 24.1 (Spectral sequence of a finite filtered complex)**.** Let $C^\bullet$ be a cochain complex with a finite decreasing filtration by subcomplexes

$$C^\bullet = F^0C^\bullet \supseteq F^1C^\bullet \supseteq \cdots \supseteq F^{N+1}C^\bullet = 0.$$

There is a natural spectral sequence with

$$E_0^{p,q} = \mathrm{gr}^p\,C^{p+q}, \qquad E_1^{p,q} = H^{p+q}(\mathrm{gr}^p\,C^\bullet),$$

which converges to the associated graded object of $H^{p+q}(C^\bullet)$ with its induced filtration. A filtered cochain map induces a morphism of spectral sequences.

This is the standard construction for a filtered complex, in the form given in [18, 12]. When $C^\bullet$ is a correction or deformation complex, the groups on its pages can be read as graded obstruction candidates. For an arbitrary filtered complex, they are cohomology classes of the associated graded problem.

> **Lifting and survival answer different questions**
>
> A depth obstruction tests whether a chosen approximation admits one further correction. A spectral sequence follows cohomology classes of a filtered complex through later differentials. Turning successive corrections into an exact solution also uses compatible choices, completeness, continuity, and separatedness.

Proposition 24.2 (Fine obstruction survival)**.** *A class on the first page contributes to the total obstruction only if it is represented by a permanent cycle, so it is neither killed nor identified as a boundary on any later page. If the spectral sequence degenerates at the first page, then $H^n(C^\bullet) = 0$ exactly when $E_1^{p,n-p} = 0$ for every p.*

Proof. Convergence identifies the last page with the associated graded object of the total cohomology. A first page class contributes to that object exactly when it determines a nonzero class on the last page. If the sequence degenerates at the first page, the associated graded pieces of $H^n(C^\bullet)$ are the groups $E_1^{p,n-p}$. A finite filtered module is zero exactly when all its associated graded pieces are zero. □

Corollary 24.3 (Direct packet complexes)**.** *Suppose*

$$C^\bullet = \bigoplus_{\alpha \in I} C_\alpha^\bullet$$

is a finite direct sum of subcomplexes. Then

$$H^n(C^\bullet) \cong \bigoplus_{\alpha \in I} H^n(C_\alpha^\bullet).$$

A total cohomology class vanishes if and only if every packet class vanishes.

Proof. The differential acts independently on the summands. Kernels and images therefore split as finite direct sums, and so does their quotient. □

Example 24.4 (A fine class killed at the next page)**.** Let $C^0 = \Bbbk x$ and $C^1 = \Bbbk y$, with differential $d(x) = y$. Give C^0 the filtration

$$F^0C^0 = \Bbbk x, \qquad F^1C^0 = 0,$$

and give C^1 the filtration

$$F^0C^1 = F^1C^1 = \Bbbk y, \qquad F^2C^1 = 0.$$

The associated graded differential is zero in the relevant positions, so both x and y give nonzero classes on the next page. The next differential sends the class of x to the class of y. The total complex is acyclic. Thus nonzero fine obstruction classes can cancel through a later refinement differential.

Remark 24.5. The spectral sequence does not guarantee that a total obstruction vanishes. It records which fine classes survive and which are removed by compatibility across depths.

25 Nonlinear elimination

Let U, V, Y_P, Y_Q be Banach spaces. On an open neighborhood of $(u_0, v_0) \in U \times V$, split the variables and equations as

$$x = (u, v), \qquad F(u, v) = \begin{pmatrix} F_P(u, v) \\ F_Q(u, v) \end{pmatrix}.$$

Assume F takes values in $Y_P \times Y_Q$, is twice continuously Fréchet differentiable near (u_0, v_0), satisfies $F_Q(u_0, v_0) = 0$, and

$$A = D_v F_Q(u_0, v_0)$$

is a Banach space isomorphism from V to Y_Q. The implicit function theorem gives a local map $v = \varphi(u)$, with $\varphi(u_0) = v_0$, satisfying

$$F_Q(u, \varphi(u)) = 0.$$

Define the reduced equation

$$R(u) = F_P(u, \varphi(u)).$$

Theorem 25.1 (Exact local elimination). *After the neighborhoods are chosen small enough, the equation*

$$F(u, v) = 0$$

is equivalent to

$$v = \varphi(u) \qquad \text{and} \qquad R(u) = 0.$$

Thus eliminating the hidden variable loses no local solutions.

Proof. The hidden equation $F_Q(u, v) = 0$ has the unique local solution $v = \varphi(u)$ by the implicit function theorem. After this identity is imposed, the remaining component of F is exactly $R(u)$. □

Theorem 25.2 (Nonlinear Schur derivative). *At (u_0, v_0),*

$$DR = D_u F_P - D_v F_P A^{-1} D_u F_Q.$$

Proof. Differentiate $F_Q(u, \varphi(u)) = 0$:

$$D_u F_Q + A D\varphi = 0.$$

Thus

$$D\varphi = -A^{-1} D_u F_Q.$$

Substitute this into

$$DR = D_u F_P + D_v F_P D\varphi.$$

□

The hidden variable still contributes

The reduced derivative includes the response of the hidden variable to a retained variation. That response travels through the inverse hidden derivative and then back into the retained equation.

The Schur term records this contribution. Its use depends on the hidden derivative being invertible, as required in the local elimination setup.

The second derivative records how this response enters the first nonlinear feedback from the hidden variables.

Theorem 25.3 (Second Schur derivative)**.** *For retained directions h, k, set*

$$\hat{h} = (h, D\varphi h), \qquad \hat{k} = (k, D\varphi k).$$

Then

$$D^2\varphi[h,k] = -A^{-1}D^2F_Q[\hat{h},\hat{k}]$$

and

$$D^2R[h,k] = D^2F_P[\hat{h},\hat{k}] - D_vF_PA^{-1}D^2F_Q[\hat{h},\hat{k}].$$

All derivatives on the right are evaluated at (u_0, v_0).

Proof. Differentiate the identity $F_Q(u, \varphi(u)) = 0$ twice. The chain rule gives

$$D^2F_Q[\hat{h},\hat{k}] + AD^2\varphi[h,k] = 0.$$

This proves the first formula. Differentiate $R(u) = F_P(u, \varphi(u))$ twice and substitute the first formula. □

Corollary 25.4 (Packet jets of the reduced equation)**.** *If the retained and hidden spaces carry packet systems, then the first and second packet jets of R are obtained by projecting the formulas in theorems 25.2 and 25.3. Hidden packet feedback is carried by the factors involving A^{-1}.*

Proof. Apply the packet projectors to the input and output slots of the two derivative formulas. □

Example 25.5 (Two retained variables and one hidden variable)**.** Suppose the derivative of F at a solution has block form

$$DF = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where D is invertible. Eliminating the hidden variable replaces the retained derivative A by

$$A - BD^{-1}C.$$

The feedback term is exact. Keeping only A would ignore the path from retained variables into the hidden variable and back.

Remark 25.6. Elimination and obstruction reduction address different situations. The Schur formula assumes that the hidden derivative is invertible. Lyapunov–Schmidt reduction allows a nontrivial kernel and cokernel, then leaves a reduced obstruction equation on the kernel directions.

26 Finite transport and holonomy

Let $\Gamma = (V, E)$ be a finite connected graph and fix a vector space X. At every vertex take one labeled copy of X. Every oriented edge $u \to v$ carries an invertible transport

$$T_{vu} \in \mathrm{GL}(X), \qquad T_{uv} = T_{vu}^{-1}.$$

Equivalently, one may begin with separate but isomorphic vertex fibers and choose one frame in each fiber. The matrices below are then the transports in those frames. For a path $\gamma = (v_0, \dots, v_r)$, set

$$T_\gamma = T_{v_r v_{r-1}} \cdots T_{v_1 v_0}.$$

For a loop, this product is its holonomy. In the based loop group, $\gamma_2\gamma_1$ means first traverse γ_1 and then γ_2, so $T_{\gamma_2\gamma_1} = T_{\gamma_2} T_{\gamma_1}$. Standard graph terminology may be found in [5].

An invertible map on each edge supplies the local transport data. A global frame must make these maps agree around every cycle. A spanning tree has no cycles, so a gauge can make every tree transport the identity. Each remaining edge closes a loop and retains that loop's holonomy.

Definition 26.1 (Gauge change)**.** A gauge is a family $G_v \in \mathrm{GL}(X)$. It changes the edge transports to

$$T'_{vu} = G_v T_{vu} G_u^{-1}.$$

Proposition 26.2 (Gauge covariance)**.** *For a loop γ based at v_0,*

$$\mathrm{Hol}'(\gamma) = G_{v_0} \mathrm{Hol}(\gamma) G_{v_0}^{-1}.$$

Thus the conjugacy class of each loop holonomy is gauge invariant.

Proof. Every intermediate gauge factor cancels with its inverse. The initial and terminal factors remain at the base vertex. □

Theorem 26.3 (Spanning tree normalization)**.** *Choose a root and a spanning tree $\mathcal{T}$ of Γ. There is a gauge in which every tree edge transport is the identity. In that gauge, the transport on each edge outside the tree is the holonomy of its fundamental cycle, with the orientation used to define that cycle.*

Proof. Set the gauge at the root to the identity. Propagate gauges along the unique tree paths so that every tree edge becomes the identity. An edge outside the tree closes one fundamental cycle. Since every tree factor is now the identity, the remaining edge transport is the cycle product. □

Theorem 26.4 (Cycle gluing criterion)**.** *There are vertex coordinates in which every edge transport is the identity if and only if every fundamental cycle holonomy is the identity.*

Proof. If all edge transports are identity maps, every loop product is the identity. Conversely, use theorem 26.3. Identity fundamental holonomy forces every edge outside the tree to become the identity as well. □

Corollary 26.5 (Number of independent cycle tests)**.** *For a connected graph with $|V|$ vertices and $|E|$ unoriented edges, a spanning tree leaves*

$$|E| - |V| + 1$$

fundamental cycle holonomies.

Proof. A spanning tree has $|V| - 1$ edges. Every remaining edge creates one fundamental cycle. □

26.1 Intertwining two transport systems

Let A be a transport system on copies of a fixed vector space V_0, and let B be a transport system on copies of a fixed vector space W_0. Fix a root r and an isomorphism

$$U_r : V_0 \longrightarrow W_0.$$

Theorem 26.6 (Fundamental holonomy intertwining criterion)**.** *The following conditions are equivalent.*

1. *There is a family of isomorphisms $U_v : V_0 \to W_0$ with the prescribed root value U_r and satisfying*

$$U_v A_{vu} = B_{vu} U_u$$

   for every oriented edge $u \to v$.

2. *For every fundamental loop γ based at r,*

$$U_r \operatorname{Hol}_A(\gamma) = \operatorname{Hol}_B(\gamma) U_r.$$

3. *The same identity holds for every loop based at r.*

When these conditions hold, the family (U_v) is uniquely determined by U_r.

Proof. If the edge intertwiners exist, multiply their equations around a loop. Every intermediate vertex map cancels. This proves that the first condition implies the third, and the third implies the second.

Conversely, choose a spanning tree. Starting from U_r, propagate U_v along the unique tree path by the rule

$$U_v = B_{vu} U_u A_{vu}^{-1}$$

on every tree edge. This is the only possible extension satisfying the tree edge equations. An edge outside the tree closes one fundamental loop. Its intertwining equation holds exactly when U_r intertwines the two corresponding fundamental holonomies. The second condition therefore supplies every remaining edge equation. Uniqueness follows from propagation along the tree. □

Corollary 26.7 (Scalar holonomy bridge)**.** *Suppose every vertex space is one dimensional. Two transport systems are gauge equivalent if and only if their holonomies agree on a fundamental cycle basis.*

Proof. Choose a nonzero root isomorphism. Scalar loop maps commute with it, so theorem 26.6 reduces to equality of the cycle scalars. □

Theorem 26.8 (Gauge classification of transports)**.** *Let $G \subseteq \mathrm{GL}(X)$ be the chosen structure group and fix a root r. Restrict both edge transports and gauge values to G. Gauge equivalence classes of G valued transport systems are naturally identified with*

$$\mathrm{Hom}(\pi_1(\Gamma, r), G)/G,$$

where G acts by conjugation. After a spanning tree and orientations of the remaining edges are chosen, this set is identified with simultaneous conjugacy classes of

$$b_1 = |E| - |V| + 1$$

tuples in G.

Proof. A transport system assigns a group element to every oriented edge, with inversion under reversal. The product along a based loop gives a homomorphism from the free group $\pi_1(\Gamma, r)$ to G. A gauge change conjugates this homomorphism by the gauge value at the root. Conversely, normalize a transport system on a spanning tree. The remaining edge values prescribe the images of a free basis of $\pi_1(\Gamma, r)$, and every homomorphism arises in this way. A further gauge preserving the tree normalization is constant at all vertices and acts by simultaneous conjugation. The number of non-tree edges is b_1. □

Corollary 26.9 (Abelian cohomology class)**.** *When the structure group A is abelian, gauge classes of transports are naturally*

$$H^1(\Gamma, A) \cong \mathrm{Hom}(H_1(\Gamma, \mathbb{Z}), A).$$

After a basis of $H_1(\Gamma, \mathbb{Z})$ is chosen, this group is identified with A^{b_1}.

Proof. An edge assignment is an A valued one cochain. A gauge change multiplies it by a coboundary. A graph has no two cells, so every one cochain is a cocycle. The usual computation of the first cohomology of a connected graph gives A^{b_1}. □

Corollary 26.10 (Projective transport)**.** *The same normalization and gluing statements hold for transports valued in a projective linear group. A projective transport is globally trivial exactly when all fundamental holonomies are identity elements in that projective group.*

Proof. Apply theorems 26.3 and 26.4 with the projective linear group as the structure group. □

Example 26.11 (A square loop)**.** A square graph has four vertices and four edges, so it has one fundamental cycle. After a tree gauge, three edges are identities and the fourth is the full loop holonomy. The system glues to one global frame exactly when that remaining matrix is the identity.

27 Smooth projector transport

Finite graph transport has a continuous counterpart. Let H be a Hilbert space and let

$$\mathcal{P}(t) = (P_\alpha(t))_{\alpha \in I}$$

be a complete orthogonal packet system with finitely many labels. Assume that it is continuously differentiable in operator norm for $t \in [0, T]$, where $T \geq 0$. Here every $P_\alpha(t)$ is a bounded self-adjoint projection, so $P_\alpha(t)^* = P_\alpha(t) = P_\alpha(t)^2$.

Zero projections are permitted in this section and contribute zero to all generator sums. When $T = 0$, every partition product is the empty product I. Completeness and orthogonality give

$$P_\alpha(t)P_\beta(t) = 0 \quad \text{for } \alpha \neq \beta, \qquad \sum_\alpha P_\alpha(t) = I.$$

Define the canonical generator

$$K(t) = \sum_\alpha \dot{P}_\alpha(t)P_\alpha(t).$$

Lemma 27.1 (Complete packet commutator identity)**.** *For every packet label β,*

$$\dot{P}_\beta(t) = [K(t), P_\beta(t)].$$

The generator is also skew-adjoint:

$$K(t)^* = -K(t).$$

Proof. Differentiating $P_\alpha^2 = P_\alpha$ gives

$$P_\alpha \dot{P}_\alpha P_\alpha = 0.$$

Differentiating $P_\beta P_\alpha = 0$ for $\alpha \neq \beta$ gives

$$P_\beta \dot{P}_\alpha = -\dot{P}_\beta P_\alpha.$$

Hence

$$KP_\beta = \dot{P}_\beta P_\beta$$

and

$$P_\beta K = -\dot{P}_\beta (I - P_\beta).$$

Their difference is $\dot{P}_\beta$. Also,

$$K + K^* = \sum_\alpha (\dot{P}_\alpha P_\alpha + P_\alpha \dot{P}_\alpha) = \sum_\alpha \dot{P}_\alpha = 0.$$

□

Theorem 27.2 (Canonical packet transport)**.** *Let $U(t)$ solve*

$$\dot{U}(t) = K(t)U(t), \qquad U(0) = I.$$

Then $U(t)$ is unitary and transports every packet exactly:

$$P_\alpha(t)U(t) = U(t)P_\alpha(0)$$

for all α.

Proof. The continuous bounded generator has an evolution inverse. Indeed, solve $\dot{V} = -VK$ with $V(0) = I$. Differentiation gives $(VU)' = 0$, hence $VU = I$. The product $W = UV$ satisfies $\dot{W} = KW - WK$ and $W(0) = I$. Uniqueness of this bounded linear equation gives $UV = I$. Thus U is invertible.

Skew-adjointness of $K(t)$ gives

$$\frac{d}{dt}(U(t)^*U(t)) = 0,$$

so $U(t)$ is unitary. By lemma 27.1,

$$\frac{d}{dt}(U(t)^{-1}P_\alpha(t)U(t)) = U(t)^{-1}(\dot{P}_\alpha - [K, P_\alpha])U(t) = 0.$$

The expression equals $P_\alpha(0)$ at time zero. □

Corollary 27.3 (One moving subspace)**.** *For one orthogonal projection $P(t)$ and its complement, the canonical generator is*

$$K(t) = [\dot{P}(t), P(t)].$$

The transport in theorem 27.2 maps $\operatorname{im} P(0)$ *unitarily onto* $\operatorname{im} P(t)$.

Proof. Apply the complete packet formula to $P(t)$ and $I - P(t)$, then simplify the sum. □

Definition 27.4 (Direct rotation of nearby packet systems)**.** Let $(P_\alpha)_{\alpha\in I}$ and $(Q_\alpha)_{\alpha\in I}$ be complete orthogonal packet systems on the same Hilbert space. Put

$$S(\mathcal{Q}, \mathcal{P}) = \sum_{\alpha\in I} Q_\alpha P_\alpha.$$

When $S(\mathcal{Q}, \mathcal{P})$ is invertible, define

$$\mathcal{U}(\mathcal{Q}, \mathcal{P}) = S(\mathcal{Q}, \mathcal{P})\big(S(\mathcal{Q}, \mathcal{P})^*S(\mathcal{Q}, \mathcal{P})\big)^{-1/2}.$$

Proposition 27.5 (Exact direct packet rotation)**.** *If*

$$\max(\{0\} \cup \{\|P_\alpha - Q_\alpha\| : \alpha \in I\}) < 1,$$

then $S(\mathcal{Q}, \mathcal{P})$ is invertible. The operator $\mathcal{U}(\mathcal{Q}, \mathcal{P})$ is unitary and satisfies

$$Q_\alpha \mathcal{U}(\mathcal{Q}, \mathcal{P}) = \mathcal{U}(\mathcal{Q}, \mathcal{P})P_\alpha$$

for every α.

Proof. The identities

$$Q_\alpha S(\mathcal{Q}, \mathcal{P}) = S(\mathcal{Q}, \mathcal{P})P_\alpha = Q_\alpha P_\alpha$$

follow from pairwise orthogonality. Also

$$S^*S = \sum_\alpha P_\alpha Q_\alpha P_\alpha, \qquad SS^* = \sum_\alpha Q_\alpha P_\alpha Q_\alpha.$$

Put $\delta = \max(\{0\} \cup \{\|P_\alpha - Q_\alpha\| : \alpha \in I\})$. On $P_\alpha H$,

$$P_\alpha Q_\alpha P_\alpha = P_\alpha - P_\alpha(P_\alpha - Q_\alpha)P_\alpha,$$

so this block is bounded below by $1-\delta > 0$. The same argument on $Q_\alpha H$ applies to $Q_\alpha P_\alpha Q_\alpha$. Thus both S^*S and SS^* are invertible. It follows that S is injective and surjective. The polar factor $\mathcal{U} = S(S^*S)^{-1/2}$ is unitary. Since S^*S commutes with every P_α, the intertwining identity for S passes to $\mathcal{U}$. □

Lemma 27.6 (One-step direct rotation estimate)**.** *Assume $\mathcal{P}(t)$ is twice continuously differentiable in operator norm on $[0,T]$. There are constants $h_0 > 0$ and $C > 0$ such that, whenever $t, t+h \in [0,T]$ and $|h| \le h_0$,*

$$\|\mathcal{U}(\mathcal{P}(t+h), \mathcal{P}(t)) - I - hK(t)\| \le Ch^2.$$

The constants can be chosen uniformly in t.

Proof. Taylor's theorem in the Banach space of bounded operators gives

$$P_\alpha(t+h) = P_\alpha(t) + h\dot{P}_\alpha(t) + E_\alpha(t,h), \qquad \|E_\alpha(t,h)\| \le C_1 h^2,$$

uniformly in t and α. Hence

$$S(\mathcal{P}(t+h), \mathcal{P}(t)) = I + hK(t) + E(t,h), \qquad \|E(t,h)\| \le C_2 h^2.$$

Since $K(t)^* = -K(t)$,

$$S^*S = I + E_2(t,h), \qquad \|E_2(t,h)\| \le C_3 h^2.$$

For small h, the spectrum of S^*S lies in a fixed compact subinterval of $(0,\infty)$. The functional calculus for $z^{-1/2}$ gives

$$(S^*S)^{-1/2} = I + E_3(t,h), \qquad \|E_3(t,h)\| \le C_4 h^2.$$

Multiplying the last two expansions proves the estimate. □

Theorem 27.7 (Direct rotations converge to packet transport)**.** *Assume the complete packet family $\mathcal{P}(t)$ is twice continuously differentiable in operator norm on $[0,T]$. For a partition*

$$0 = t_0 < t_1 < \cdots < t_N = T$$

with sufficiently small mesh, define

$$V_\pi = \mathcal{U}(\mathcal{P}(t_N), \mathcal{P}(t_{N-1})) \cdots \mathcal{U}(\mathcal{P}(t_1), \mathcal{P}(t_0)).$$

Then

$$V_\pi \longrightarrow U(T)$$

in operator norm, where U is the canonical transport from theorem 27.2. On a fixed compact interval the error is bounded by a constant times the mesh.

Proof. Let $U(t,s)$ be the exact propagator generated by K. Since K is continuously differentiable, the integral equation for $U(t+h,t)$ gives a uniform estimate

$$\|U(t+h,t) - I - hK(t)\| \le C_1 h^2.$$

By lemma 27.6, the direct rotation V_j from t_j to t_{j+1} therefore satisfies

$$\|V_j - U(t_{j+1}, t_j)\| \le C_2 h_j^2, \qquad h_j = t_{j+1} - t_j.$$

Both V_j and the exact propagator are unitary. A telescoping product comparison gives

$$\|V_\pi - U(T,0)\| \le \sum_j \|V_j - U(t_{j+1},t_j)\| \le C_2 \sum_j h_j^2 \le C_2 T|\pi|.$$

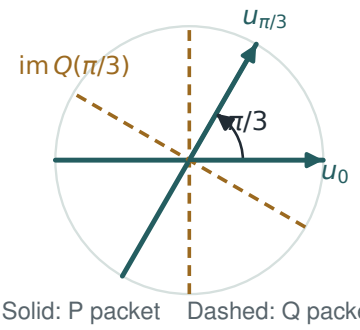
This proves norm convergence and the stated first-order global bound. □

Exact packet matching at each sample

Each direct rotation maps a sampled packet exactly to the corresponding packet at the next sample. The product therefore preserves packet labels throughout the partition. Convergence compares this discrete product with the canonical continuous transport. The uniform estimate uses twice continuous differentiability in operator norm, as stated in the theorem.

A rotating line and its complementary packet

$u_\theta = (\cos\theta, \sin\theta)$, $P(\theta) = u_\theta u_\theta^\mathsf{T}$, $Q(\theta) = I - P(\theta)$.



Canonical continuous transport

$$K = [P', P] = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$U'(\theta) = KU(\theta), \quad U(0) = I$$

$$U(\theta) = R(\theta)$$

$$S = \cos\delta\, R(\delta), \qquad S^\mathsf{T}S = \cos^2\delta\, I$$

For $|\delta| < \pi/2$: polar$(S) = R(\delta)$ exactly.

Steps N	Each increment	Ordered product
2	$\pi/6$	$R(\pi/3)$
4	$\pi/12$	$R(\pi/3)$
8	$\pi/24$	$R(\pi/3)$

Every mesh product is exact in this finite model.
The general theorem concerns convergence for broader projector paths.

Figure 6. A rotating rank-one packet and its orthogonal complement. With $u(\theta) = (\cos\theta, \sin\theta)$ and $P(\theta) = u(\theta)u(\theta)^\mathsf{T}$, the canonical generator is $[P', P] = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. For a positive step $\delta < \pi/2$, the polar unitary is the usual rotation $R(\delta)$. On each displayed mesh, the direct rotation product equals $R(\pi/3)$ exactly. This special planar example illustrates the construction. The general theorem supplies convergence for families whose step products need not already equal the continuous transport.

Theorem 27.8 (Convergence of sampled transports)**.** *Assume K is Lipschitz on $[0,T]$. For a partition*

$$0 = t_0 < t_1 < \cdots < t_N = T,$$

define

$$U_\pi = \exp\left((t_N - t_{N-1})K(t_{N-1})\right) \cdots \exp\left((t_1 - t_0)K(t_0)\right).$$

As the mesh $|\pi|$ tends to zero,

$$U_\pi \longrightarrow U(T)$$

in operator norm. There is a constant C, depending only on T, the supremum norm of K, and its Lipschitz constant, such that

$$\|U_\pi - U(T)\| \le C|\pi|.$$

Proof. On each interval, compare the exact propagator with the frozen exponential. Variation of constants and the Lipschitz bound give a local error bounded by a constant times the square of the interval length. Every factor is unitary, so multiplication does not amplify norms. Summing the local errors gives a bound by a constant times

$$\sum_j (t_{j+1} - t_j)^2 \le T|\pi|.$$

□

Corollary 27.9 (Continuous holonomy from graph products)**.** *Assume the complete packet family is twice continuously differentiable in operator norm on $[0,T]$. If $P_\alpha(0) = P_\alpha(T)$ for every α, then $U(T)$ preserves every initial packet. Its restriction to each packet is the holonomy of the closed packet path. The products in theorems 27.7 and 27.8 converge to these holonomies. The direct rotation product has the added property that every factor maps each sampled packet exactly to its successor.*

Proof. The transport identity in theorem 27.2 gives packet preservation at the endpoint. The two convergence statements follow from theorems 27.7 and 27.8. □

Example 27.10 (A rotating line)**.** In $\mathbb{R}^2$, let

$$v(t) = (\cos t, \sin t)$$

and let $P(t)$ be the orthogonal projection onto the line spanned by $v(t)$. The canonical transport is the ordinary rotation matrix

$$U(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}.$$

It carries the initial line and its orthogonal complement to the two moving packets. Products of short rotations converge to the same transport.

Standard Fact 27.11 (Curvature of a projected connection)**.** Let P be a smooth orthogonal projector-valued map on a parameter manifold. The connection

$$\nabla = P\,d$$

on the range bundle has curvature

$$\mathcal{F} = P(dP \wedge dP)P.$$

Proof. For a section s satisfying $Ps = s$,

$$\nabla^2 s = P\, d(P\, ds) = P(dP) \wedge ds.$$

Since $ds = (dP)s + Pds$ and $P(dP)P = 0$, the surviving term is

$$P(dP \wedge dP)Ps.$$ □

28 Realization data and limits of the calculus

The smooth transport theorem starts with a declared projector family. Applying it to a physical model requires a construction of that family and, where relevant, a separate continuum limit. Triviality of holonomy is a further geometric question. The curvature formula in standard fact 27.11 is classical. Relating it to global holonomy requires geometric information as in [7, 8, 10].

An application begins by specifying the following objects:

1. an ambient nonlinear equation or continuation problem
2. the allowed correction space
3. a linear correction map or a correction complex
4. the defect whose class is to be tested
5. a packet decomposition or filtration compatible with the correction data
6. transition maps between refinement levels
7. transport maps when local solutions must be glued.

Once these objects are fixed, the cokernel and cone constructions are functorial by theorems 21.3 and 21.11. Packet splitting and descent follow from theorem 21.4, proposition 21.6, and corollary 21.12. Relative loss under coarsening is measured by theorem 21.7.

Filtered lifting follows from theorems 23.2, 23.6 and 23.8. Gluing and comparison of transport systems follow from theorems 26.4 and 26.6. Smooth packet transport follows from theorems 27.2 and 27.7.

Exact packet coordinates give a representation of the data. Solving the associated problem still requires the correction and lifting hypotheses. The correction map belongs to the problem data, and its cokernel is canonical relative to that choice. The cokernel may be infinite dimensional. In the Hilbert setting, the orthogonal obstruction projector requires closed range.

Coarse vanishing can erase a nonzero fine obstruction, and higher nonlinear terms can obstruct a problem whose linear class vanishes. A spectral sequence records which classes survive the later differentials. An application of finite graph holonomy or smooth projector transport to a continuum field theory requires the additional realization data just described.

Whenever a nonlinear continuation problem supplies a correction map, its residual can be recorded in an exact quotient or cohomology object. Under the stated hypotheses, these objects admit packet decomposition, filtered refinement, natural transport, and holonomy tests. They identify the obstruction that a correction or gluing step must address.

A Visual atlas

The plates give exact finite models for the algebraic constructions in this volume. Their stable labels run from Plate III.1 to Plate III.24. Each plate names the related statements in the chapter. These models make the hypotheses and map directions visible. They do not extend the finite Lean contract to the analytic results of Part II.

> **Three routes through the atlas**
>
> For typed transfer, read Plate III.3 to Plate III.8. For coordinates, connections, and gauge, read Plate III.9 to Plate III.18, followed by Plate III.23. For candidate witnesses and trees, read Plate III.19 to Plate III.22, followed by Plate III.24. Plate III.1 and Plate III.2 introduce the objects and the questions they answer.

Products act in traversal order. The first edge traversed is the rightmost operator. Rows represent targets and columns represent sources.

Three objects, three different questions

Chapter III, mathematical text: Section 1, Section 4, Section 5

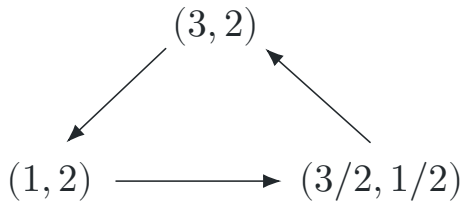
An edge operator moves data between named packets. An exact coordinate change describes the same data in another complete chart. A connection supplies an additional choice of transport. The three can use similar arrows while making different mathematical assertions.

EDGE ACTION

0	0	0
2	0	0
0	0	0

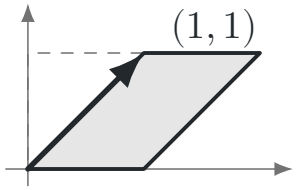
$e_0 \longmapsto 2e_1$

EXACT COORDINATES



one unchanged ambient vector

CHOSEN CONNECTION



$H = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) \neq I$

Three actual models used below: a single supported matrix entry, one ambient vector in three coordinate systems, and a shear obtained as chosen loop transport. The coordinates in the middle are not three different ambient vectors.

One chapter, three tests

For an edge word, ask whether the target of each step matches the source of the next. For exact coordinates, ask whether analysis and synthesis are inverses. For a connection, ask whether its transports through two steps agree with its declared direct transports. None of these tests can replace another.

This guide follows explicit integer and rational matrices through these tests. Each example has an inspectable input, calculation, and conclusion. Vector drawings display those calculations rather than replace them. A graph with two paths can have zero net transfer. A loop with trace two and determinant one can still differ from the identity.

Scope of the worked models

The worked models illustrate the finite algebraic core. The main text also develops rooted tree expansions, nonlinear defects, obstruction objects, filtered lifting, and analytic transport under their stated hypotheses. Those ordinary mathematical developments have a wider scope than the finite Lean contract. The present models explain the finite mechanisms and provide a route into the broader chapter.

> **Read this as:** Follow the object, not the arrow shape. A typed transfer can vanish, a complete coordinate loop must telescope, and an independently chosen connection can have genuine algebraic holonomy.

An atlas of the objects

Chapter III, mathematical text: Definition 2.1, Definition 5.5, Definition 7.4

Rows are targets and columns are sources throughout. If E_{ji} is a matrix unit, then $E_{ji}e_i = e_j$. The notation e_i means a basis vector. A named graph edge is written a, b, c, d or e, f when its role is explicit.

Object	What it records	First worked page
$\mathrm{src}, \mathrm{tgt} : E \to V$	The ordered endpoints of each distinct edge, including parallel edges.	Plate III.3 to Plate III.4
$L_e = c(e)E_{\mathrm{tgt}(e),\mathrm{src}(e)}$	One operator supported on a packet.	Plate III.3
$L_w = L_{e_n} \cdots L_{e_1}$	Execute the written word from left to right.	Plate III.4 to Plate III.5
$T = \sum_e L_e$	Add all edge operators before taking powers.	Plate III.4 to Plate III.7
F_a	A synthesis frame from chart a into the ambient space.	Plate III.9
$\mathcal{A}_a, \mathcal{S}_a$	Inverse analysis and synthesis maps, which may be maps of sets.	Plate III.10
$J_{b\leftarrow a}$	Canonical coordinate transport, forced by the exact charts.	Plate III.9 to Plate III.10
$\Gamma_{b\leftarrow a}$	Additional connection data, with reversibility stated explicitly.	Plate III.11 to Plate III.12
$\Delta_{c,b,a}$	The difference between transport through two steps and direct transport, a map from C_a to C_c.	Plate III.12
$H_{c,b,a}$	Loop transport based at a, an automorphism of C_a.	Plate III.13
G_a	An invertible gauge change at each chart.	Plate III.14 to Plate III.15
$\Pi_\mu \, \mathrm{Ob} \, \Pi_\nu$	Block from input to output of $\mathrm{Ob} = H - I$.	Plate III.16 to Plate III.18
$r(x, p)$	A residual at packet p for the named candidate x.	Plate III.19
$t, \mathrm{inputs}(t)$	A finite operation tree and its syntactically used input labels.	Plate III.20

Reading routes

For transfer, read Plate III.3 to Plate III.8. For the finite geometry models, read Plate III.9 to Plate III.18. For candidate certificates and nonlinear scope, read Plate III.19 to Plate III.22. Plate III.23 puts the matrix mechanisms together. Plate III.24 gives the formal boundary and replay guide.

> **Read this as:** A $C_a \to C_c$ defect and a $C_a \to C_a$ holonomy are not the same typed map. Returning the defect to the base fiber requires a specified inverse transport.

Read the product in execution order

Chapter III, mathematical text: Lemma 2.2, Theorem 2.3, Theorem 2.4

Work over $\mathbb{Z}$ with packets $0, 1, 2$. Let $e : 0 \to 1$ have coefficient 2, and let $f : 1 \to 2$ have coefficient 3. The elementary operators are

$$L_e = 2E_{10}, \qquad L_f = 3E_{21}.$$

The location of the nonzero entry records the type of the transfer, not an arbitrary matrix convention.

L_e

	0	1	2
0	0	0	0
1	2	0	0
2	0	0	0

L_f

	0	1	2
0	0	0	0
1	0	0	0
2	0	3	0

$L_f L_e$

	0	1	2
0	0	0	0
1	0	0	0
2	6	0	0

row = target / column = source



e first, f second; $\qquad L_e L_f = 0$

Each grid is an actual 3×3 matrix. The rightmost matrix factor acts first. In $L_f L_e$, packet 1 is the shared intermediate index. In $L_e L_f$ the interface is broken.

On a vector $(x_0, x_1, x_2)^{\mathsf{T}}$, the first step produces $(0, 2x_0, 0)^{\mathsf{T}}$, and the second produces $(0, 0, 6x_0)^{\mathsf{T}}$. Thus

$$L_f L_e = 6E_{20}, \qquad L_e L_f = 0.$$

For general edges, a summand in the entrywise product can survive only when the same intermediate packet is both $\mathrm{tgt}(e)$ and $\mathrm{src}(f)$:

$$(L_f L_e)_{yx} = \sum_z (L_f)_{yz} (L_e)_{zx}.$$

This is why a broken interface annihilates the entire product, even if both edge coefficients are nonzero.

What the zero says

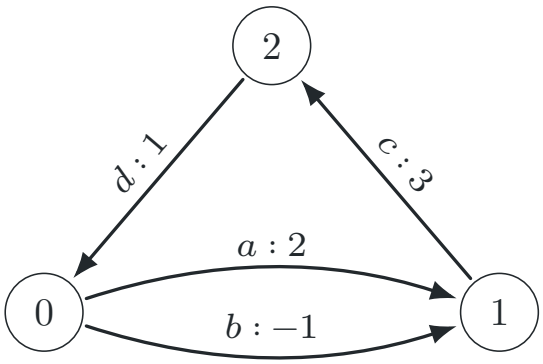
It says that this ordered composition does not pass data through the declared packet types. It does not say that either edge is individually zero. For a longer word, finding one broken adjacent interface is sufficient to prove that the whole word vanishes.

> **Read this as:** Write the traversed edges first. Then reverse their order when writing the operator product: e followed by f is $L_f L_e$.

All sixteen words, only five walks

Chapter III, mathematical text: Theorem 3.8, Corollary 3.9

Add four edges on three packets: $a : 0 \to 1$ with coefficient 2, $b : 0 \to 1$ with coefficient -1, $c : 1 \to 2$ with coefficient 3, and $d : 2 \to 0$ with coefficient 1. The parallel edges a and b are separate labels even though their source and target agree.



$$(T^2)_{20} = 6 - 3 = 3$$

first executed edge = column

	a	b	c	d
a	×	×	×	2
b	×	×	×	−1
c	6	−3	×	×
d	×	×	3	×

second executed edge = row

×: broken packet interface

The board tests all ordered pairs. Columns are the first executed edge and rows are the second. A shaded cell gives the ordered coefficient of a valid walk. A cross means the product is zero by type mismatch.

The surviving words are ac, bc, cd, da, db. Their coefficients are $6, -3, 3, 2, -1$, respectively. Their endpoints determine where these numbers enter the matrix:

$$T = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}, \qquad T^2 = \begin{pmatrix} 0 & 3 & 0 \\ 0 & 0 & 1 \\ 3 & 0 & 0 \end{pmatrix}.$$

At row 2, column 0, the two parallel choices give $6 - 3 = 3$. Deduplicating the edge labels before summing would alter the operator. On the other hand, adding the parallel coefficients in the already assembled matrix is legitimate: both act in the same supported matrix position.

The same calculation extends from pairs to all finite words by distributivity. The included check enumerates every word through length six and independently accumulates the endpoint coefficients. The empty word is treated separately as the identity.

> **Read this as:** The matrix power is a signed, ordered walk sum. It is not simply the number of paths, and a graph edge is not identified solely by its endpoints.

Coefficients need not commute

Chapter III, mathematical text: Lemma 3.7, Section 9

The edge theorem allows a noncommutative coefficient semiring. Use $R = M_2(\mathbb{Q})$ and let the two coefficients themselves be matrices:

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Attach A to $0 \to 1$ and B to $1 \to 2$. The single surviving endpoint coefficient is BA, not AB.

A

1	1
0	1

B

1	0
1	1

BA

1	1
1	2

AB

2	1
1	1

$e : 0 \to 1$ carries A; $\quad f : 1 \to 2$ carries B

$$BA(1,0)^{\mathsf{T}} = (1,1)^{\mathsf{T}}, \qquad AB(1,0)^{\mathsf{T}} = (2,1)^{\mathsf{T}}$$

All entries are exact integers. The two products differ on the diagonal and act differently on the same input vector. This example is within the scalar model with semiring coefficients by taking the scalar ring to be $M_2(\mathbb{Q})$.

The distinction is visible before any abstraction:

$$BA \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad AB \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

For a written walk $(e_1, \dots, e_n)$, its coefficient must therefore be

$$c^{\leftarrow}(w) = c(e_n)c(e_{n-1}) \cdots c(e_1).$$

A commutative example can hide an orientation mistake because reversing the coefficients changes nothing. This one cannot.

One ring is not a dependent family of fibers

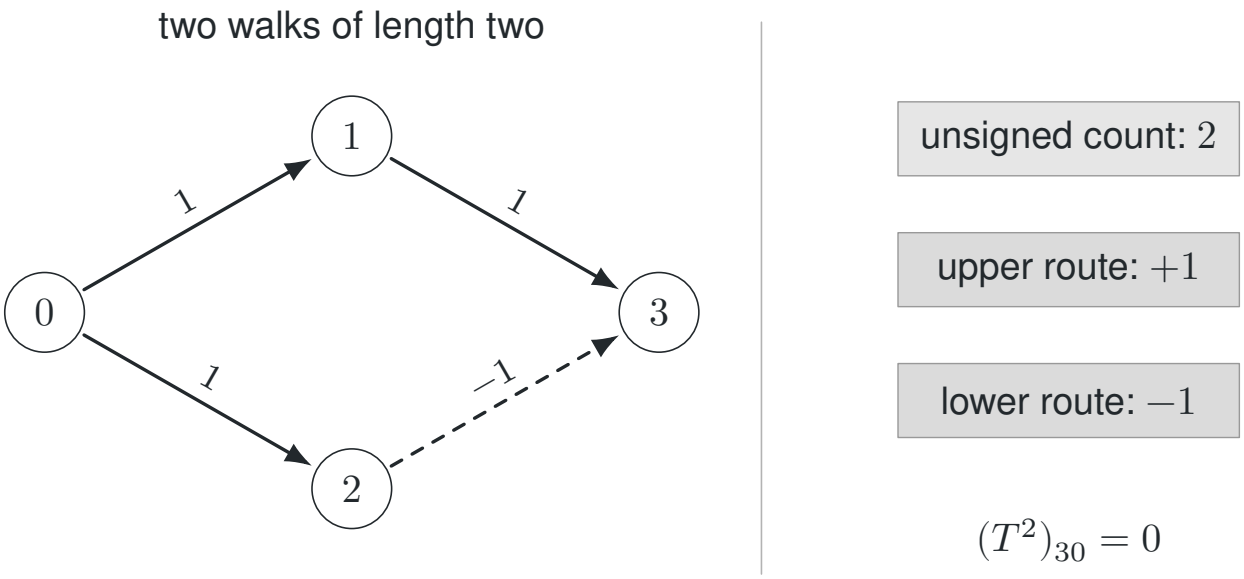
The 3×3 matrix over $M_2(\mathbb{Q})$ can be displayed as an ordinary 6×6 matrix with 2×2 blocks. This is a concrete representation of the example over one ring. It does not establish an arbitrary family of differently typed edge fibers, which is outside the finite formal contract's scalar interface for edge systems.

> **Read this as:** The order rule carries mathematical information. It survives exact tests over a noncommutative ring and must not be inferred from pictures drawn over scalar real numbers alone.

A reachable packet can receive zero

Chapter III, mathematical text: Corollary 3.9, Corollary 3.10

Consider a diamond: two routes of length two from packet 0 to packet 3. Three edge coefficients are 1 and the lower final coefficient is -1. Both routes are correctly typed. Neither is annihilated by an interface mismatch.



The dashed edge carries the coefficient -1. It is an actual edge, not an uncertain or missing one. Both paths exist. The unsigned count is two, while their exact operator contributions sum to zero.

The endpoint identity gives

$$(T^2)_{30} = (1)(1) + (-1)(1) = 0.$$

Thus the chapter's reachability implication has a definite direction:

$$\text{no compatible walk} \implies (T^n)_{ji} = 0.$$

Its converse is false. A zero entry may come from absent paths, zero coefficients, zero divisors in the coefficient ring, or cancellation among nonzero terms. The diamond exhibits the last possibility over $\mathbb{Z}$, where no zero divisor is needed.

Three different support records

The declared graph records permitted routes. The support of an individual word records the endpoints of that word when its coefficient is nonzero. The support of the sum records what remains after all contributions have been added. Passing from either of the first two records to the last requires the coefficients, not just adjacency.

A deliberately false test in the companion script asserts that reachability forces nonzero transfer. This diamond makes the test fail. That is a check against overreading the theorem, not a defect in the theorem.

> **Read this as:** Do not interpret a visible route as an occupied output. Graph support is a useful upper bound. Signed operator support can be strictly smaller.

Reachability is only an upper bound

Chapter III, mathematical text: Corollary 3.11

Keep the diamond and add an isolated packet 4. Start with the single seed $e_0 \in \mathbb{Q}^5$. Direct multiplication gives

$$Te_0 = e_1 + e_2, \qquad T^2 e_0 = 0.$$

Every later orbit vector is zero. The orbit hull generated by T is therefore exactly the span of the first two displayed columns.

$[e_0 \;\; Te_0 \;\; T^2 e_0]$ a basis of the orbit hull

0	1	0	0
1	0	1	0
2	0	1	0
3	0	0	0
4	0	0	0

1	0
0	1
0	1
0	0
0	0

$\text{Reach}(0) = \{0, 1, 2, 3\}$

$\text{Hull}_T(e_0) = \{(a, b, b, 0, 0)\}$

e_3 : reachable but absent

e_4 : unreachable and absent

rank $= 2$

The first matrix contains the actual orbit columns. The second supplies a basis of their span. The orbit hull lies inside the reachable subspace on four coordinates, but it has dimension two, not four.

A vector belongs to this hull precisely when it has the form

$$(a, b, b, 0, 0)^{\mathsf{T}}, \qquad a, b \in \mathbb{Q}.$$

Packet 4 is excluded because it is unreachable. Packet 3 is also absent, but for a different reason: the two contributions cancel. Even the reachable packets 1 and 2 remain linked. Powers of the single total operator do not separate them.

The admissible operator grammar matters

Allowing each individual edge operator is a different generation problem. For instance, applying the edge $0 \to 1$ and then $1 \to 3$ individually produces e_3. It does not follow that e_3 belongs to the orbit of the summed operator T. The chapter's theorem on orbit support concerns the explicitly stated operator action.

> **Read this as:** Unreachable means impossible under the declared action. Reachable means only that graph geometry has not ruled it out. Linear dependence and cancellation still have to be checked.

Paths through changing packet lists

Chapter III, mathematical text: Proposition 8.1

Here the packet lists have sizes $2, 3, 2, 1$ in succession. Let the actual maps be

$$\mathbb{Q}^2 \xrightarrow{A} \mathbb{Q}^3 \xrightarrow{B} \mathbb{Q}^2 \xrightarrow{C} \mathbb{Q}.$$

Unlike a power of one matrix, each step has its own entries and its own source and target list.

$A : \mathbb{Q}^2 \to \mathbb{Q}^3$

1	0
2	1
0	3

$B : \mathbb{Q}^3 \to \mathbb{Q}^2$

1	−1	2
0	1	1

$C : \mathbb{Q}^2 \to \mathbb{Q}$

2	−1

CBA

−4	6

	$(j,k)=(0,0)$	$(0,1)$	$(0,2)$	$(1,0)$	$(1,1)$	$(1,2)$	
$i=0$	2	−4	0	0	−2	0	$=-4$
$i=1$	0	−2	12	0	−1	−3	$=6$

The lower board expands one final row into all six choices of intermediate indices for each input packet. The entries are $C_{0j}B_{jk}A_{ki}$, with $k \in \{0,1,2\}$ and $j \in \{0,1\}$. Zero terms remain visible.

The intermediate and total products are

$$BA = \begin{pmatrix} -1 & 5 \\ 2 & 4 \end{pmatrix}, \qquad CBA = \begin{pmatrix} -4 & 6 \end{pmatrix}.$$

For input packet 0, the six contributions sum to $2 - 4 + 0 + 0 - 2 + 0 = -4$. For input packet 1, they sum to $0 - 2 + 12 + 0 - 1 - 3 = 6$. These are literal matrix entries, not probabilities or path counts.

The formal scope is deliberately precise

The finite formal contract's forms with two and three steps allow changing packet types. Its heterogeneous product theorem for arbitrary lengths allows changing *factors* over one common finite packet index. That formal statement does not assert a fully dependent chain of changing index types at arbitrary lengths. The finite rectangular illustration here uses exactly three factors.

For a single factor, the list of intermediate indices is empty and contributes that matrix entry once. For zero factors, the result is the identity on the common space. These conventions must not be replaced by an empty sum equal to zero.

> **Read this as:** Heterogeneous can mean different matrices or different spaces. State which one is allowed before using a path formula.

One vector, three exact frames

Chapter III, mathematical text: Theorem 4.1, Theorem 4.2

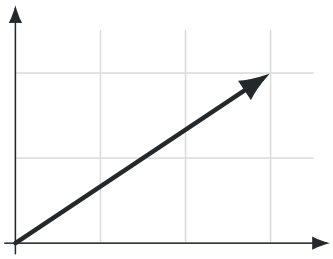
Let F_a synthesize chart coordinates into one ambient $\mathbb{Q}^2$. Choose

$$F_0 = I, \qquad F_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad F_2 = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix}.$$

The same ambient vector $x = (3,2)^{\mathsf{T}}$ has chart coordinates $u_a = F_a^{-1}x$.

chart 0

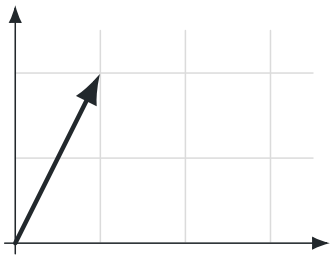
$(3,2)$



$$F_0 u_0 = (3,2)^{\mathsf{T}}$$

chart 1

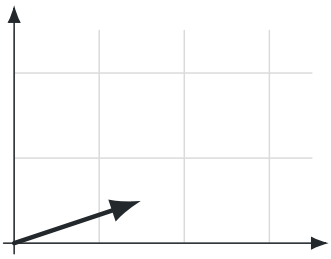
$(1,2)$



$$F_1 u_1 = (3,2)^{\mathsf{T}}$$

chart 2

$(\frac{3}{2}, \frac{1}{2})$



$$F_2 u_2 = (3,2)^{\mathsf{T}}$$

Each panel is a coordinate plane, not a different physical location. The numerical arrows differ because the frames differ. Multiplying each coordinate vector by its corresponding F_a recovers the same x.

Transport from chart a to chart b is forced:

$$J_{b \leftarrow a} = F_b^{-1} F_a.$$

The route $0 \to 1 \to 2$ agrees with the direct route because the middle frame cancels:

$$(F_2^{-1}F_1)(F_1^{-1}F_0) = F_2^{-1}F_0.$$

Returning to chart 0 cancels every remaining frame. The final coordinate vector is $(3,2)^{\mathsf{T}}$, not a sheared or rotated variant.

No metric is required

The frames need not be orthogonal. The second one is a shear, and the third has determinant two. Exact invertibility, rather than preservation of angles or lengths, gives the cocycle and loop identities. The same telescoping argument works for every finite chart path.

> **Read this as:** A canonical coordinate loop is a sequence of changes in description of one object. Nonidentity holonomy requires additional transport data, not merely nonorthogonal charts.

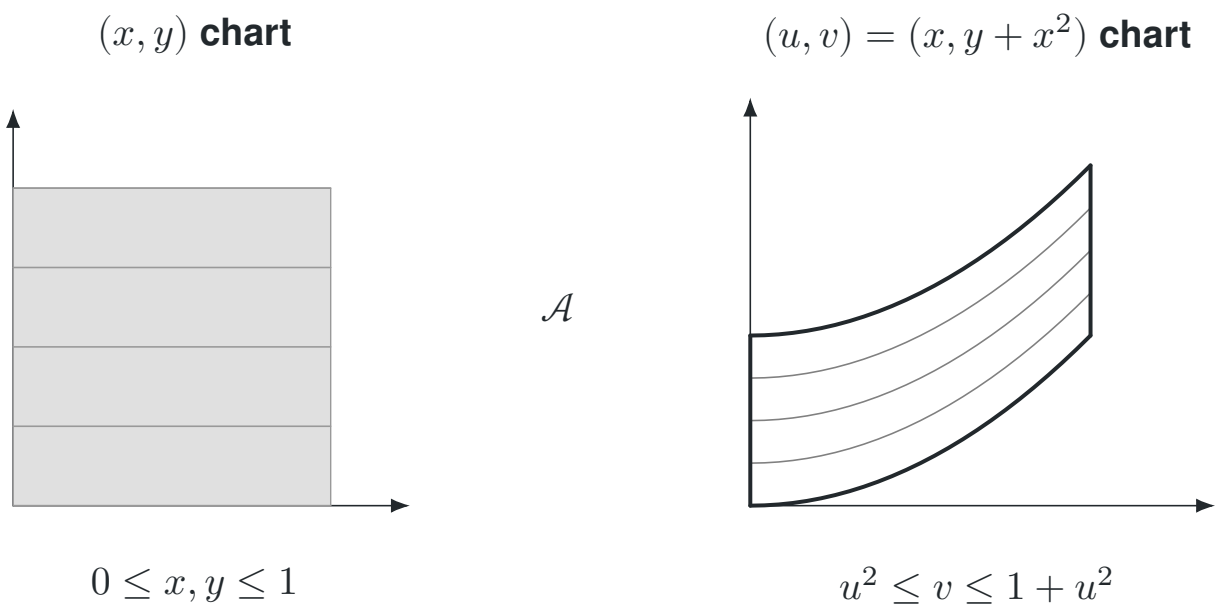
Nonlinear charts still cancel exactly

Chapter III, mathematical text: Proposition 10.1

Exact coordinate equivalences can be nonlinear. On $\mathbb{Q}^2$, use

$$\mathcal{A}(x, y) = (x, y + x^2), \qquad \mathcal{S}(u, v) = (u, v - u^2).$$

Both composites are the identity by direct substitution. The real drawing below is a geometric view of these same polynomial formulas.



Horizontal segments become parabolas, but no information is discarded. The drawing samples the displayed polynomial boundaries for rendering. Inverse identities and route equality are checked symbolically, not inferred from the curves.

Now represent the nonlinear operation $F(x, y) = (x + y, xy)$ in the new chart. Set $w = u + v - u^2$. Its correct coordinate formula is

$$\widehat{F}_b(u, v) = \mathcal{A}F\mathcal{S}(u, v) = (w,\ u(v - u^2) + w^2).$$

For $(x, y) = (1, 2)$, applying F first gives $(3, 2)$ and then $(3, 11)$ in the new chart. Changing the input first gives $(u, v) = (1, 3)$, and $\widehat{F}_b(1, 3) = (3, 11)$ as well.

The equality holds as a polynomial identity for all inputs. It is the naturality theorem for maps of sets: only the inverse laws for the coordinate maps are needed. No Taylor expansion, derivative, or higher chain rule is involved.

> **Read this as:** Nonlinearity alone does not create a coordinate defect. Reusing the old formula after changing coordinates would be an extra modeling choice, not exact conjugation.

Reversibility is a real hypothesis

Chapter III, mathematical text: Definition 5.1, Definition 5.5

Choose the forward scalar transports $\Gamma_{1\leftarrow 0} = 2$, $\Gamma_{2\leftarrow 1} = 3$, and $\Gamma_{2\leftarrow 0} = 5$. Their multiplicative comparison between transport through two steps and direct transport is $D^{\times} = 6/5$. A loop also needs a declared return map $\Gamma_{0\leftarrow 2}$.

connection data	$\Gamma_{0\leftarrow 2}$	H	$D^{\times}$	$\Gamma_{0\leftarrow 2}\Gamma_{2\leftarrow 0}$
self normalization only	$\frac{1}{6}$	1	$6/5$	$\frac{5}{6}$
reversible	$\frac{1}{5}$	$\frac{6}{5}$	$6/5$	1

$$\Gamma_{1\leftarrow 0} = 2, \quad \Gamma_{2\leftarrow 1} = 3, \quad \Gamma_{2\leftarrow 0} = 5$$

The first row has H = identity at this triangle, but fails its cocycle.

All transports are invertible rational scalars and transport from each chart to itself is one. The first row is permitted by the minimal interface. The second uses the inverse of the direct forward edge and therefore satisfies that part of reversibility.

In the first row, choose the return scalar $1/6$. Then

$$H = (1/6) \cdot 3 \cdot 2 = 1, \qquad \Gamma_{2\leftarrow 1}\Gamma_{1\leftarrow 0} - \Gamma_{2\leftarrow 0} = 1.$$

This triangle loop is the identity even though its cocycle equation fails. The return is not the inverse of the direct transport: $(1/6)5 \neq 1$. Thus one cannot transfer the reversible equivalence between defect and holonomy to the minimal interface.

In the reversible row, the return is $1/5$. Now the loop is $6/5$, and the additive defect of one is converted into the loop obstruction $1/5$ by that inverse return map.

Two safeguards

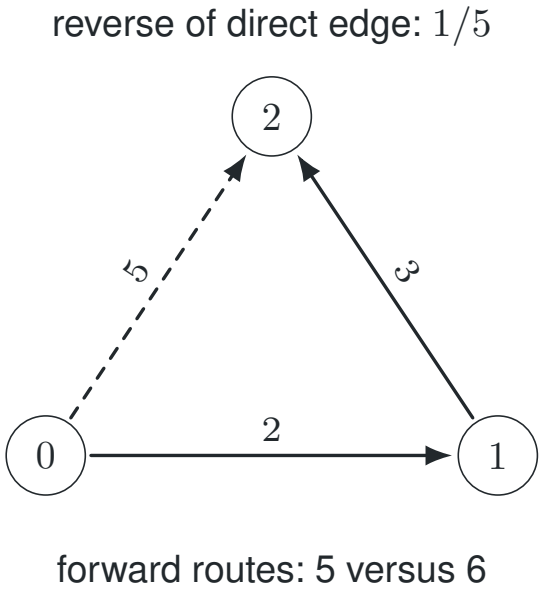
Global flatness in the minimal interface implies trivial loops by using its cocycle on the whole route, including the return. Separately, in the reversible interface, triviality of each triangle loop is equivalent to that triangle's cocycle. Neither assertion says that one isolated loop certifies all triangles.

> **Read this as:** Self normalization means $\Gamma_{a\leftarrow a} = I$. Reversible additionally means $\Gamma_{a\leftarrow b} = \Gamma_{b\leftarrow a}^{-1}$. These are different assumptions.

Measure the mismatch at its base

Chapter III, mathematical text: Theorem 5.10, Section 9

Retain the reversible scalar model with forward values $2, 3, 5$ and all reverse values given by inverses. Two routes from chart 0 to chart 2 multiply the input by 6 and 5, respectively. The additive discrepancy equals the original input.



loop input 1 returns $6/5$

loops	returned value
0	1
1	$\frac{6}{5}$
2	$\frac{36}{25}$
3	$\frac{216}{125}$
4	$\frac{1296}{625}$

The dashed direct edge is a specified comparison map, not a missing edge. The table shows the exact effect of traversing the same loop repeatedly. These are rational values of this example, not an analytic convergence claim.

The defect lives at the target of the forward routes:

$$\Delta_{2,1,0} = 3 \cdot 2 - 5 = 1.$$

The holonomy lives back at chart 0:

$$H_{2,1,0} = 5^{-1} \cdot 3 \cdot 2 = \frac{6}{5}, \qquad H - I = \frac{1}{5} = \Gamma_{0 \leftarrow 2} \Delta.$$

For input 5, the direct output and the output after two steps are 25 and 30. Returning by $1/5$ gives 5 and 6. The target discrepancy of 5 has become a base discrepancy of 1. The types and locations of these differences matter even in an example in one dimension.

Because the return map is invertible, it cannot turn a nonzero defect into zero. This is the mechanism behind $\Delta = 0 \iff H = I$ in the reversible interface.

> **Read this as:** Holonomy compares paths after both have been returned to the same base fiber. It is the transported mismatch, not an untyped difference between outputs in different charts.

The identity can hide in the spectrum

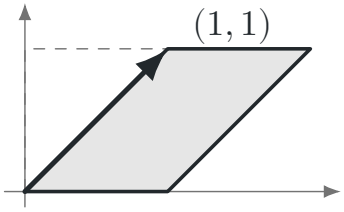
Chapter III, mathematical text: Definition 5.9, Definition 7.4

Put $\Gamma_{1\leftarrow 0} = H$, $\Gamma_{2\leftarrow 1} = I$, and $\Gamma_{2\leftarrow 0} = I$, with reverse maps given by inverses, where

$$H = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

The loop $0 \to 1 \to 2 \to 0$ then has exactly this shear as its holonomy.

The unit square under H



H

1	1
0	1

$H - I$

0	1
0	0

$$\operatorname{tr} H = 2, \quad \det H = 1, \quad (H - I)e_1 = e_0$$

The dashed square is the original set. The solid parallelogram is its exact image. The second coordinate basis vector moves to $(1, 1)^{\mathsf{T}}$. Its nonzero displacement is the first basis vector.

The usual scalar invariants can be misleading:

$$\operatorname{tr} H = 2, \qquad \det H = 1, \qquad \chi_H(t) = (t-1)^2.$$

These are also the invariants of the 2×2 identity matrix. Nevertheless,

$$\mathrm{Ob} = H - I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0, \qquad \mathrm{Ob}\, e_1 = e_0.$$

A complete obstruction test must therefore retain the operator or a certificate strong enough to detect its nonzero action. Trace, determinant, and eigenvalue lists alone do not distinguish this loop from the identity.

What the picture certifies, and what it does not

The plotted corners are the images of the four corners of the unit square under the displayed matrix. The equation $\mathrm{Ob}\, e_1 = e_0$ is the exact witness. No smooth connection, curvature tensor, or continuum limit is used in this model. Its holonomy is the finite algebraic loop product. The mathematical text treats smooth transport separately under additional hypotheses.

> **Read this as:** Flatness is the equality $H = I$, not agreement of a short list of spectral summaries. A single explicit moved vector can disprove that equality.

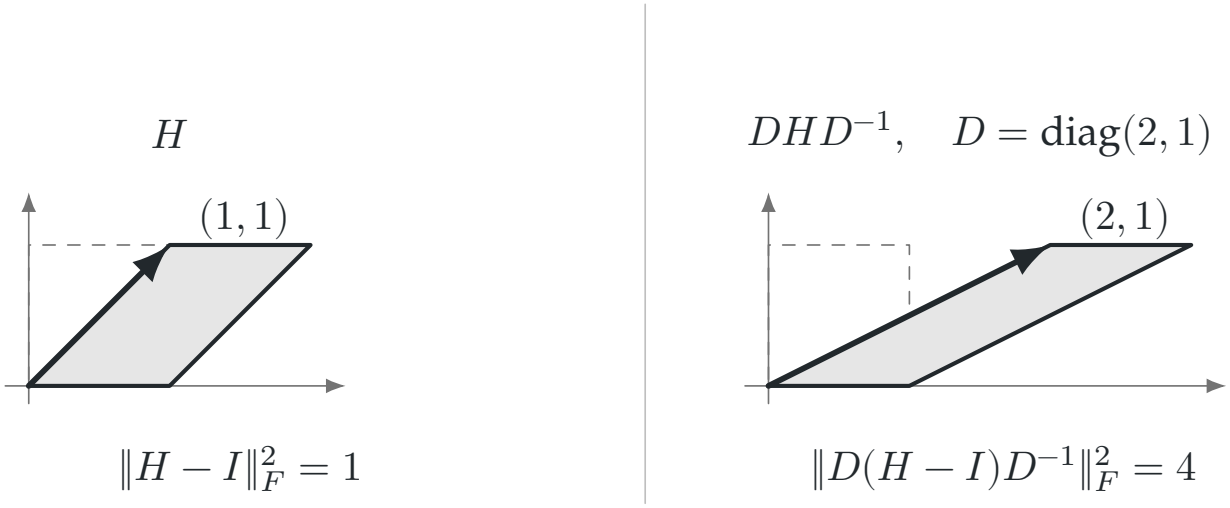
Gauge changes the picture, not flatness

Chapter III, mathematical text: Theorem 6.2

At the base chart, change coordinates by $D = \mathrm{diag}(2, 1)$. Transform every connection edge by the prescribed endpoint gauges. The loop transforms by conjugation, giving

$$H^D = DHD^{-1} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

The shear looks stronger in these fixed numerical axes, but the verdict on flatness is unchanged by gauge.



Both figures are exact images of a unit square under their displayed matrices. The Frobenius norms are computed in the indicated fixed coordinate bases. A nonorthogonal gauge need not preserve such a norm.

The rank of the obstruction is one on both sides. Trace, determinant, and characteristic polynomial are preserved under this conjugation. Conjugation also gives

$$D(H - I)D^{-1} = 0 \quad \Longleftrightarrow \quad H - I = 0.$$

An invertible coordinate change cannot turn a nonidentity loop into the identity. The fact that a particular residual norm changes does not contradict the invariance of exact vanishing.

The base chart determines the outer gauge

In a triangle product, adjacent intermediate gauges cancel:

$$(G_a \Gamma_{a \leftarrow c} G_c^{-1})(G_c \Gamma_{c \leftarrow b} G_b^{-1})(G_b \Gamma_{b \leftarrow a} G_a^{-1}) = G_a H G_a^{-1}.$$

Only the base gauge survives around the loop. For the additive defect, the surviving outer gauges instead belong to its different source and target fibers.

> **Read this as:** Exact nonvanishing is gauge invariant. The numerical size of a matrix entry or a chosen norm generally is not.

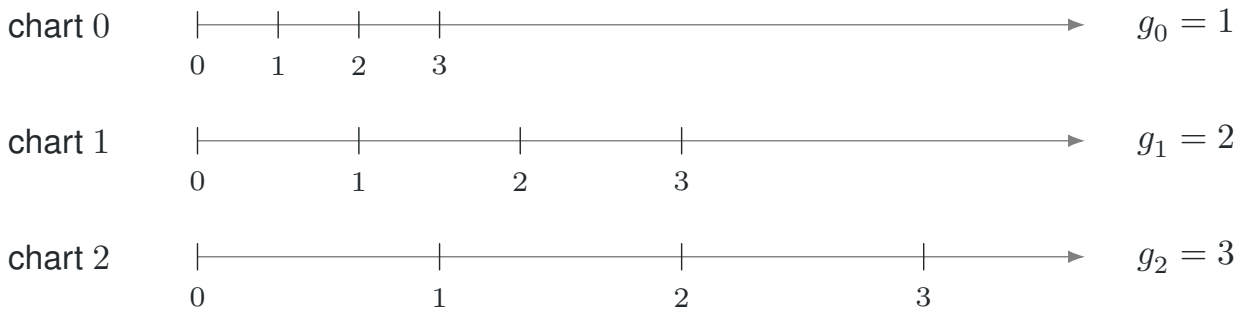
Flat is not the same as calibrated

Chapter III, mathematical text: Theorem 6.3

Make all three canonical charts literally identical, so $J_{b\leftarrow a} = 1$ over $\mathbb{Q}$. Choose scalar gauges $g_0 = 1$, $g_1 = 2$, and $g_2 = 3$, and define

$$\Gamma_{b\leftarrow a} = g_b g_a^{-1}.$$

The edge $0 \to 1$ is multiplication by two, although its canonical counterpart is the identity.



$$\Gamma_{1\leftarrow 0} = 2, \quad \Gamma_{2\leftarrow 1} = 3/2, \quad \Gamma_{2\leftarrow 0} = 3$$

The rulers display the chosen gauge scales: the marks are $g_a n$ for integer n. They visualize the supplied scalar factors, not an intrinsic metric. Relative transport is the ratio of the target and source scales.

The route through two steps $0 \to 1 \to 2$ has multiplier $(3/2)2 = 3$, equal to the direct route $0 \to 2$. More generally,

$$(g_c g_b^{-1})(g_b g_a^{-1}) = g_c g_a^{-1}$$

for every triple. Thus this connection is flat and reversible, but it is not calibrated to the original J, whose values are identities.

Recovering the gauge from one root

Choose a base chart o. The theorem constructs

$$G_a = \Gamma_{a\leftarrow o} J_{o\leftarrow a}, \qquad \Gamma_{b\leftarrow a} = G_b J_{b\leftarrow a} G_a^{-1}.$$

With $o = 0$ and $J = 1$, the recovered G_a are exactly $1, 2, 3$. The statement concerns the complete pairwise interface and its cocycle. It is not a topological claim about an arbitrary graph with missing direct comparisons.

> **Read this as:** Flatness makes the chosen transports independent of path. Calibration identifies them with one particular canonical transport before any gauge change.

Resolve the entire obstruction

Chapter III, mathematical text: Theorem 7.5, Corollary 7.6

Use three coordinate packets with projectors P_0, P_1, P_2, and set

$$H = I + 2E_{01} + 3E_{12}, \qquad \mathrm{Ob} = H - I.$$

Here H is invertible, so it can be the loop product of a reversible connection on three charts. Its entries outside the diagonal encode two effects that mix packets.

H

1	2	0
0	1	3
0	0	1

$\mathrm{Ob} = H - I$

0	2	0
0	0	3
0	0	0

	$\nu = 0$	$\nu = 1$	$\nu = 2$
$\mu = 0$	0	$2E_{01}$	0
$\mu = 1$	0	0	$3E_{12}$
$\mu = 2$	0	0	0

$$\mathrm{Ob}(4,5,6)^{\mathsf{T}} = (10,18,0)^{\mathsf{T}}$$

sum all nine blocks

The board on the right contains all nine actual embedded blocks $P_\mu \,\mathrm{Ob}\, P_\nu$. Row labels are output packets and column labels are input packets. Only the addresses $(0,1)$ and $(1,2)$ carry nonzero maps.

Completeness on both sides supplies the reconstruction:

$$\mathrm{Ob} = (P_0 + P_1 + P_2)\,\mathrm{Ob}\,(P_0 + P_1 + P_2) = \sum_{\mu,\nu} P_\mu \,\mathrm{Ob}\, P_\nu.$$

For $x = (4,5,6)^{\mathsf{T}}$, the input in packet 1 contributes $10e_0$ and the input in packet 2 contributes $18e_1$. Their sum is the full displacement $\mathrm{Ob}\, x$.

A block becomes a witness when it acts

The nonzero block $2E_{01}$ has the explicit source vector e_1, supported entirely in packet 1, and $P_0 \,\mathrm{Ob}\, e_1 = 2e_0 \neq 0$. Similarly, $P_1 \,\mathrm{Ob}\, e_2 = 3e_1 \neq 0$. The general theorem supplies such a witness classically when a nonzero block is known. The example displays it concretely.

> **Read this as:** An obstruction packet is not just a colored matrix cell. It consists of a specific label for a pair of packets and an exact nonvanishing statement about the corresponding map.

Why diagonal checks are insufficient

Chapter III, mathematical text: Theorem 7.5

Return to $\mathrm{Ob} = E_{01}$. With the two standard coordinate projectors, both diagonal blocks vanish:

$$P_0 \,\mathrm{Ob}\, P_0 = P_1 \,\mathrm{Ob}\, P_1 = 0.$$

The obstruction nevertheless survives in the block outside the diagonal $P_0 \,\mathrm{Ob}\, P_1 = E_{01}$.

Ob	$\sum_i P_i \,\mathrm{Ob}\, P_i$	Q_0	Q_1
$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix}$
nonzero	diagonal test misses it	$Q_0 + Q_1 = I, \quad Q_i^2 = Q_i$	

$$\sum_{i,j} Q_i \,\mathrm{Ob}\, Q_j = \mathrm{Ob}; \qquad Q_0^{\mathsf{T}} \neq Q_0$$

The test of diagonal blocks alone on the left misses the whole obstruction. On the right, the displayed oblique projectors still form a complete exact resolution. They need not be orthogonal projections for the Euclidean inner product.

There is a second failure mode. Supplying only P_0 gives a family of idempotents but not a complete one. The single tested block is zero, while the omitted packet carries the missing input. The implication "all tested blocks vanish, therefore the operator vanishes" requires the completeness hypothesis.

Exact reconstruction does not require right angles

For the displayed Q_0, Q_1,

$$Q_0 + Q_1 = I, \qquad Q_i^2 = Q_i, \qquad Q_0 Q_1 = Q_1 Q_0 = 0,$$

but $Q_0^{\mathsf{T}} \neq Q_0$. Expanding both copies of I still proves

$$\sum_{i,j} Q_i \,\mathrm{Ob}\, Q_j = \mathrm{Ob}\,.$$

The block reconstruction identity itself uses the condition that the projectors sum to the identity. Idempotence is additionally used when constructing a witness that remains supported in one packet after projection. The general formal reconstruction statement does not require pairwise annihilation.

> **Read this as:** Test every block from input to output of a complete resolution. Testing only the diagonal, or only the packets supplied so far, answers a weaker question.

A gauge can move every packet label

Chapter III, mathematical text: Proposition 7.7

The diagonal gauge on Plate III.14 commuted with each standard projector. Now use the gauge that mixes packets

$$G = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \mathrm{Ob} = E_{01}.$$

The total obstruction transforms by conjugation, but its pattern in the *fixed* standard packets changes.

Ob		G		$G\,\mathrm{Ob}\,G^{-1}$	
0	1	1	0	-1	1
0	0	1	1	-1	1
one nonzero entry		$GP_0 \neq P_0G$		four nonzero entries	

fixed axes: different packet pattern / moving axes: conjugate labeled blocks

The first obstruction has one nonzero entry. Its conjugate has four. Both have rank one and are nonzero. The change in fixed block pattern is not a change in whether the connection is flat.

Direct multiplication gives

$$G\,\mathrm{Ob}\,G^{-1} = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}.$$

For example, the original $(0,0)$ block vanishes, while the new $(0,0)$ block relative to the old coordinate projectors does not. This is why the theorem about preservation of each block's zero status requires $GP_i = P_iG$.

Transporting the packet frame as well

Define $Q_i = GP_iG^{-1}$. Then the identities cancel exactly:

$$Q_i(G\,\mathrm{Ob}\,G^{-1})Q_j = G(P_i\,\mathrm{Ob}\,P_j)G^{-1}.$$

Relative to these transported packets, the corresponding labeled blocks again have matching vanishing status. These are the same oblique projectors displayed on Plate III.17. One must distinguish moving the operator alone in a fixed resolution from moving the entire resolved description.

> **Read this as:** The total obstruction is intrinsic under gauge. Its location in a fixed packet grid is not, unless the gauge preserves that grid.

An obstruction rejects one candidate

Chapter III, mathematical text: Theorem 7.2

Let the candidates be rational numbers and the packet labels be $0, 1$. Define

$$r(x, 0) = x^2 - 1, \qquad r(x, 1) = x - 1.$$

A candidate is a solution when both residuals vanish. The five candidate columns below are a display of this explicit residual function, not a search over an unspecified class.

candidate x	-2	-1	0	1	2
$r(x, 0) = x^2 - 1$	3	0	-1	0	3
$r(x, 1) = x - 1$	-3	-2	-1	0	1
	rejected	rejected	rejected	solution	rejected

$$x = -1 : \quad r(-1, 0) = 0, \quad r(-1, 1) = -2$$

A shaded cell is a nonzero exact residual. At $x = -1$, the first equation passes but the second supplies an obstruction. At $x = 1$, both pass. The rejected candidate does not obstruct the successful one.

The statement proved by the generic certificate has the form

$$\neg(\forall p,\ r(x, p) = 0) \quad \Longleftrightarrow \quad \exists p,\ r(x, p) \neq 0.$$

The same candidate x appears on both sides. A certificate consists of the packet label and its nonvanishing proof. The general converse uses classical reasoning and does not itself provide an algorithm to find the label.

What would be required to rule out all solutions?

One would need the stronger assertion $\forall x\, \exists p,\ r(x, p) \neq 0$. The value $x = 1$ disproves that assertion in this example. A witness for $x = -1$ supplies no evidence for it. Likewise, finding an obstruction block for one triangle of one chosen connection does not classify every possible connection.

The term *obstruction* in this generic interface does not automatically mean a cokernel class, a cohomology group, a norm bound, or a minimal certificate. Those require additional constructions and theorems.

> **Read this as:** Keep the candidate attached to the witness. The theorem's strength is its exact logical scope, not a universal existence or nonexistence claim.

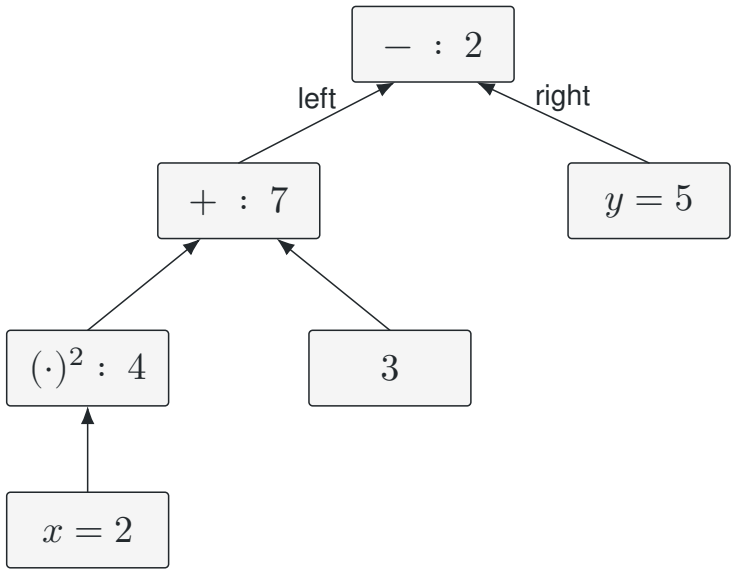
A tree reads its named leaves

Chapter III, mathematical text: Definition 11.1, Theorem 11.2

Use the grammar's constant, input, unary, and binary constructors to build

$$t = (x^2 + 3) - y.$$

Here "square," "add," and "subtract" are explicitly supplied meanings of operation symbols. At $x = 2, y = 5$, every node value can be evaluated exactly.



$$\text{inputs}(t) = \{x, y\}$$

unused input: -100 or 100

same value: 2

$x - x : \{x\}$ syntactically

but always zero

The rooted tree records the order of input slots. Arrows indicate evaluation from children to parent, not chart transport. An environment may contain additional variables, but this tree does not read them.

The input set is $\text{inputs}(t) = \{x, y\}$. If two environments agree at x and y, structural induction proves that the two evaluations agree: input leaves agree, constants agree, and every parent applies the same supplied operation to equal child values.

Syntactic locality is not minimal dependence

The tree $x - x$ has input set $\{x\}$, although its value is always zero under ordinary subtraction. Locality provides a sufficient agreement condition. It does not compute the smallest set of semantically relevant variables. Nor does it prove that two different trees are equivalent.

The semantics may act on arbitrary values. For example, a binary concatenation symbol applied to strings "ab" and "cd" gives "abcd." No linearity or differentiability is required. Conversely, swapping the ordered inputs of the subtraction root above changes the value from 2 to -2.

> **Read this as:** The proved tree theorem controls which input labels are read. It does not supply a nonlinear chain rule, a stability estimate, or a solution of a tree equation.

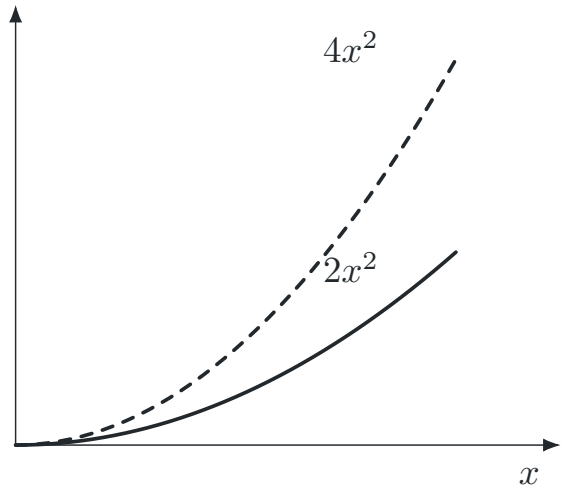
Where a nonlinear defect can enter

Chapter III, mathematical text: Proposition 10.1, Formalization target 11.7, Theorem 20.2

Take the unary operation $f(x) = x^2$ over $\mathbb{Q}$. Suppose both input and output transport are chosen as multiplication by two, while the same numerical formula f is reused at the target. The two routes give

$$2f(x) = 2x^2, \qquad f(2x) = 4x^2.$$

Their difference is $-2x^2$, which is generally nonzero.



x	0	1	2
$2f(x)$	0	2	8
$f(2x)$	0	4	16
$2f(x) - f(2x)$	0	-2	-8

correct chart formula: $f_b(v) = v^2/2$

Solid: $2x^2$. Dashed: $4x^2$. The curves are a real visualization of exact polynomial formulas. The table contains exact evaluations over the rationals. The mismatch is caused by the specified combination of transport and formula.

For a canonical coordinate change with input and output chart $v = 2x$, the target formula is not still v^2. It is

$$\hat{f}_b(v) = 2f(v/2) = \frac{v^2}{2}.$$

Consequently $\hat{f}_b(2x) = 2x^2$, and the exact coordinate defect vanishes. This is the same distinction illustrated in the example in two dimensions on Plate III.10.

An example is not a general propagation theorem

The polynomial calculation here is exact and checked symbolically. It explains a chosen node defect, while the mathematical text develops composition identities, iterated defects, fixed point stability, and derivative defects under explicit hypotheses. Those broader ordinary proofs must be read on their own terms. This one example and the finite Lean locality theorem do not certify them.

Read this as: A nonzero nonlinear route difference must come from identifiable extra data: a changed transport, an approximation, a restricted model, or a reused formula that was not conjugated correctly.

Mixed terms are not optional

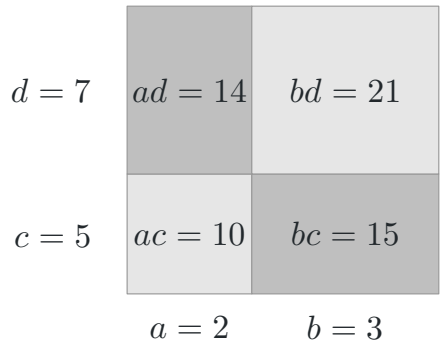
Chapter III, mathematical text: Formalization target 11.6, Theorem 18.3

A bilinear node distributes over two independently resolved inputs:

$$(a+b)(c+d) = ac + ad + bc + bd.$$

The four terms correspond to all ordered choices of one input packet from each slot. Keeping only matching packet labels loses the mixed terms.

$$(2+3)(5+7) = 60$$



Diagonal pieces are not the whole product

$$ac + bd = 10 + 21 = 31$$

$$ad + bc = 14 + 15 = 29$$

$$(2+3)^2 = 25 \neq 2^2 + 3^2 = 13$$

missing square term: $2 \cdot 2 \cdot 3 = 12$

The rectangle has width $2 + 3$ and height $5 + 7$, with differently scaled horizontal and vertical display units. Each cell is labeled by its actual product. The lighter diagonal pair contributes only 31 of the total 60.

For a unary square, inserting $a+b$ does not give a^2+b^2. The extra $2ab$ arises because the operation is nonlinear in its single argument. It can be analyzed by representing the square using a binary multiplication node with a repeated input, but that is an additional choice of syntax and semantics.

The broader tree theorem needs its hypotheses

General packet tree expansion needs correctly typed input slots, finite sets of packet labels at each node, and multilinearity in every slot used for distributive expansion. Empty inputs and unary nodes need explicit conventions. Slot order must survive even when values or operations do not commute. The mathematical text supplies rooted tree reconstruction and pruning results in its ordinary mathematical development.

This page supplies elementary exact examples of the issue. The finite algebraic Lean contract remains narrower: it contains syntactic input locality, not the general dependent tree expansion or the mathematical text's analytic convergence results.

> **Read this as:** A branching picture does not justify throwing away mixed interactions. Exact packet expansion requires the appropriate multilinearity and typing, not merely a diagram that appears finite.

From loop data to a packet witness

Chapter III, mathematical text: Theorem 5.10, Theorem 6.2, Corollary 7.6

Supply a reversible connection on three copies of $\mathbb{Q}^2$ by choosing

$$\Gamma_{1\leftarrow 0} = A, \qquad \Gamma_{2\leftarrow 1} = B, \qquad \Gamma_{2\leftarrow 0} = C = I,$$

and declaring every reverse edge to be the inverse. This time both nontrivial forward matrices are retained.

A

1	1
0	1

B

1	0
1	1

$H = BA$

1	1
1	2

$H - I$

0	1
1	1

$$e_0 = (1,0) \;\mapsto\; (1,0) \;\mapsto\; (1,1) \;\mapsto\; (1,1)$$

$$P_1(H - I)e_0 = e_1 \neq 0$$

---

replace direct edge $C = I$ by $C = BA$: then $C^{-1}BA = I$

The vector row starts with the source witness e_0 and follows it through A, then B, then C^{-1}. Returning to the base produces $(1,1)^{\mathsf{T}}$, whose displacement from the input is e_1.

The calculation gives

$$\Delta = BA - C = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad H = C^{-1}BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Choose the standard packet projectors. The $(1,0)$ block is nonzero, and the explicit candidate vector satisfies $P_1(H - I)e_0 = e_1$. This proves nonflatness of the supplied connection at the supplied triangle.

Repairing data is not the same as changing gauge

Replacing the direct map $C = I$ by $C = BA$, and updating its reverse, makes the new loop equal to I. That changes the connection. It cannot be a gauge of the original nonflat connection, because gauge preserves whether holonomy is the identity. For this complete example on three charts all cocycles then hold. Changing one edge in a larger network would require checking the other affected loops as well.

> **Read this as:** Declare the data, multiply in traversal order, form the defect, resolve all blocks, and exhibit an input on which one block acts nontrivially. Each step has a checkable mathematical object.

Read the evidence with the pictures

Chapter III, mathematical text: Section 12

The 24 worked plates and 22 exact figures connect finite models with the main text's broader nonlinear and analytic theory.

Read each evidence record in its scope

Evidence	What it supports
Formal development	The finite declarations and the additions in Sections 16 and 17 were compiled in the pinned environment. The Lean guide records the source scope, axiom audit, and kernel replay.
Ordinary mathematics	The mathematical text's expanded proofs, including nonlinear, obstruction, lifting, and smooth transport results. Inclusion in that text does not add these results to the finite Lean contract.
Exact model checks	The recorded run passed all 124 exact checks and rejected all four deliberately false controls. The scope is the displayed finite matrices, paths, polynomials, candidate tables, and trees.
Figure regeneration	All 22 editable TikZ figures are supplied with their exact model data and generators. The original figure sources and their earlier regeneration record remain preserved.

The computation record also contains 18,630 exact assertions for the examples in Sections 16 and 17. These finite calculations support the displayed mechanisms. General claims rest on their mathematical proofs and, where mapped, their Lean declarations.

Boundaries that remain in force

The formal edge system uses a single scalar coefficient semiring. The heterogeneous path theorem for arbitrary lengths uses a common packet index, with forms allowing changing types for two and three factors. General dependent tree expansion and analytic convergence lie outside this finite formal contract. The mathematical text states and proves its broader results with their own hypotheses.

The generic obstruction theorem rejects its named candidate. A nonzero holonomy block rejects flatness for its named loop data. Neither conclusion supplies a universal classification, a new arithmetic realization, or an algorithmic search guarantee.

Replay the worked layer

Check the exact examples with `python checks/verify_examples.py`.

Regenerate the figures with `python source/generate_graphics.py`.

Use a temporary copy to preserve the recorded evidence. The finite algebraic source distribution supplies the formal sources separately.

> **Read this as:** Use the figures to understand the mechanisms, the mathematical text to read the mathematical proofs, and the Lean guide to identify the exact formal contract. These are related records with distinct scopes.

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